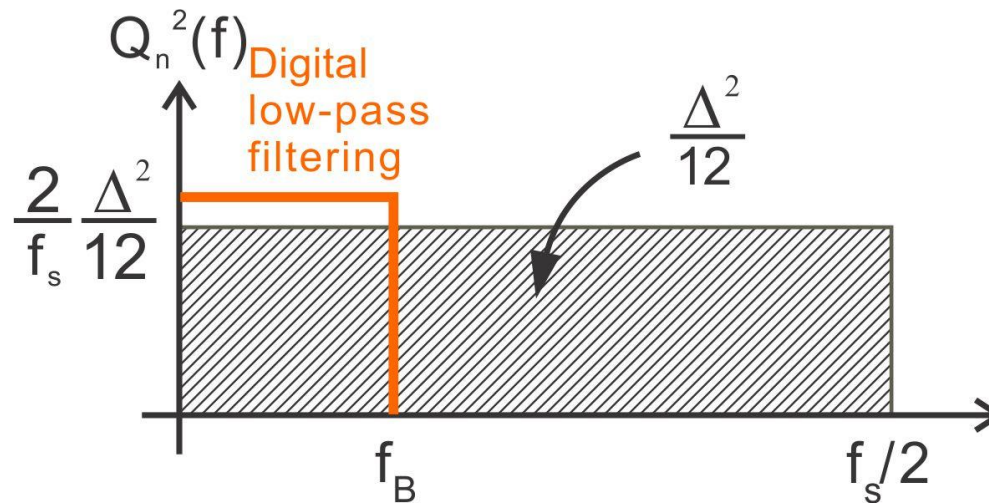


Oversampling ADCs

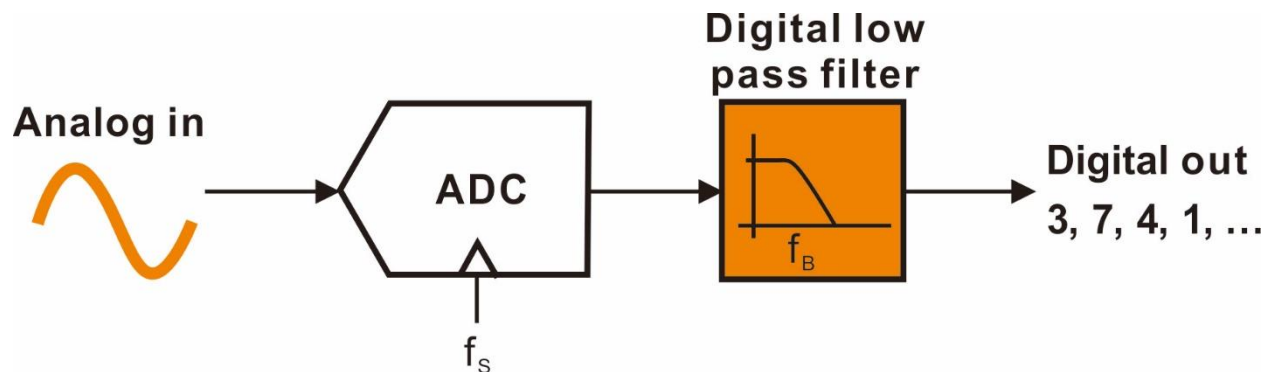
Nan Sun
Tsinghua University
nansun@Tsinghua.edu.cn

Quantization Noise



- We define oversampling ratio (OSR) = $f_s/2f_B$
- We can reduce quantization noise by oversampling
- Total in-band quantization noise = $(2f_B/f_s) (\Delta^2/12) = (\Delta^2/12) / \text{OSR}$

Reducing Quantization Noise

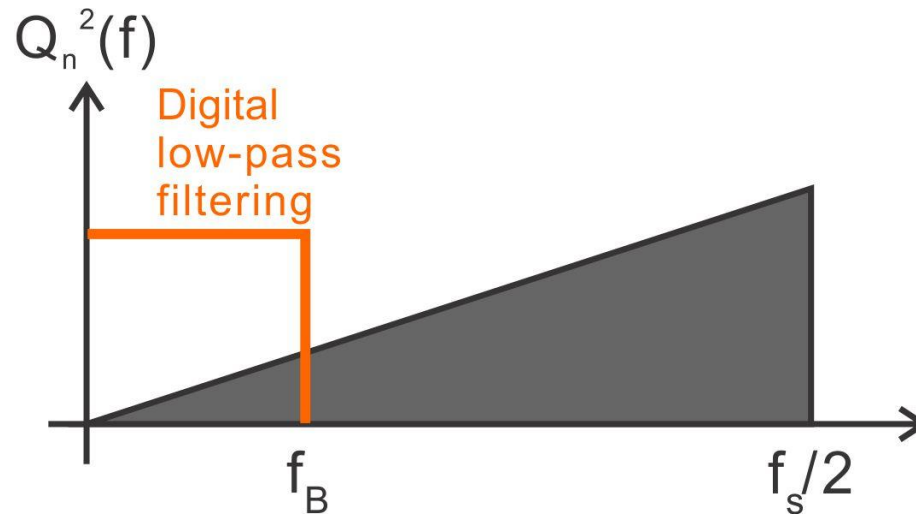


- How can we understand this from time domain?
 - Assuming $D[n] = V_{in}[n] + q_e[n]$, pass through an FIR filter $H(z) = (1+z^{-1})/2$
 - We have $D_{LPF}[n] = (V_{in}[n] + V_{in}[n-1])/2 + (q_e[n] + q_e[n-1])/2$
 - Low-frequency signal adds in amplitude while noise adds in power

Brief Summary

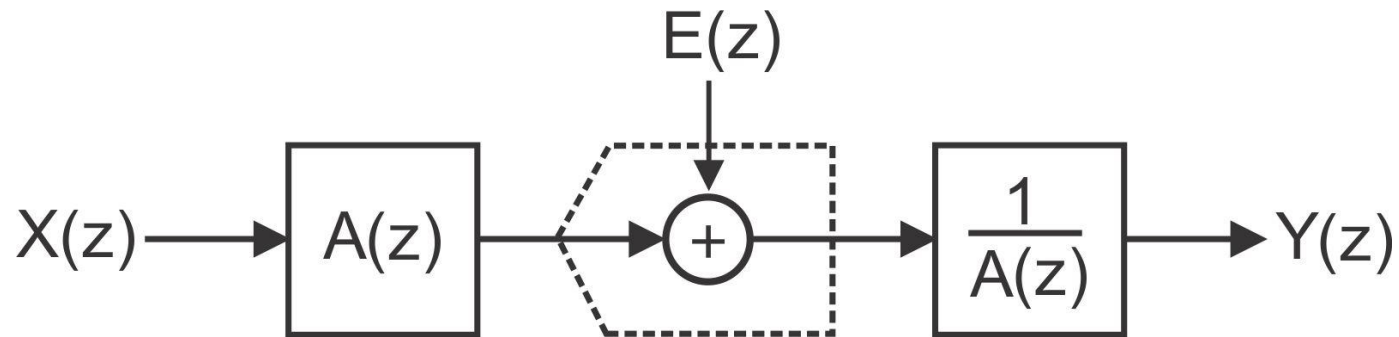
- Increasing OSR by $2x \rightarrow 2x$ reduction in quantization noise power $\rightarrow$ 0.5 bit increase in resolution
 - We can increase the ADC resolution simply by oversampling
- How useful is this?
 - For every 1 extra bit, we need to increase the sampling rate by $4x$
 - Too costly 😞
- Can we get more bits without increasing the sampling rate very much?

Noise Shaping



- It would be desirable if we can somehow shape the quantization noise
- If we can push the quantization noise to higher frequencies, the low-pass filtering would be much more effective

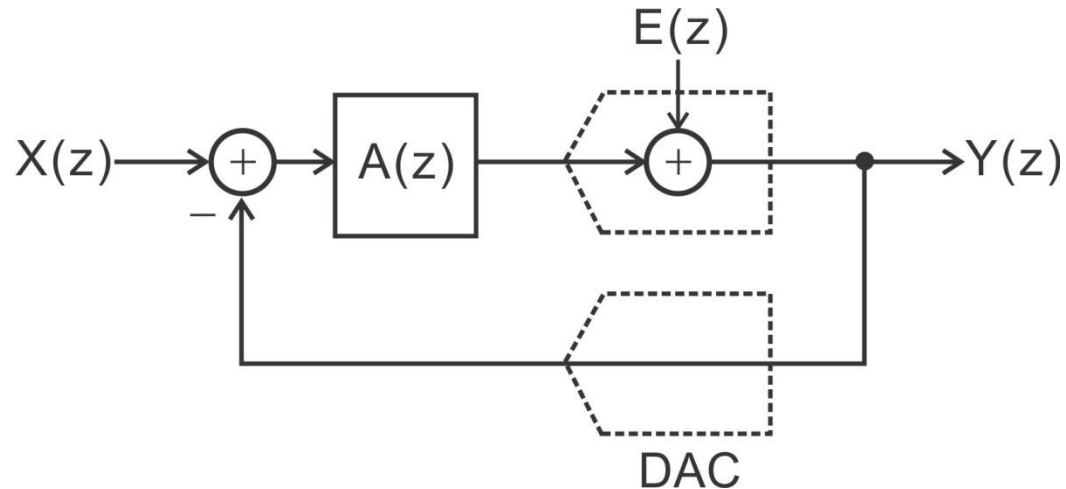
Open-Loop Noise Shaping



$$Y(z) = X(z) + E(z)/A(z)$$

- If $A(z)$ has high-gain at low frequency, the in-band quantization noise will be reduced
- However, a high-gain $A(z)$ may overload the ADC

Noise Shaping Using Feedback (1)



$$\begin{aligned} Y(z) &= X(z) \frac{A(z)}{1 + A(z)} + E(z) \frac{1}{1 + A(z)} \\ &= X(z) H_X(z) + E(z) H_E(z) \end{aligned}$$

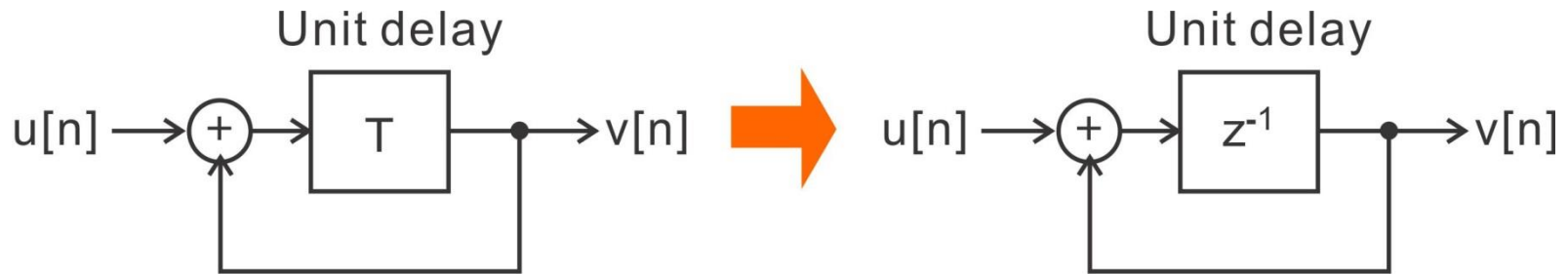
- $H_X(z)$ is called the signal transfer function (STF)
- $H_E(z)$ is called the noise transfer function (NTF)

Noise Shaping Using Feedback (2)

$$Y(z) = X(z) \frac{A(z)}{1 + A(z)} + E(z) \frac{1}{1 + A(z)}$$

- We want $\text{STF} \cong 1$ and $\text{NTF} \ll 1$ in the signal band
 - $|A(z)| \gg 1$ at low frequencies
- An integrator satisfies this requirement
 - $A(z) = z^{-1}/(1 - z^{-1})$

Discrete Time Integrator



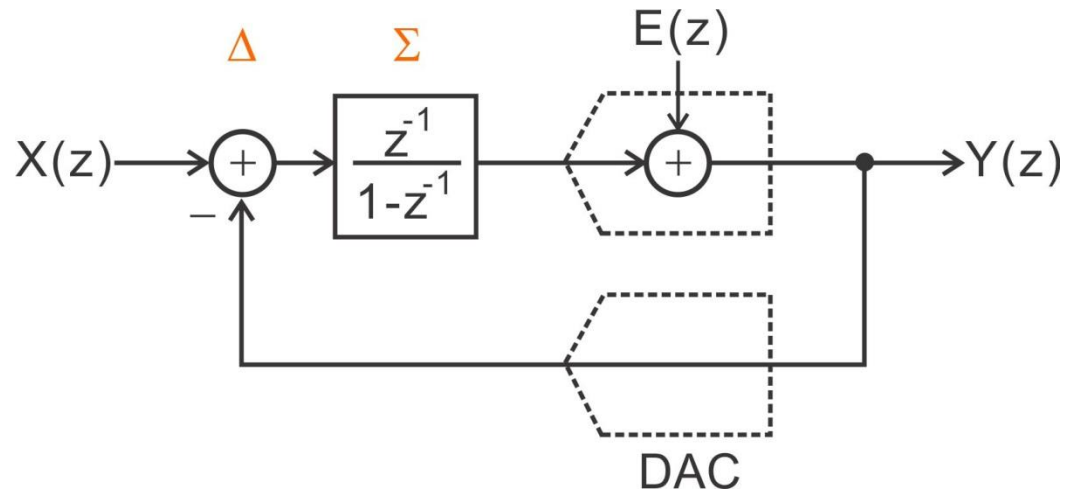
$$v[n] = u[n-1] + v[n-1]$$

$$V(z) = U(z)z^{-1} + V(z)z^{-1}$$

$$\Rightarrow \frac{V(z)}{U(z)} = \frac{z^{-1}}{1 - z^{-1}} \quad z = e^{j\omega T}$$

- An integrator provides infinite gain at DC

First Order $\Delta\Sigma$ Modulator



$$Y(z) = X(z)z^{-1} + E(z)(1 - z^{-1})$$

$$y[n] = x[n-1] + (e[n] - e[n-1])$$

- Output is equal to delayed input plus filtered quantization noise

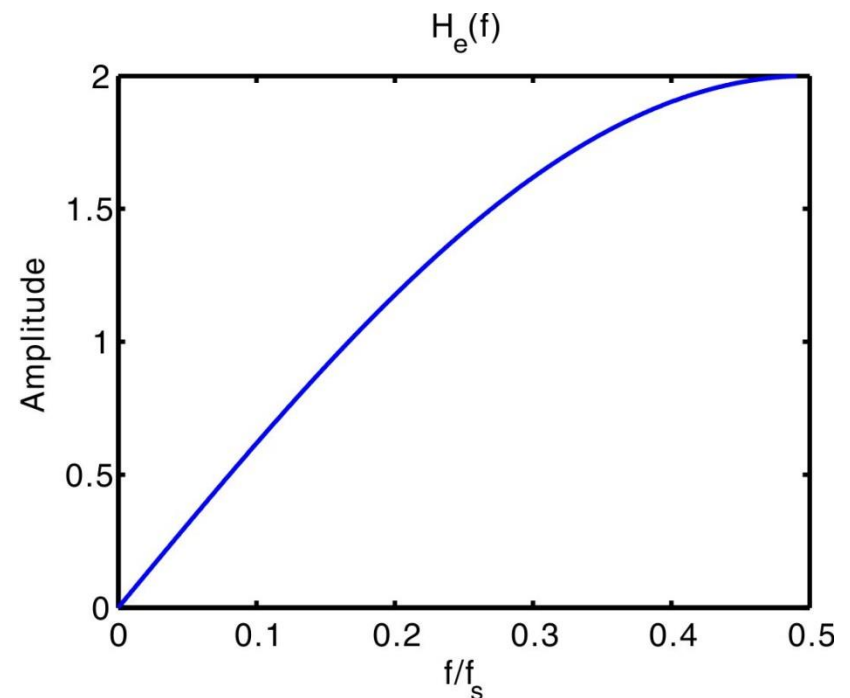
NTF Frequency Domain Analysis

$$H_e(z) = 1 - z^{-1}$$

$$H_e(\omega) = 1 - e^{-j\omega T} = 2e^{-j\omega T/2} \left(j \sin\left(\frac{\omega T}{2}\right) \right)$$

$$|H_e(\omega)| = 2 \left| \sin\left(\pi \frac{f}{f_s}\right) \right|$$

- Quantization noise is attenuated at low frequencies, but amplified at high frequencies



In-Band Quantization Noise

- The total in-band quantization noise can be calculated by

$$\begin{aligned} P_{qnoise} &= \int_0^{f_B} \frac{\Delta^2}{12} \frac{2}{f_s} |H_e|^2 df \\ &= \int_0^{f_B} \frac{\Delta^2}{12} \frac{2}{f_s} \left[2 \sin \left(\pi \frac{f}{f_s} \right) \right]^2 df \\ &\approx \int_0^{f_B} \frac{\Delta^2}{12} \frac{2}{f_s} \left[2 \left(\pi \frac{f}{f_s} \right) \right]^2 df \\ &= \frac{\Delta^2}{12} \frac{\pi^2}{3} \left(\frac{2f_B}{f_s} \right)^3 = \frac{\Delta^2}{12} \frac{\pi^2}{3} \left(\frac{1}{OSR} \right)^3 \end{aligned}$$

- What about the total amount of quantization noise between 0 and $f_s/2$? Is it smaller or larger?

SQNR

- Assuming a full-scale sinusoidal signal, we have

$$SQNR = \frac{P_{sig}}{P_{qnoise}} = \frac{\frac{1}{2} \left(\frac{(2^B - 1)\Delta}{2} \right)^2}{\frac{\Delta^2}{12} \frac{\pi^2}{3} \frac{1}{OSR^3}}$$

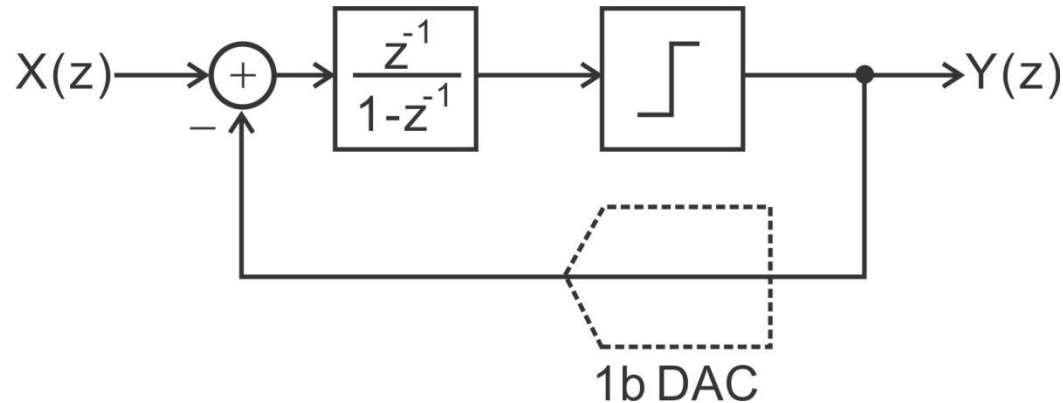
$$\approx 1.76 + 6.02B - 5.2 + 30 \log_{10}(OSR) \quad [\text{dB}]$$

- Note that this SQNR expression works only for $B > 1$, so that $2^B - 1$ can be approximated by 2^B
- 2x increase in OSR results in 8x SQNR improvement
 - 9dB (or 1.5bit) per octave oversampling

SQNR Improvement

OSR	Without Noise Shaping	With 1st-order Noise Shaping
16	12 dB (2 bits)	31dB (5 bits)
64	18 dB (3 bits)	49 dB (8 bits)
256	24 dB (4 bits)	67dB (11 bits)
1024	30 dB (5 bits)	85dB (14 bits)

Modulator with Single-Bit Quantizer

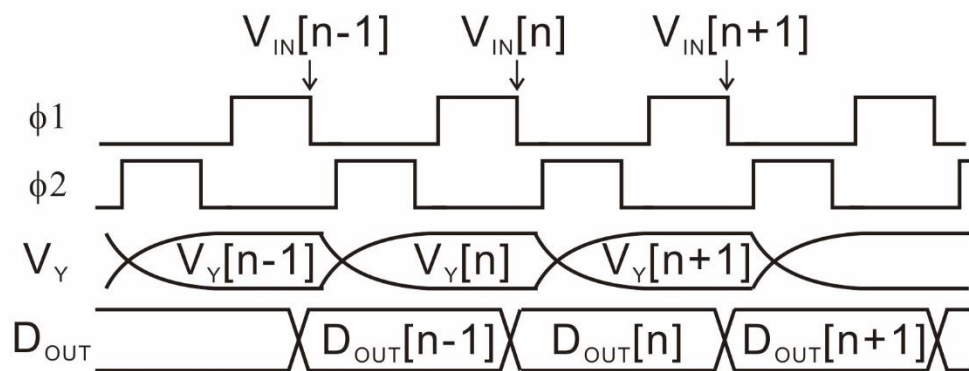
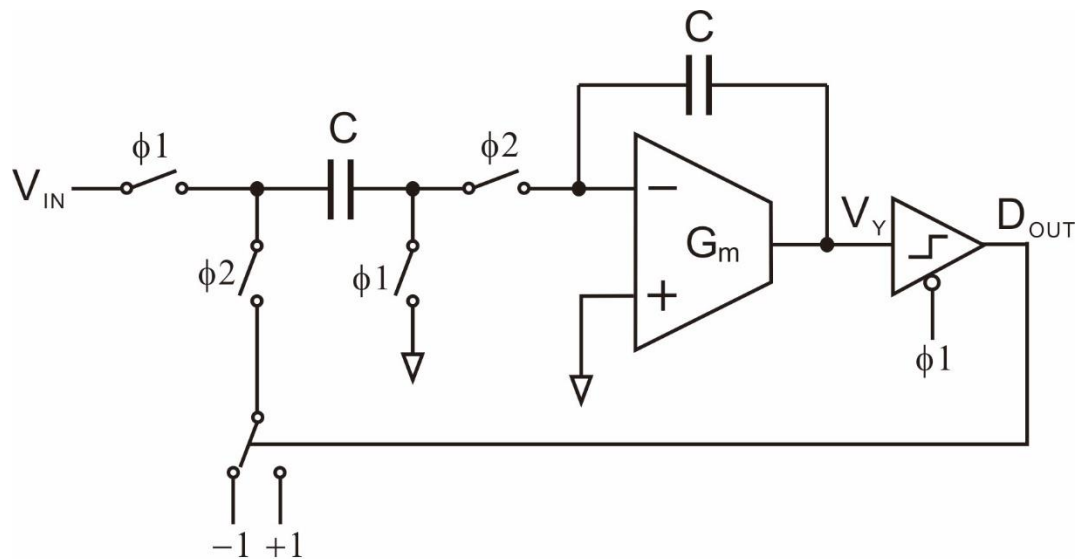


- Expected SQNR

$$SQNR \approx \frac{P_{sig}}{P_{noise}} = \frac{\frac{1}{2} \left(\frac{\Delta}{2} \right)^2}{\frac{\Delta^2}{12} \frac{\pi^2}{3} \frac{1}{OSR^3}} = -3.4 + 30 \log(OSR) \quad [\text{dB}]$$

- E.g. $OSR=128 \Rightarrow SQNR=60\text{dB}$

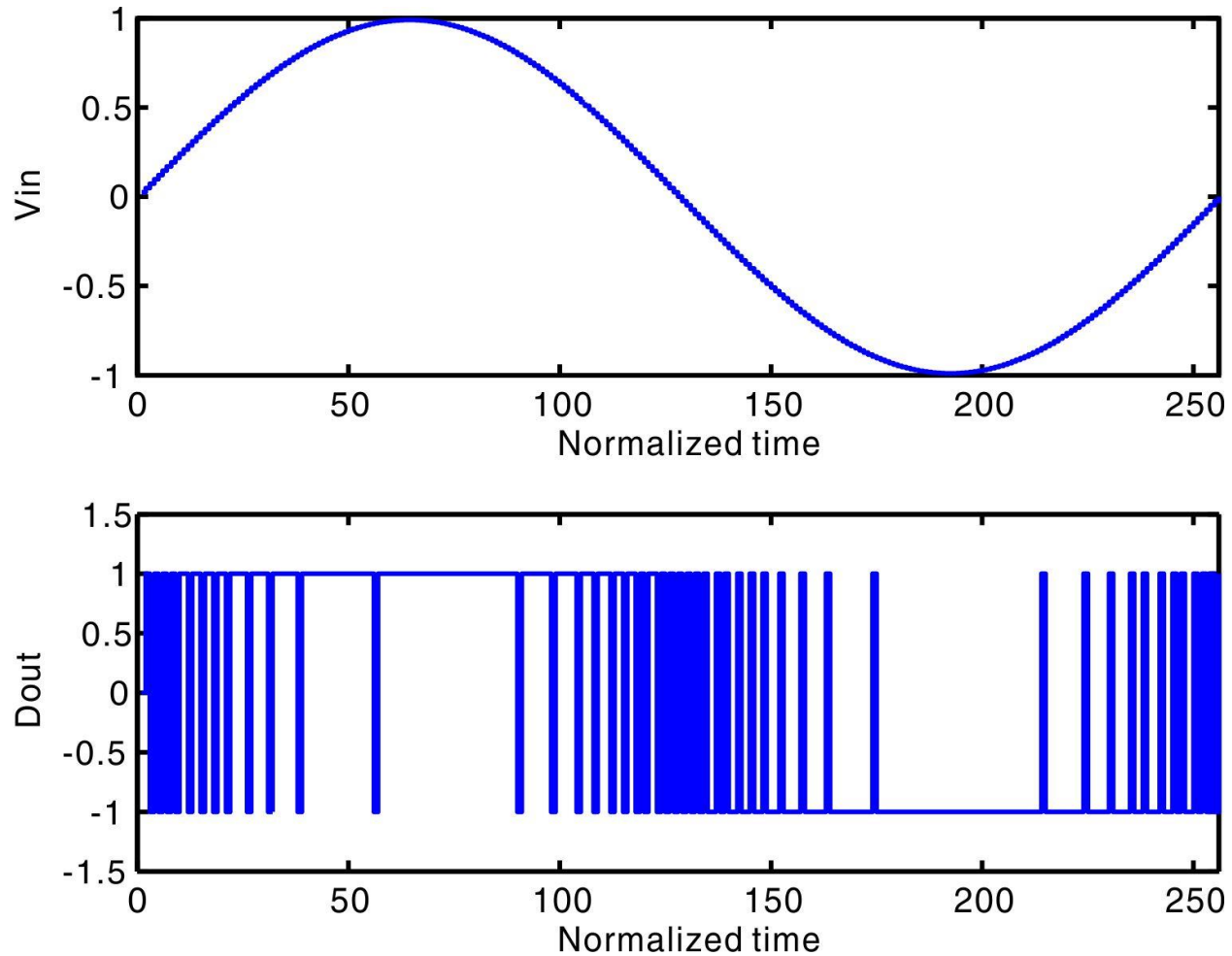
Implementation Example



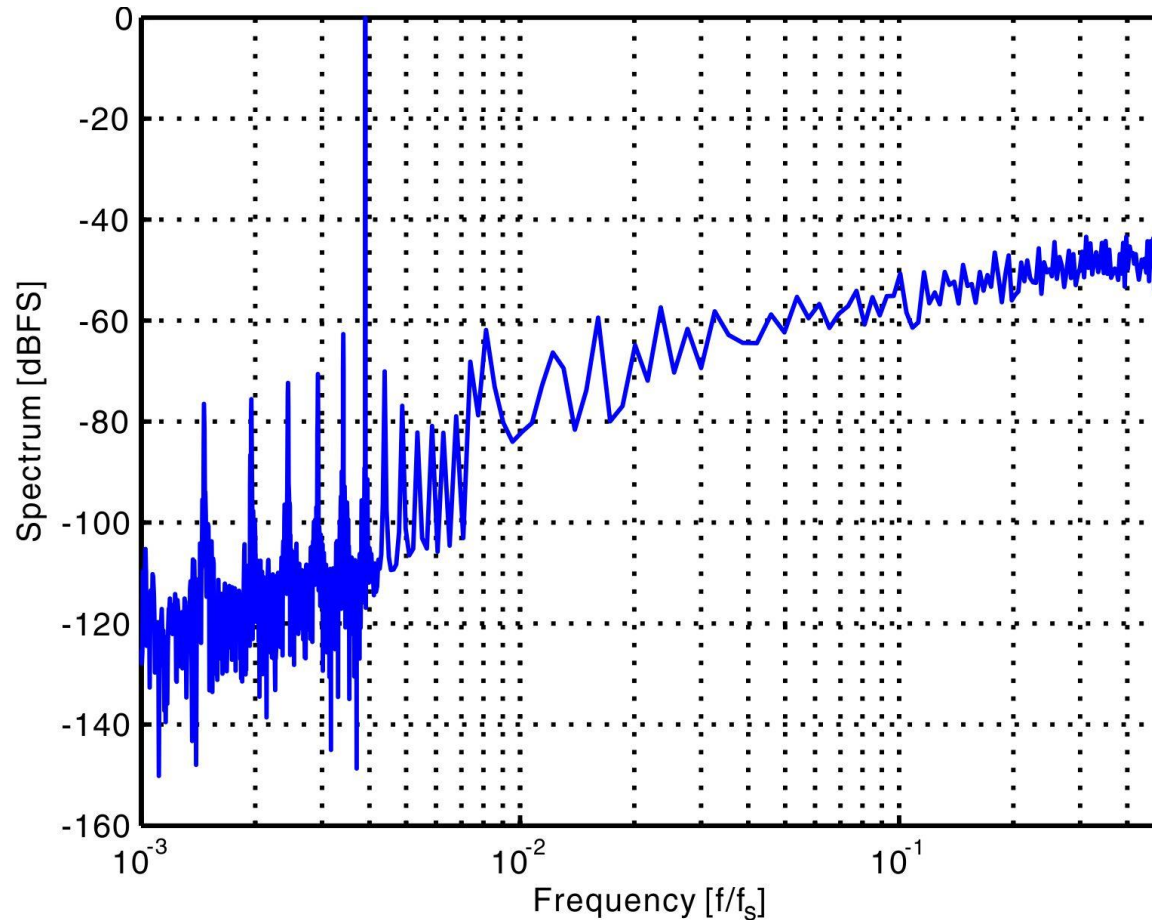
$$\begin{cases} V_Y(z) = \frac{z^{-1}}{1-z^{-1}} (V_{in}(z) - D_{out}(z)) \\ D_{out}(z) = Q(V_Y(z)) \end{cases}$$

$$\begin{cases} V_Y[n] = V_Y[n-1] + V_{in}[n-1] - D_{out}[n-1] \\ D_{out}[n] = V_Y[n] > 0 \end{cases}$$

Simulated Response

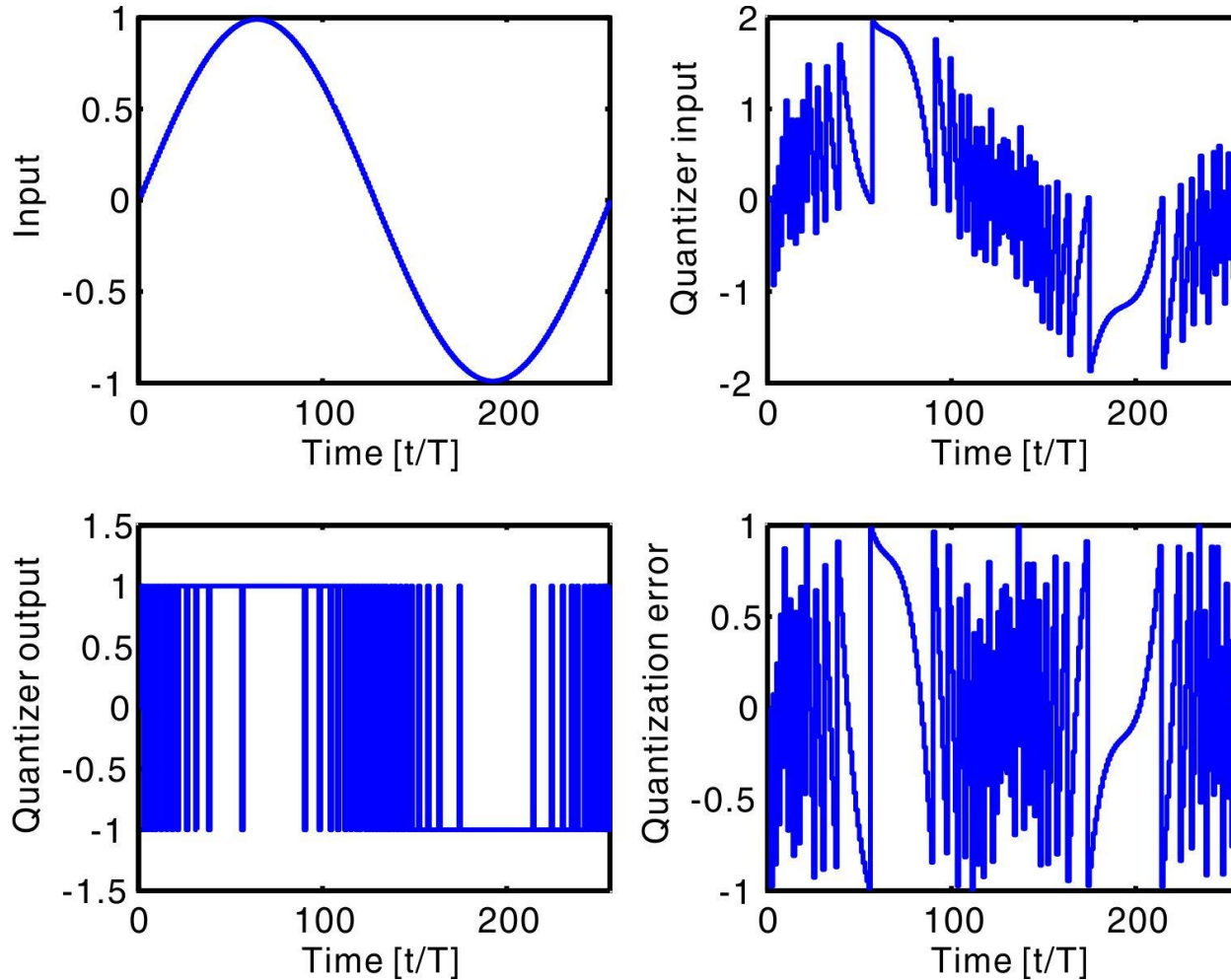


Spectrum



- SQNR=55dB < 60dB
- There are lots of spurs in the spectrum, why?

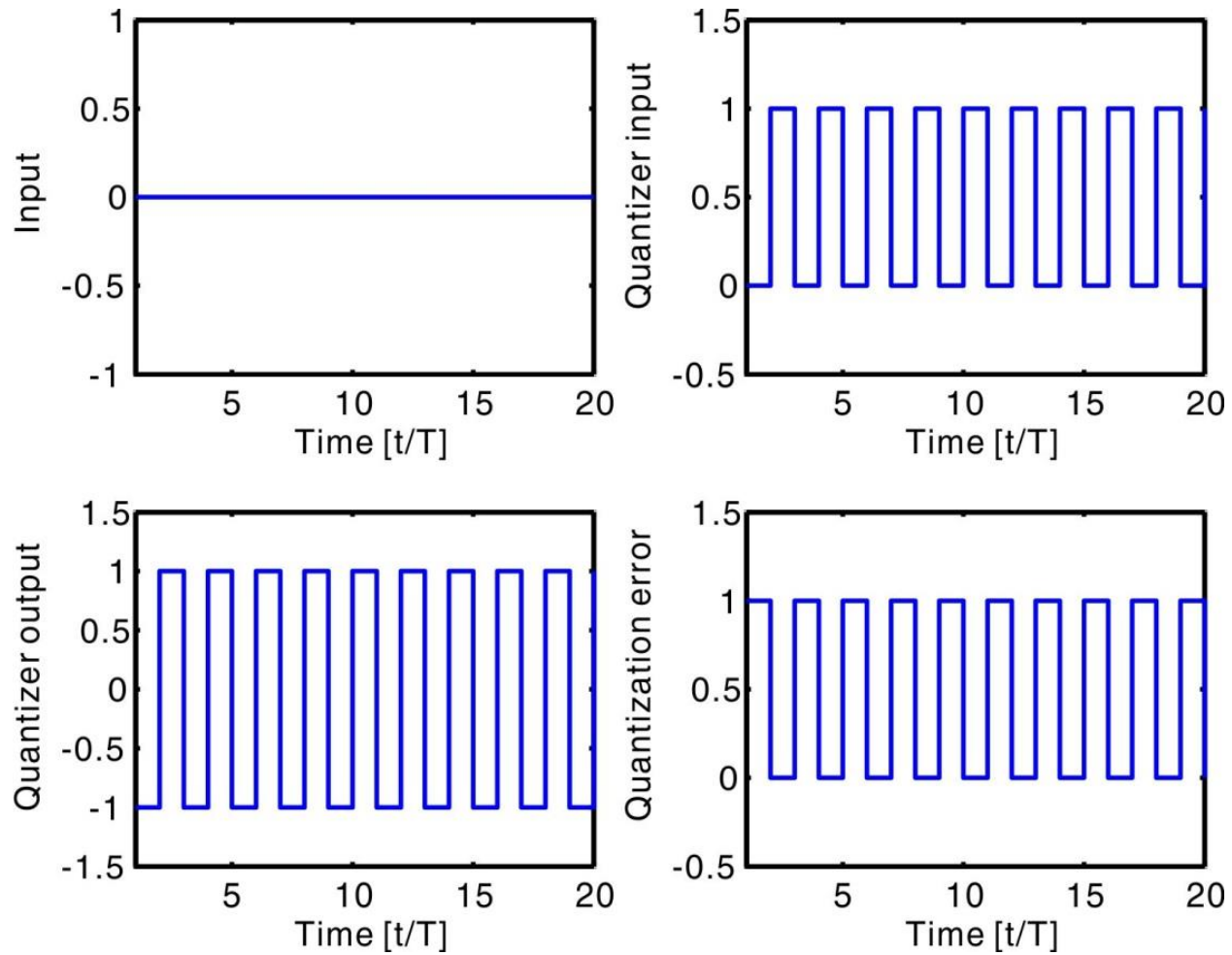
Time Domain Modulator Behavior



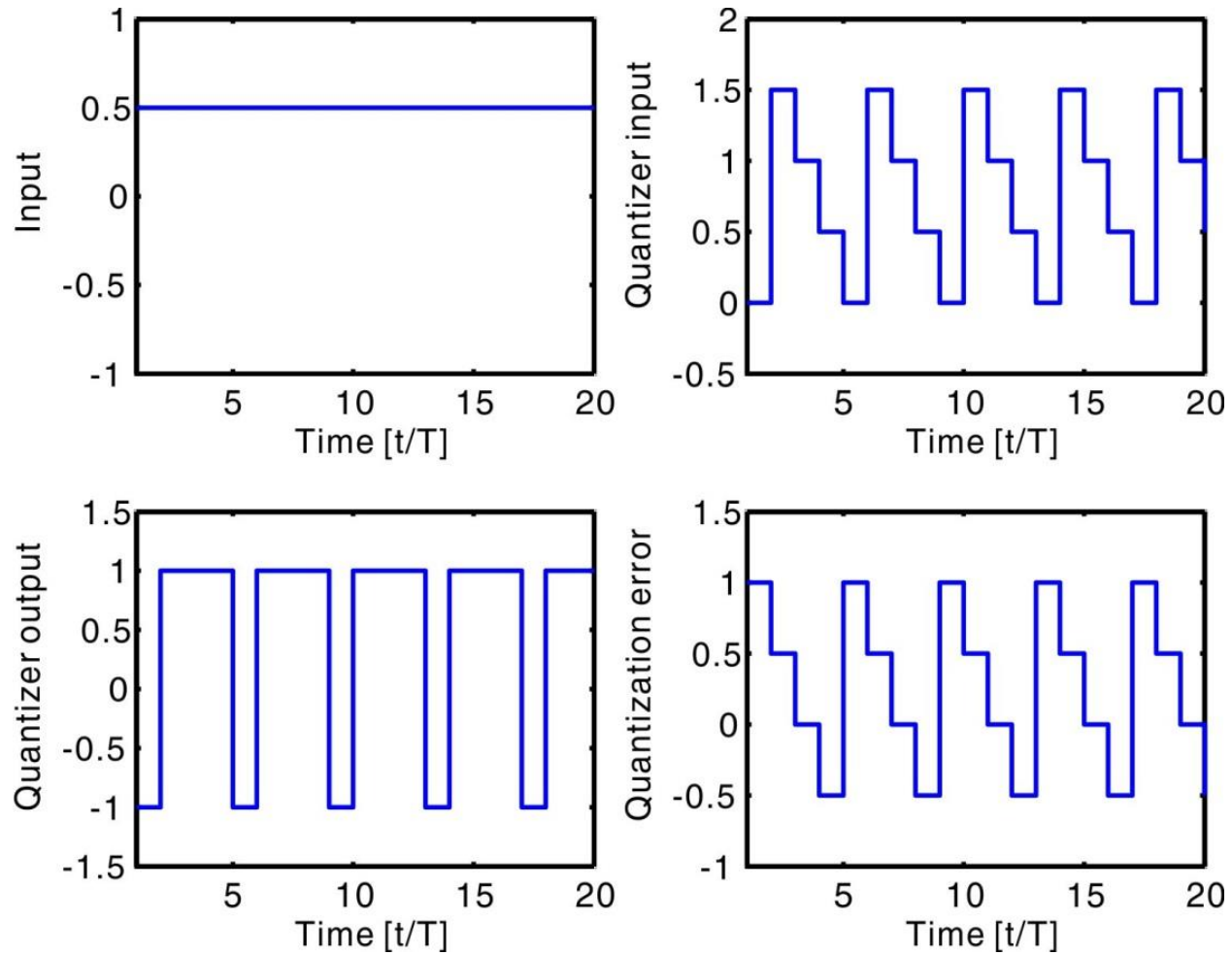
Tones

- Key observations:
 - The quantization error is deterministic, not random, if there is no randomness (e.g., dither and thermal noise) added into the loop
 - The quantization error is correlated with the input
- In a first-order delta-sigma modulator, the quantization error is highly correlated with the input, and thus, its spectrum is not white, but contains spurious tones.
- This is fundamentally due to the nonlinear behavior of the quantizer.
- It is generally difficult to analyze the tonal behavior for an arbitrary input
- Let us take a look at a simple DC input to get an intuitive understanding

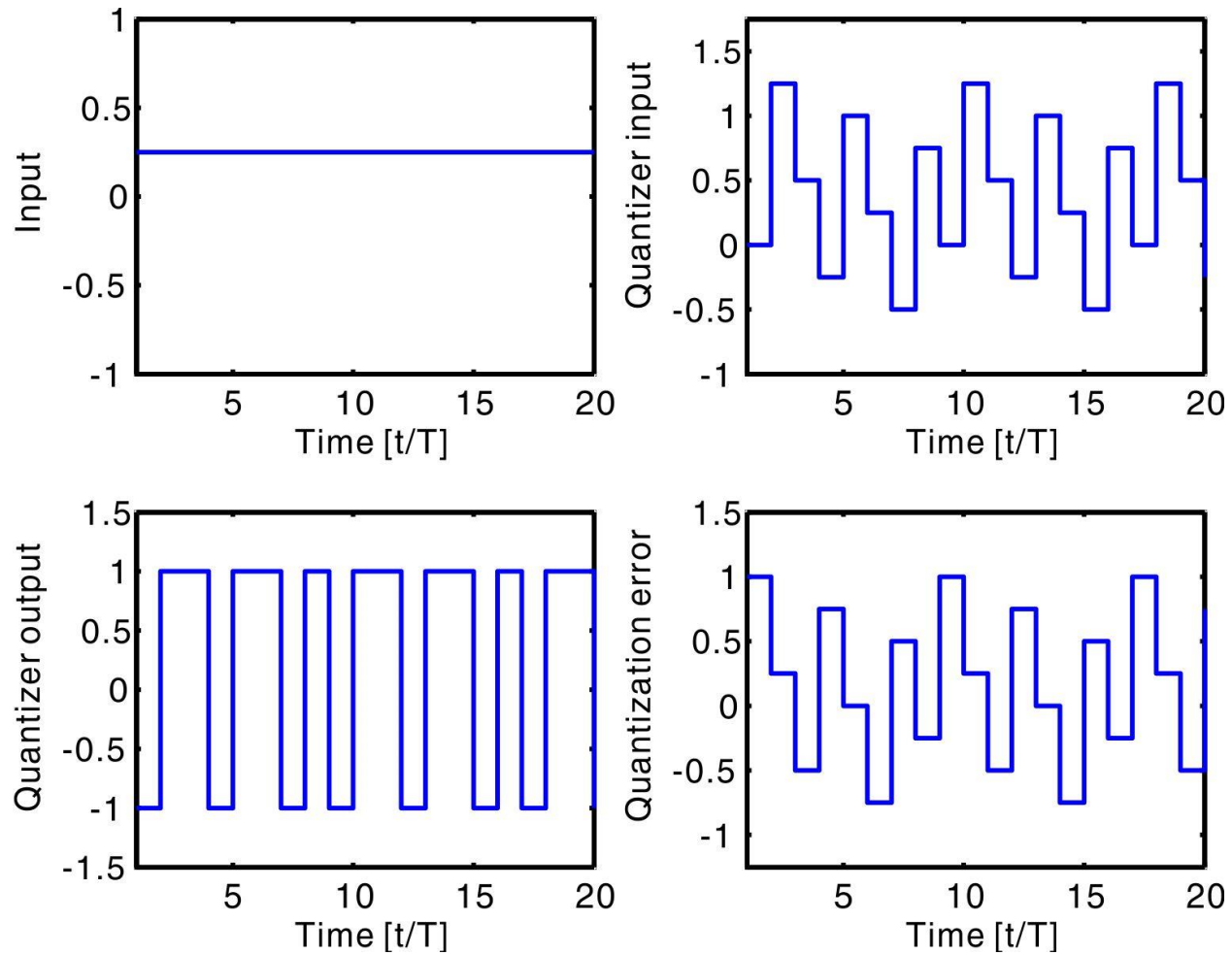
$$X[n] = 0$$



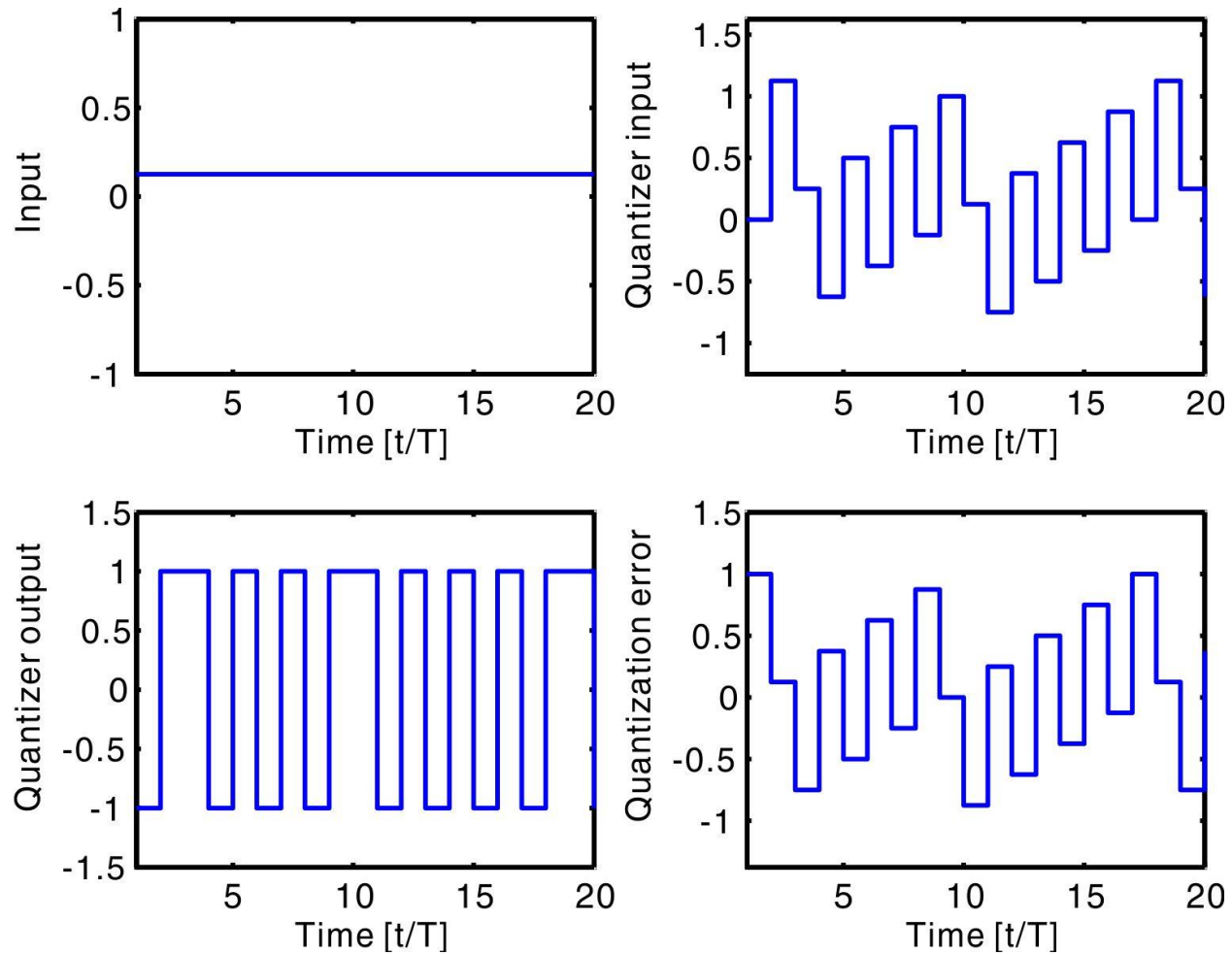
$$X[n] = 0.5$$



$$X[n] = 0.25$$



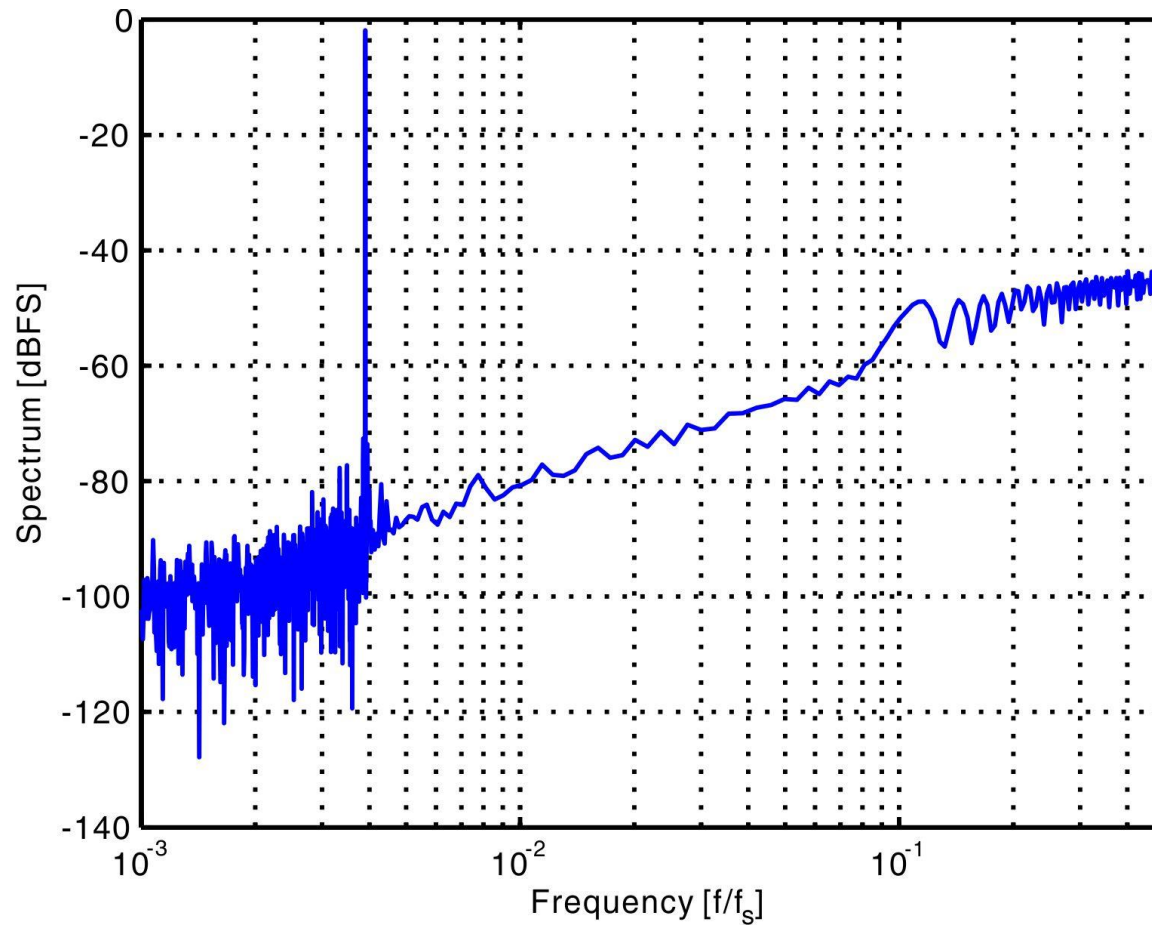
$$X[n] = 0.125$$



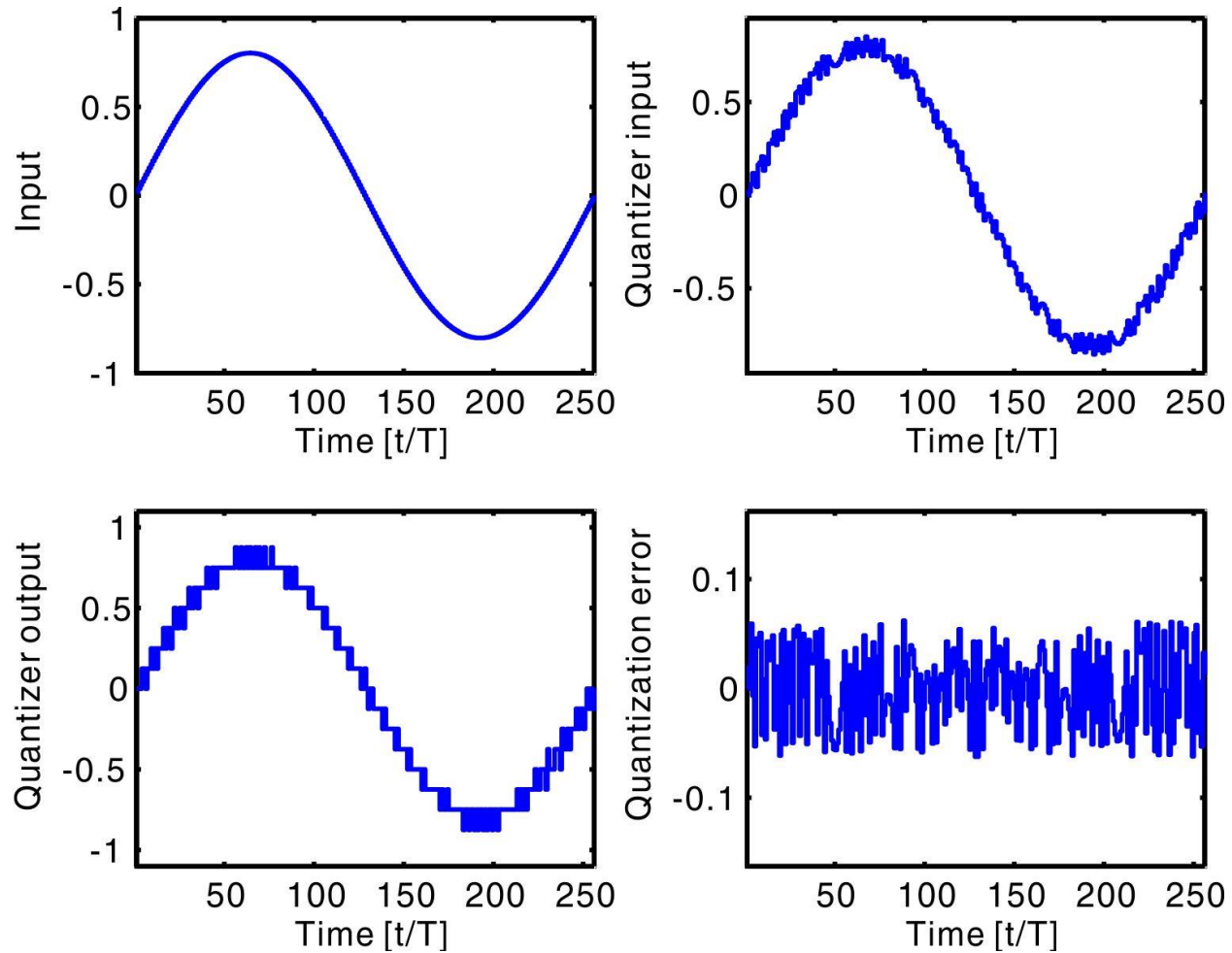
Idle Tone Considerations

- Idle tones are problematic in audio applications
 - The human ear can detect tones $\sim 20\text{dB}$ below the noise floor
- There are several way to reduce idle tones:
 - External dither
 - Add a random signal at the quantizer input to “whiten” quantization noise
 - Thermal noise dither
 - Electronic noise from integrators can dither the quantization noise for free
 - Multi-bit quantizer and DAC
 - Higher order modulators
 - Naturally produce “more random” quantization error sequences

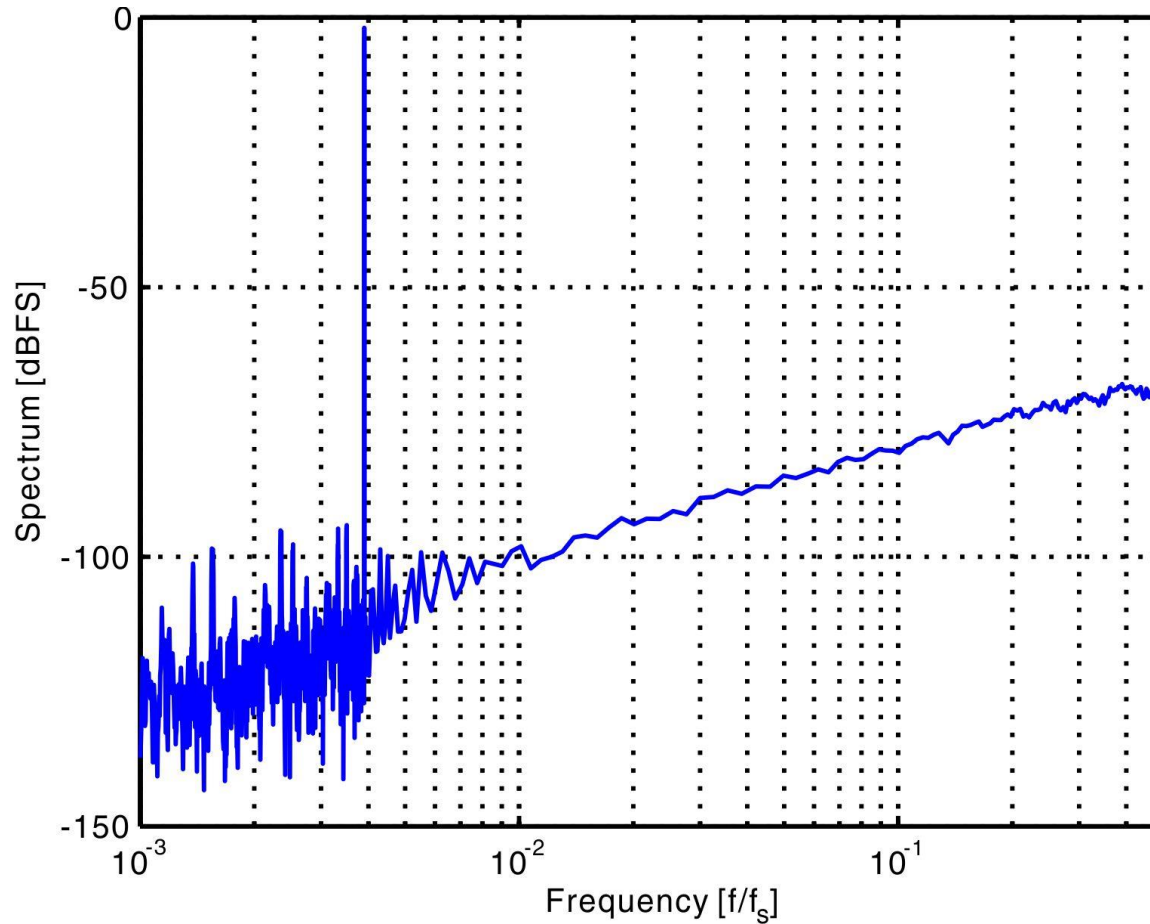
With Dither Added



With 4-bit Quantizer (1)

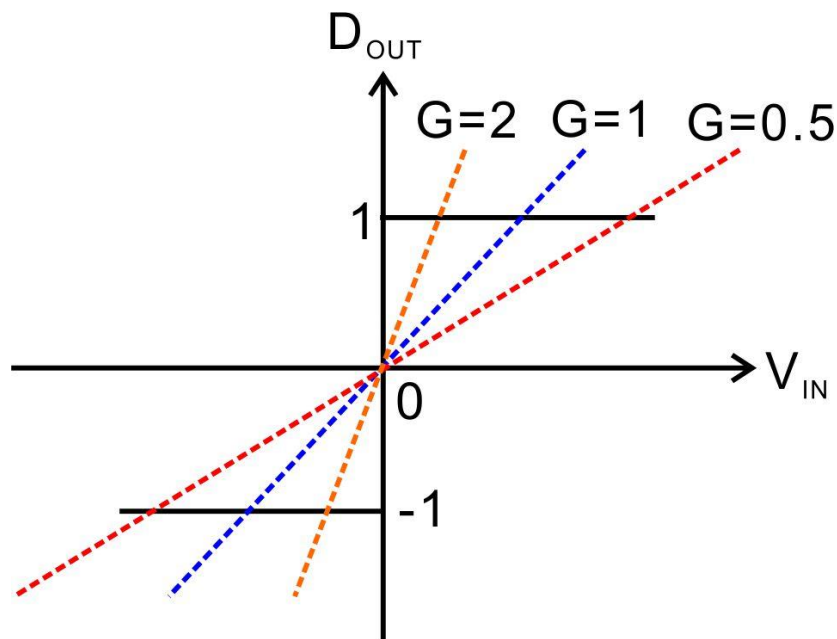


With 4-bit Quantizer (2)



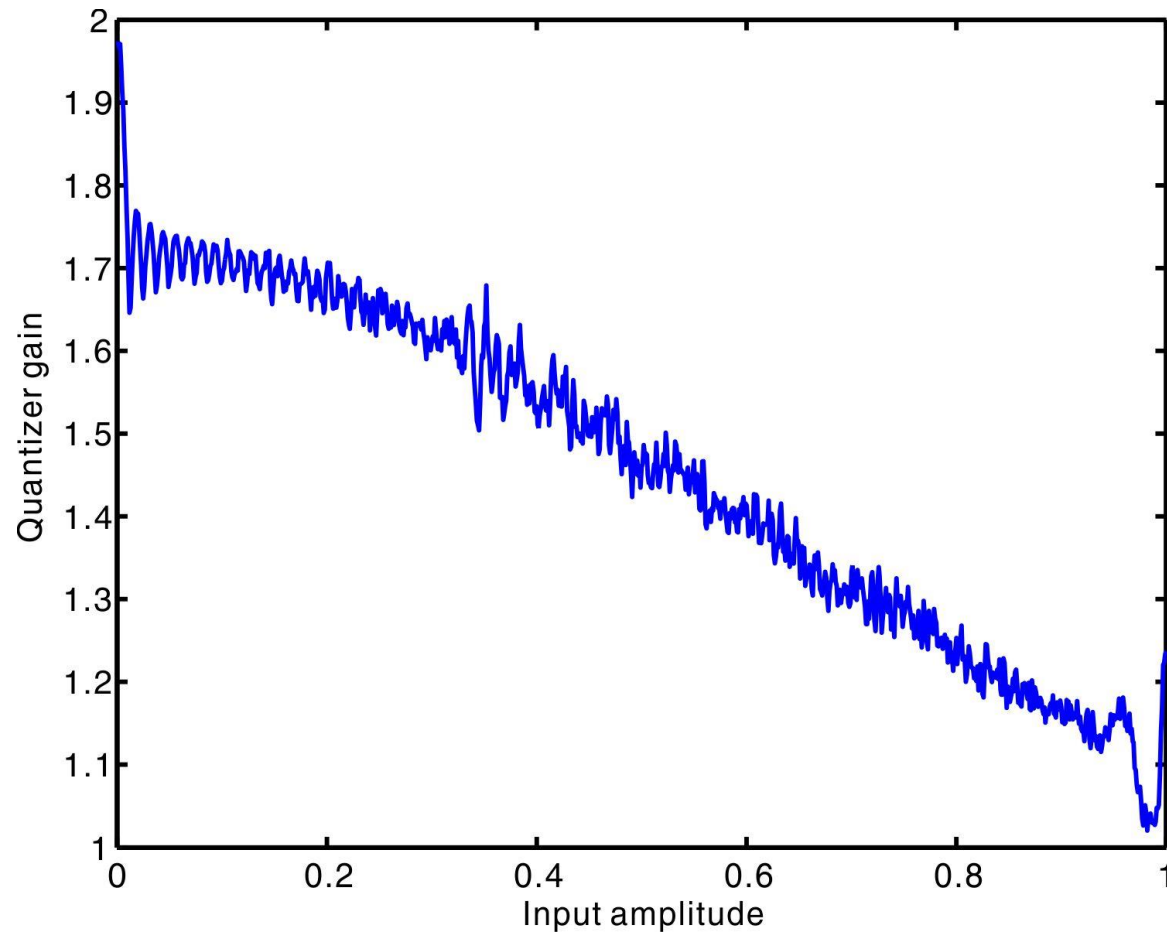
Tones lowered and SQNR increases to 81 dB ☺

Gain for 1-bit Quantizer



- Quantizer has gain!
- The gain of a 1-bit quantizer is ill-defined
- The actual closed-loop quantizer gain can be found from simulations:
 - The quantizer can be modeled as
 - $D_{out}[n] = G \cdot V_{in}[n] + q[n]$
 - q and V_{in} should be orthogonal
 - q should not depend on V_{in}
 - $G \equiv \frac{E(D_{out} \cdot V_{in})}{E(V_{in}^2)}$
- Quantizer gain affects loop stability. This is an important issue for high-order $\Delta\Sigma$ modulators

Quantizer Gain vs. Amplitude



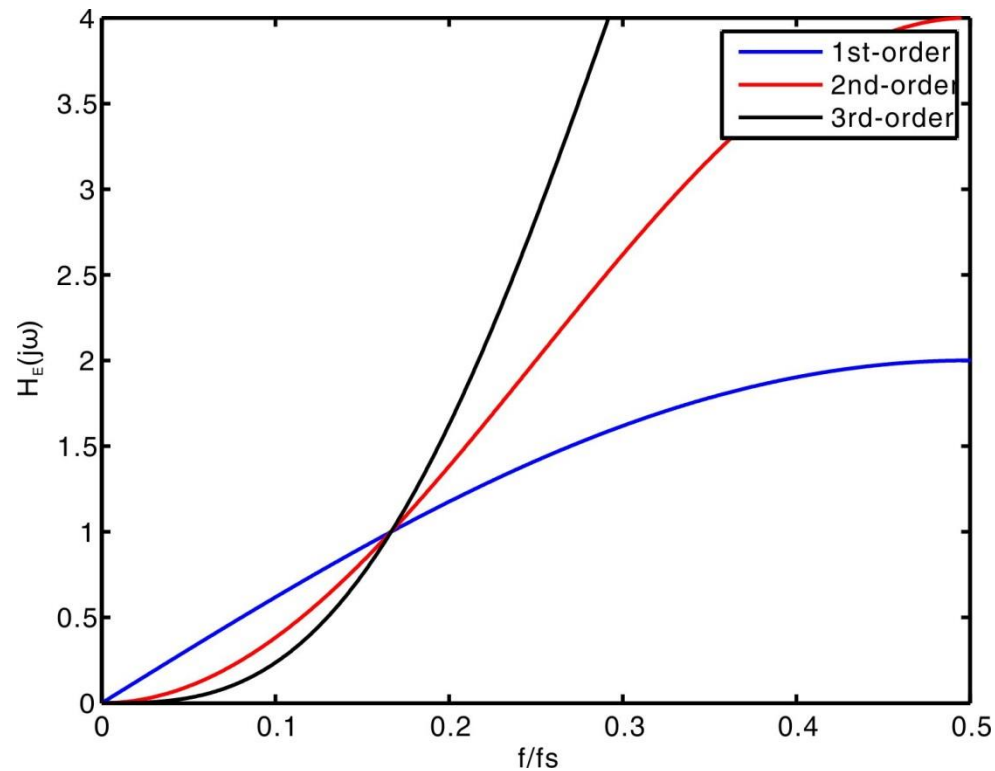
Higher Order Modulators

- Motivations:
 - Higher SQNR
 - Reduced OSR
 - Fewer and weaker idle tones
- Commonly used architectures
 - Single loop with a higher-order filter
 - Cascaded, multi-stage modulators

Higher Order Noise Shaping

- L^{th} -order noise transfer function

$$H_E(z) = (1 - z^{-1})^L$$



In-Band Quantization Noise

$$\begin{aligned} P_{qnoise} &= \int_0^{f_B} \frac{\Delta^2}{12} \frac{2}{f_s} \left[2 \sin \left(\pi \frac{f}{f_s} \right) \right]^{2L} df \approx \int_0^{f_B} \frac{\Delta^2}{12} \frac{2}{f_s} \left[2 \left(\pi \frac{f}{f_s} \right) \right]^{2L} df \\ &= \frac{\Delta^2}{12} \left(\frac{\pi^{2L}}{2L+1} \right) \left(\frac{2f_B}{f_s} \right)^{2L+1} = \frac{\Delta^2}{12} \left(\frac{\pi^{2L}}{2L+1} \right) \left(\frac{1}{OSR} \right)^{2L+1} \end{aligned}$$

- For an L^{th} -order modulator, 2x OSR leads to an SQNR increase of 6L+3dB (L+0.5bits)

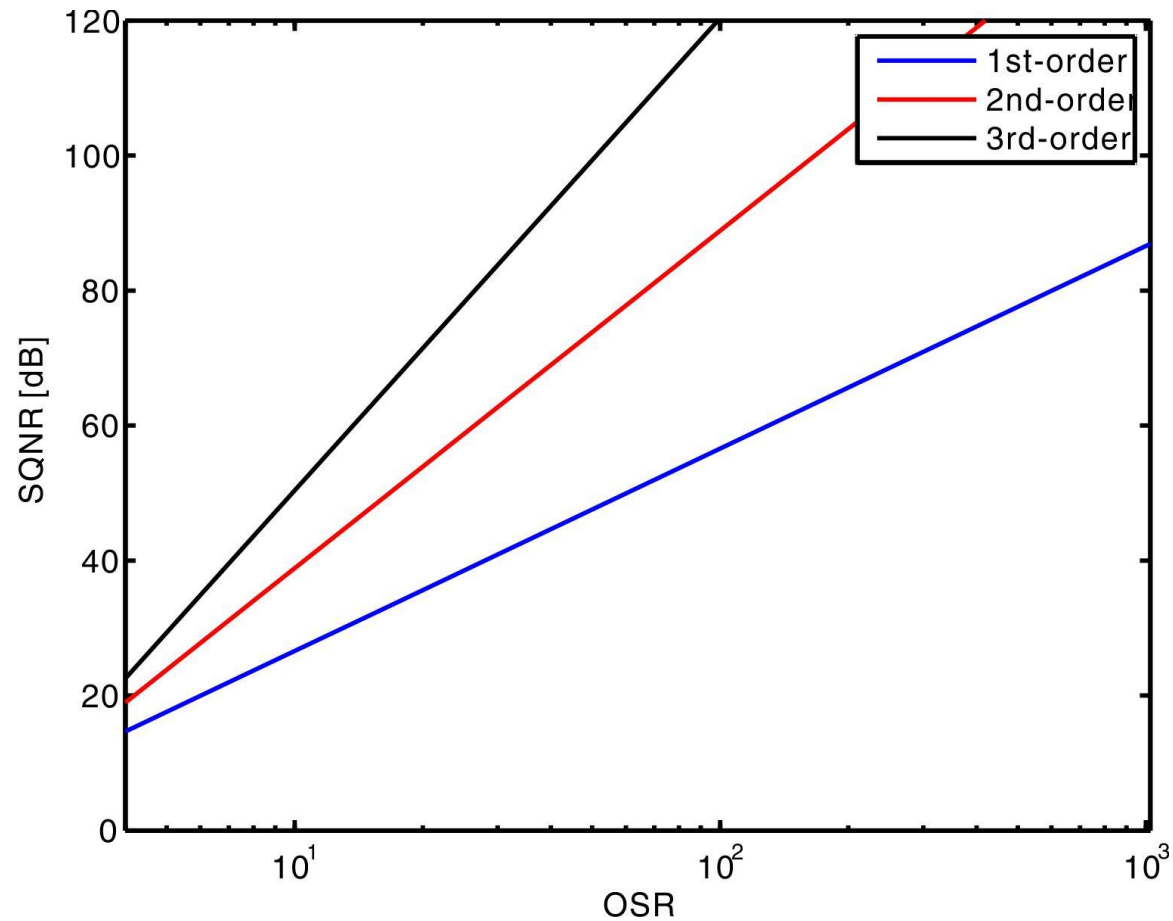
SQNR

- Assuming a full-scale sinusoidal signal, we have

$$SQNR = \frac{P_{sig}}{P_{qnoise}} = \frac{\frac{1}{2} \left(\frac{(2^B - 1)\Delta}{2} \right)^2}{\frac{\Delta^2}{12} \left(\frac{\pi^{2L}}{2L+1} \right) \left(\frac{1}{OSR} \right)^{2L+1}}$$
$$\approx 1.76 + 6.02B - 10 \log_{10} \left(\frac{\pi^{2L}}{2L+1} \right) + (20L + 10) \log_{10}(OSR) \quad [\text{dB}]$$

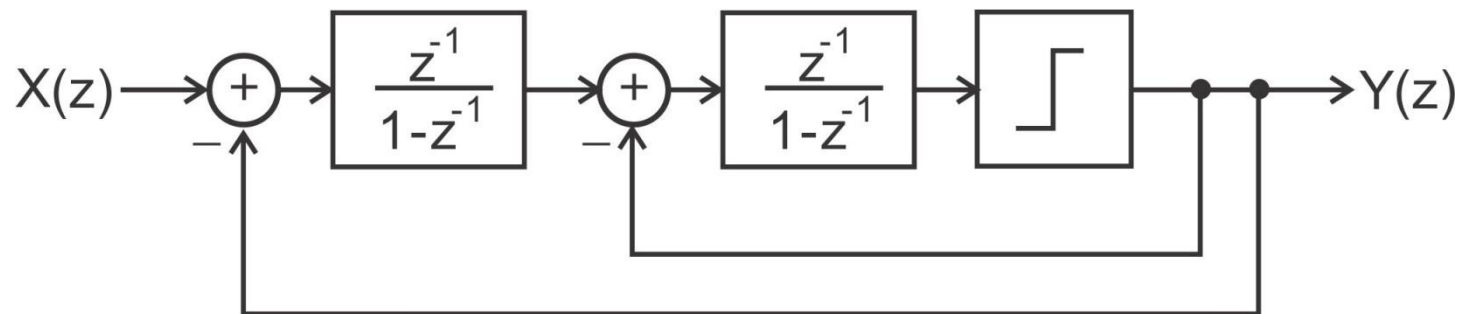
- 2x increase in OSR results in $6L+3\text{dB}$ ($L+0.5\text{bits}$) SQNR improvement

SQNR with Single-Bit Quantizer

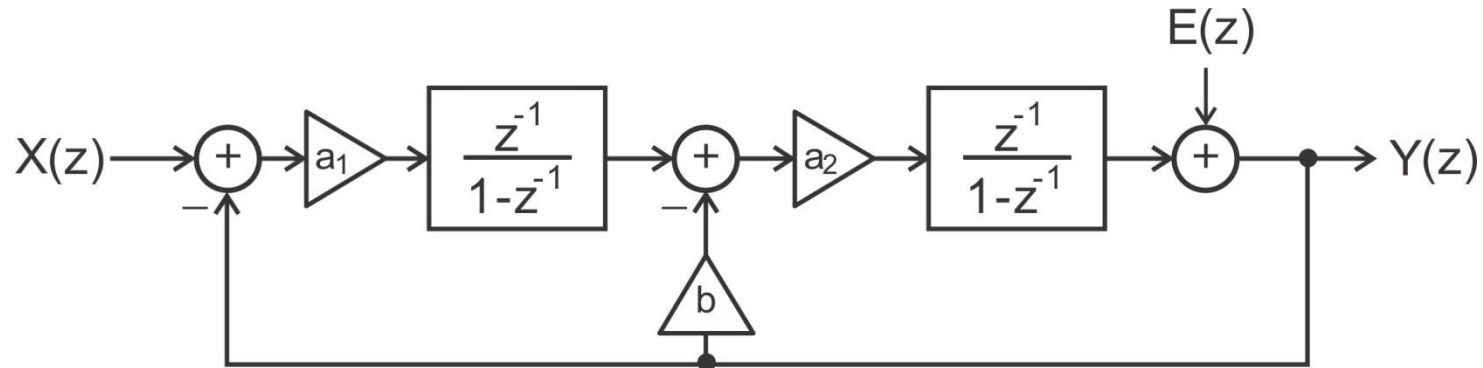


Building a Second-Order Modulator

- Cascade two integrators before the quantizer



More General Structure



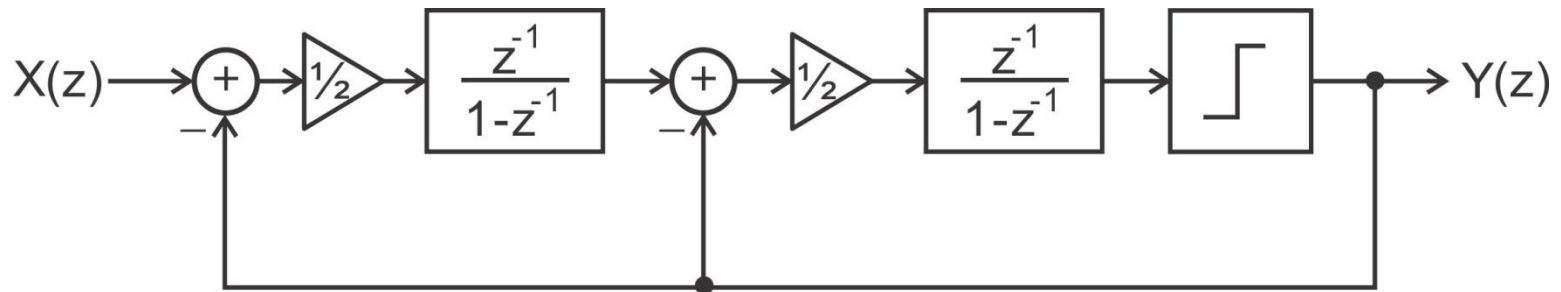
- Assuming quantizer gain of 1:

$$H_X(z) = \frac{a_1 a_2 z^{-1}}{D(z)} \quad H_E(z) = \frac{(1 - z^{-1})^2}{D(z)}$$

$$D(z) = 1 + z^{-1}(a_2 b - 2) + z^{-2}(1 - a_2 b + a_1 a_2)$$

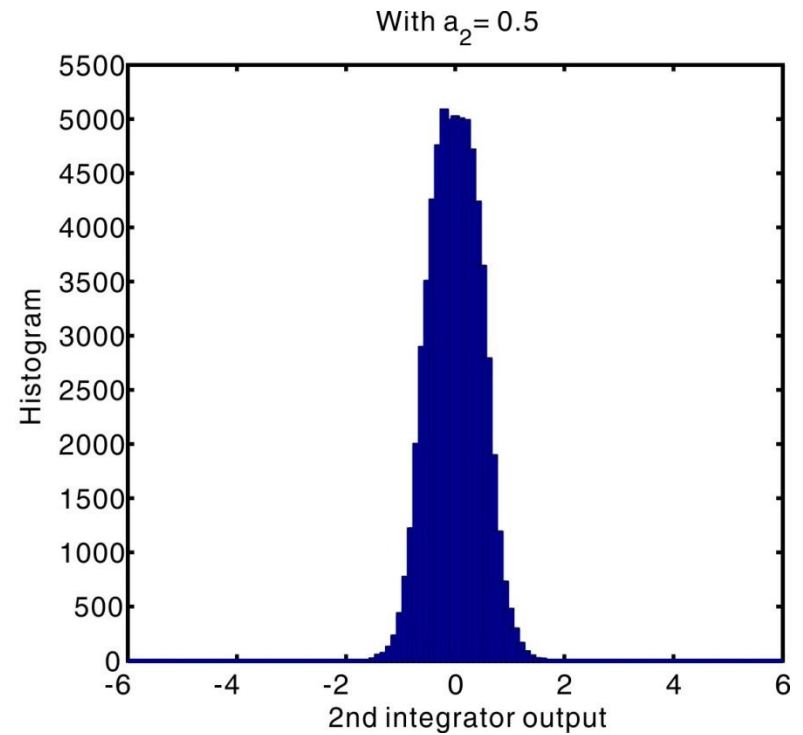
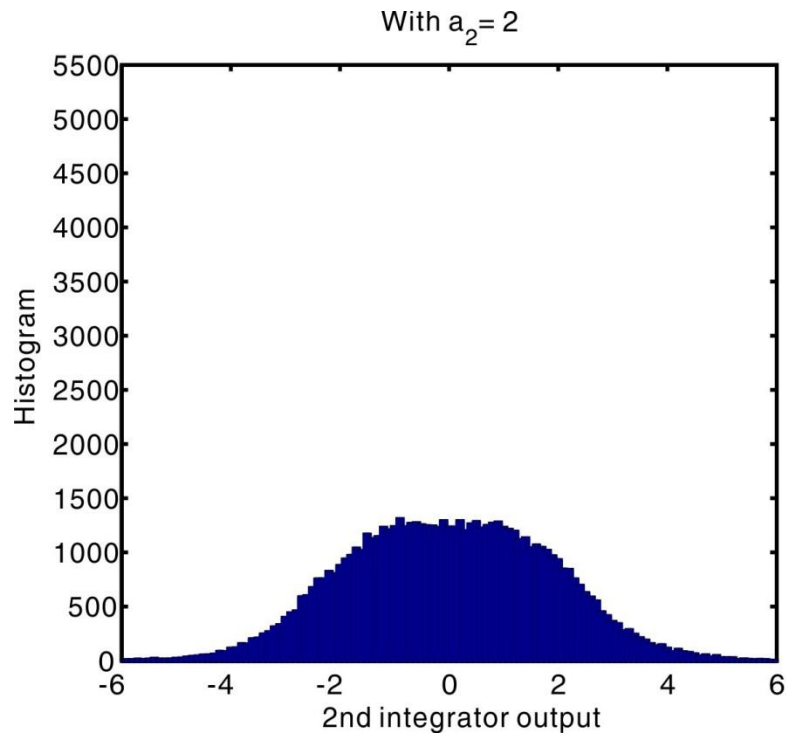
- Ideally, we want $D(z) = 1 \rightarrow a_2 b = 2, a_1 a_2 = 1$
 - $a_1 = a_2 = 1, b = 2$ or $a_1 = 0.5, a_2 = 2, b = 1$
- Note that even if $D(z) \neq 1$, the quantization noise is still shaped

Boser-Wooley Modulator (1)

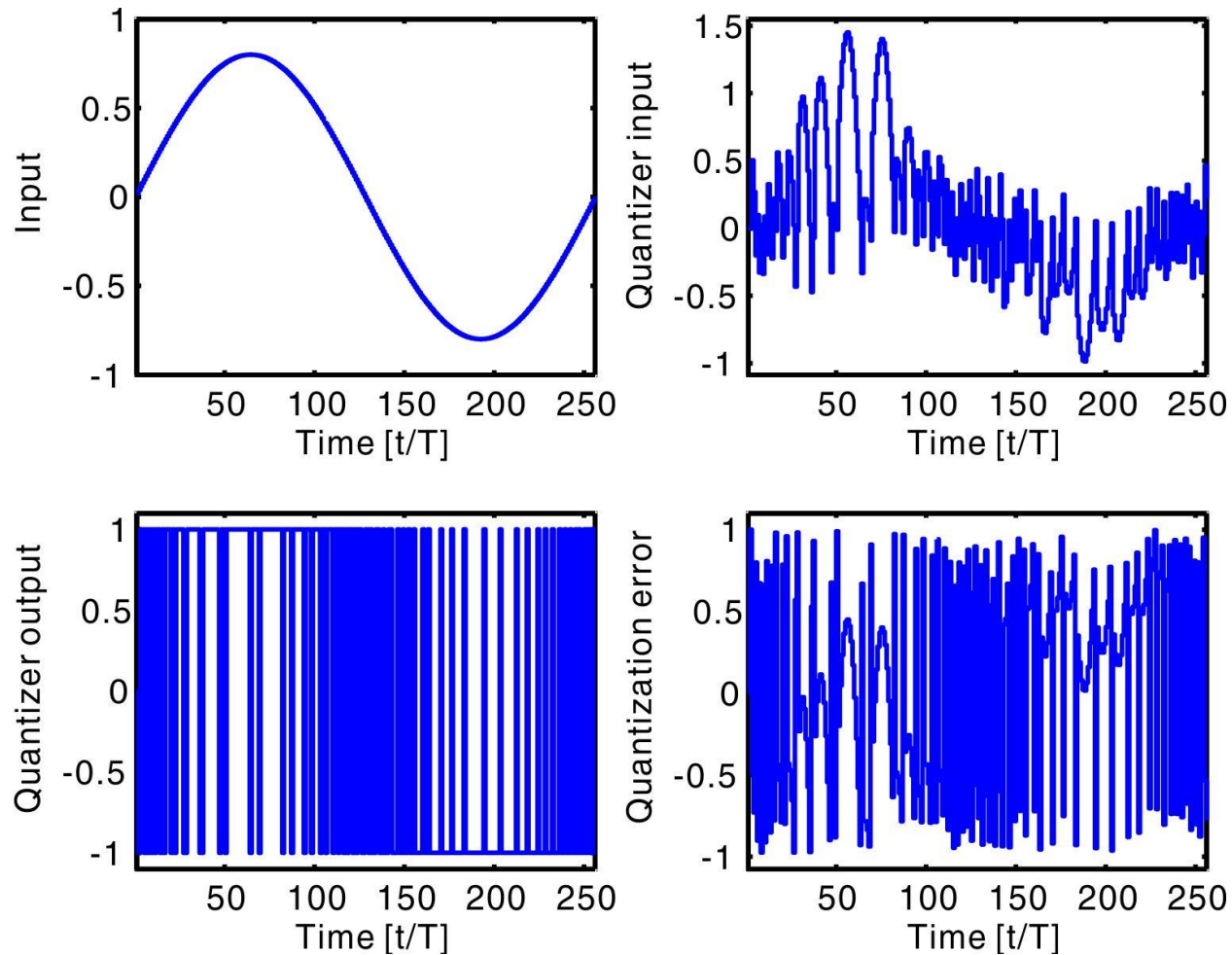


- We set $a_1=0.5$, $b=1$, but $a_2=0.5$ (instead of 2)
- This takes advantage of the property of the 1-bit quantizer
 - 1-bit quantizer only cares about sign, not amplitude
- By reducing a_2 , we lower the signal swing at the 2nd integrator output

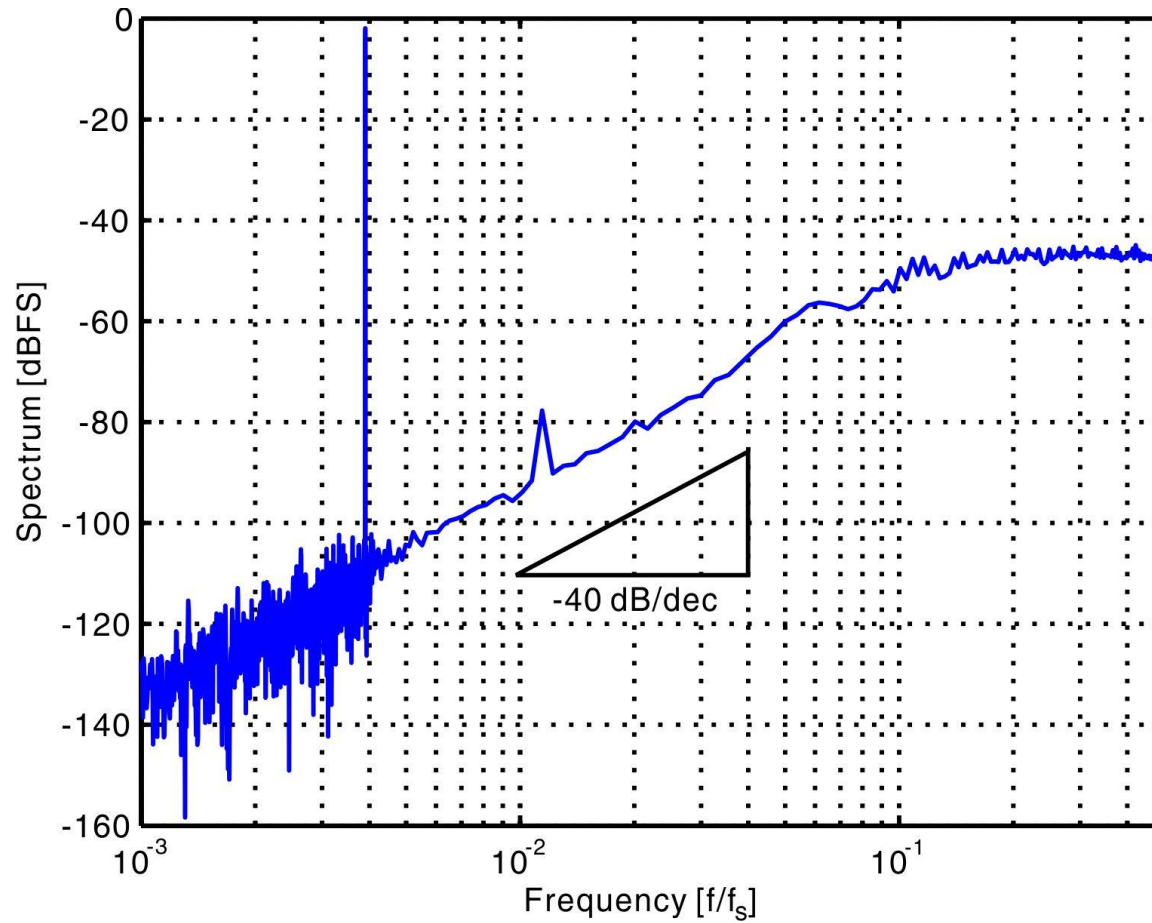
Boser-Wooley Modulator (2)



Behavior of a 2nd-Order Modulator



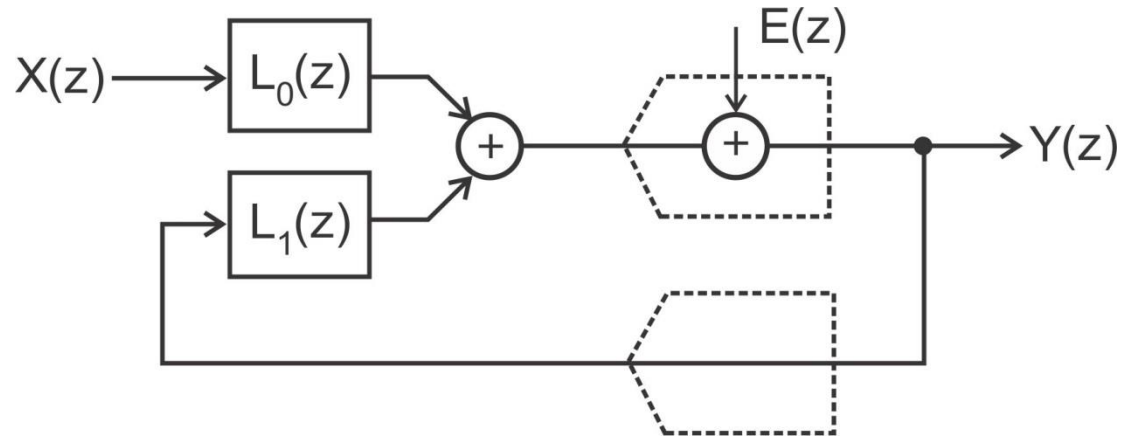
Spectrum



SQNR = 84 dB, Less spurious tones

Modulators with Order >2

- General filter decomposition



$$H_X(z) = \frac{L_0(z)}{1 - L_1(z)}$$

$$H_E(z) = \frac{1}{1 - L_1(z)}$$

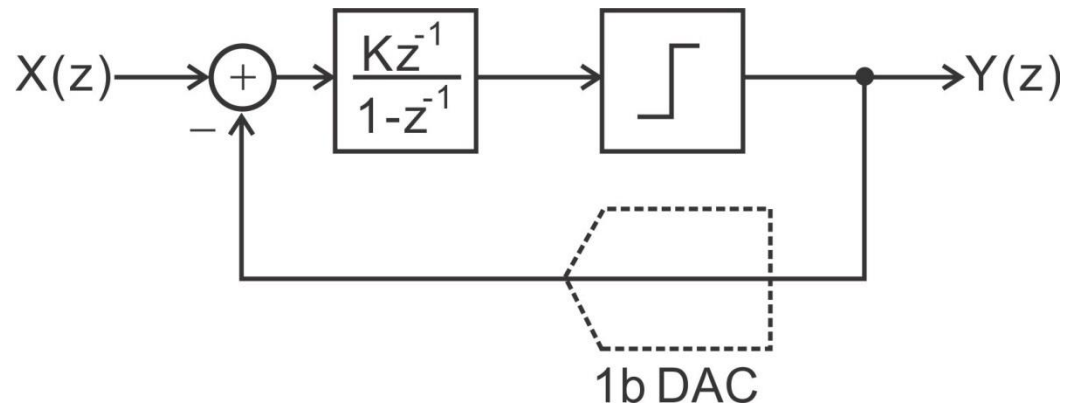
Stability

- The feedback loop can be unstable!
- Note that the quantizer is nonlinear and its gain depends on the input amplitude
 - We need to guarantee loop stability under all possible quantizer gains
- In general, it is hard to analyze the stability for a nonlinear system
- Designers usually rely on:
 - Stability analyses using the linear model
 - Heuristics
 - Intensive time-domain simulations using the nonlinear model

Stability Heuristics

- Single-bit
 - 1st-order modulator is stable with the standard integrator
 - It can go unstable if additional gain is provided in the loop
 - 2nd-order modulator is stable for inputs with magnitude $< \Delta/20$. It is “likely” to be stable for slow varying input with magnitude $< 0.8 \cdot \Delta/2$,
 - **Lee's criterion: modulator is "likely" to be stable if $\max[H_e(\omega)] < 1.5$**
- Multi-bit
 - The stability for a multibit modulator is in general much better than that for a single-bit modulator
 - A N^{th} -order modulator using an $N+1$ bit quantizer is stable for arbitrary inputs with amplitude less than half the quantizer input range

1st-order Modulator Stability



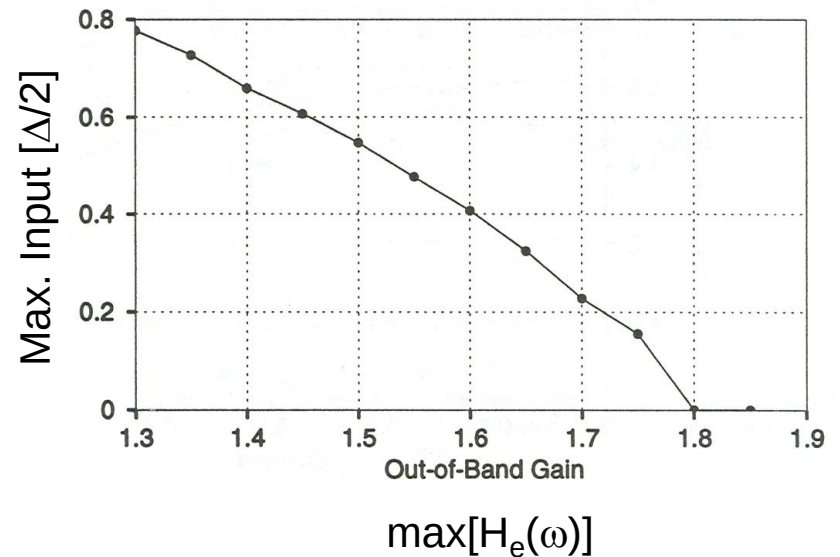
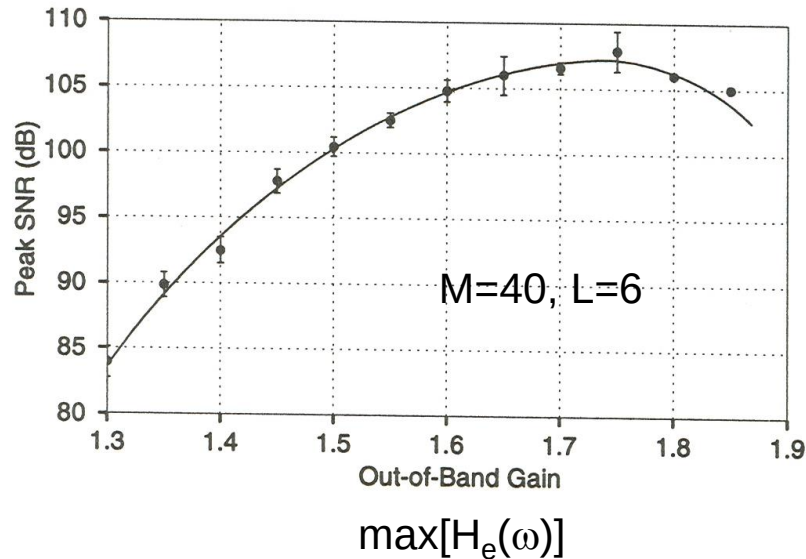
$$STF = \frac{A(z)}{1 + A(z)} = \frac{Kz^{-1}}{1 + (K-1)z^{-1}}$$

$$NTF = \frac{1}{1 + A(z)} = \frac{1 - z^{-1}}{1 + (K-1)z^{-1}}$$

- The loop is unstable for $K > 2$!

Stability and SQNR Tradeoff

[Norsworthy, pp.156]



- larger $\max[H_e(\omega)] \rightarrow$ more attenuation in the signal band $\rightarrow$ higher SQNR
- larger $\max[H_e(\omega)] \rightarrow$ lower stability

Achievable SQNR for 1-b Modulator

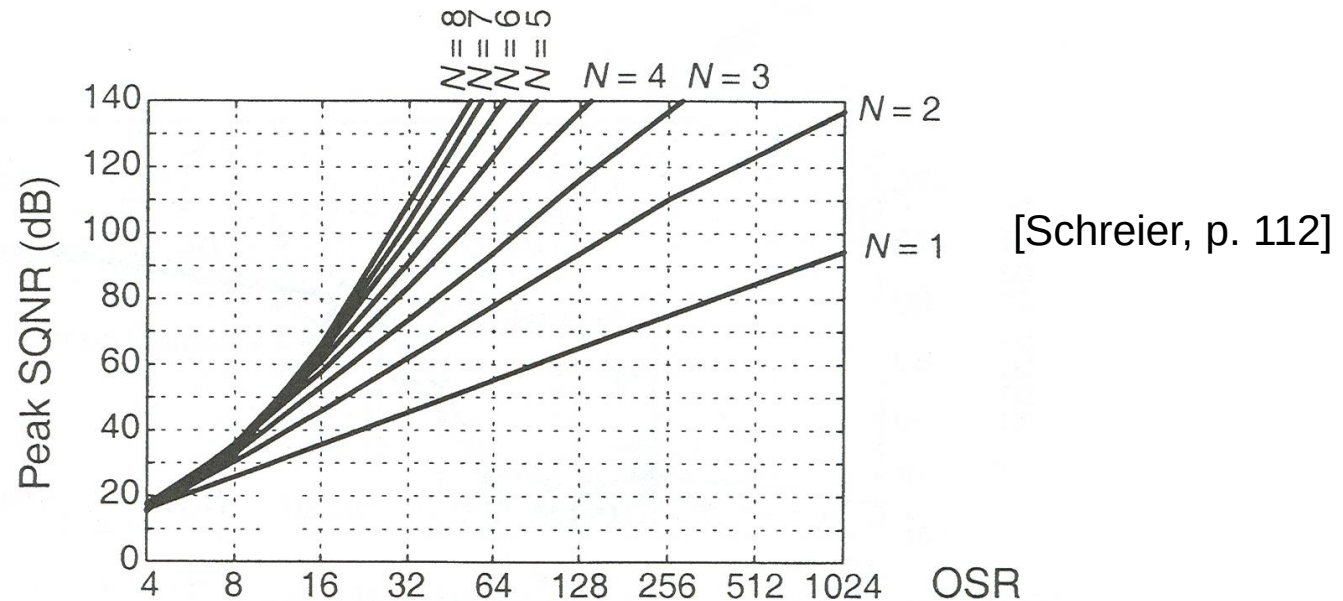
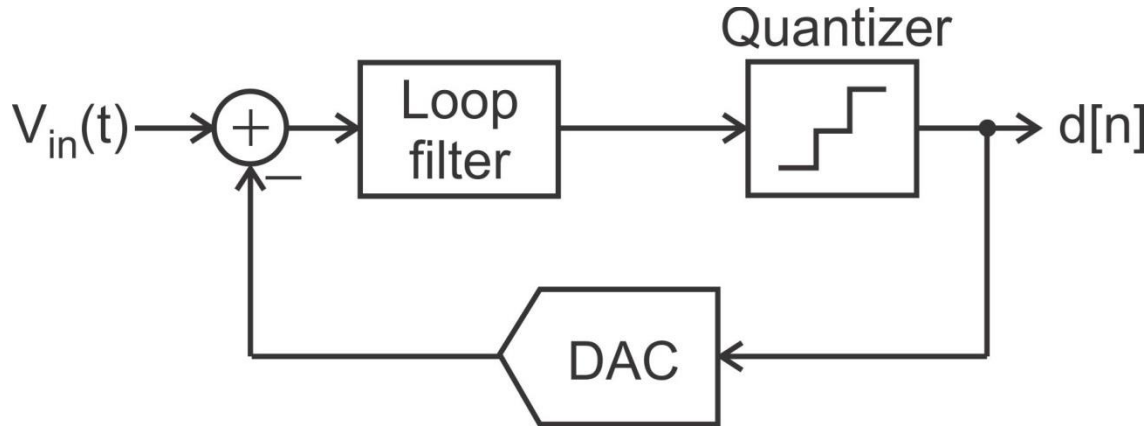


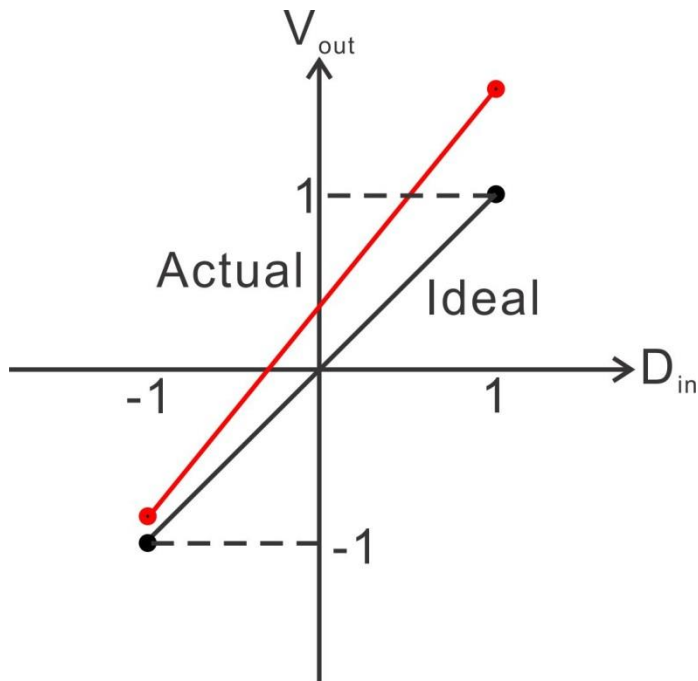
Figure 4.14: Empirical SQNR limit for 1-bit modulators of order N .

- The gain in SQNR for increasing modulator order is limited at low OSR.
- Usually, the modulator order is limited to 5.

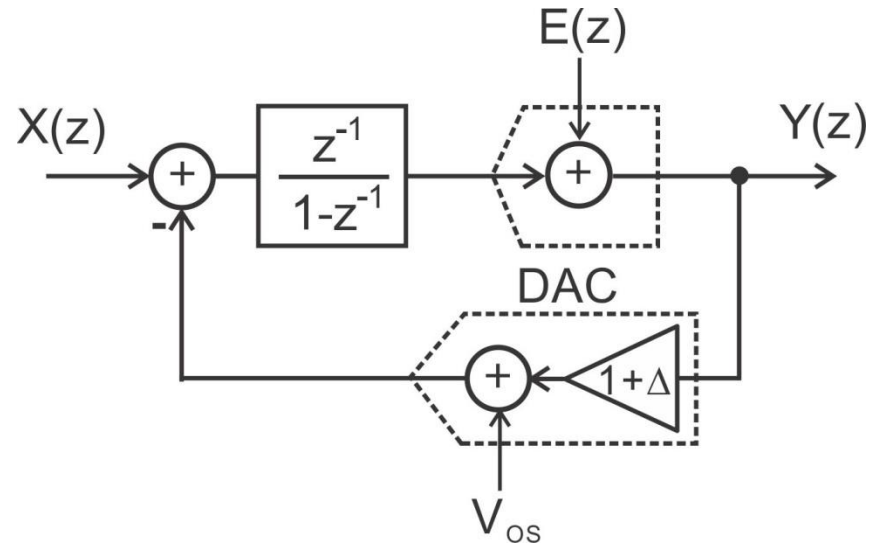
Advantages for Multibit Modulators



- Advantages for a multibit $\Delta\Sigma$ ADC:
 - Reduced quantization error
 - Well-defined quantizer gain and improved stability
 - More aggressive noise transfer function
 - Reduced requirement on slew rate and loop filter linearity, and less sensitivity to clock jitter
- Disadvantage:
 - Unlike a binary DAC, a multi-bit DAC is not guaranteed linear



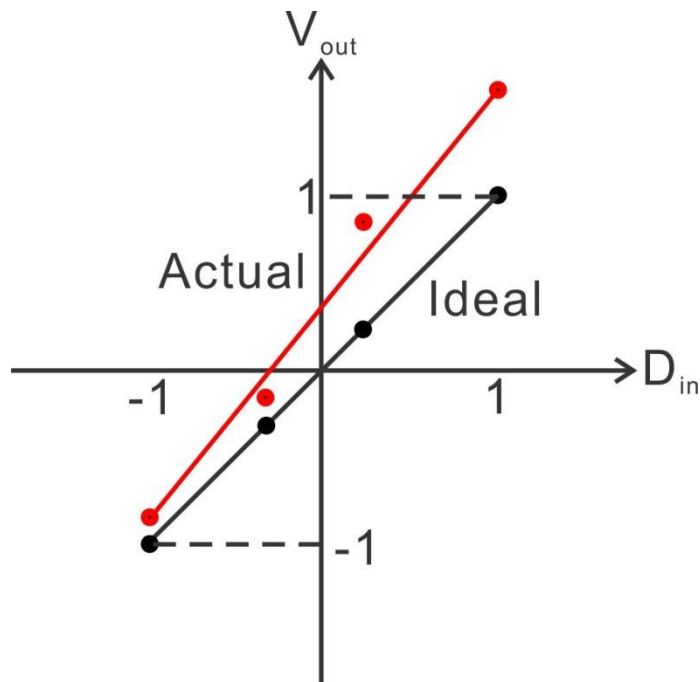
1-bit DAC is always linear



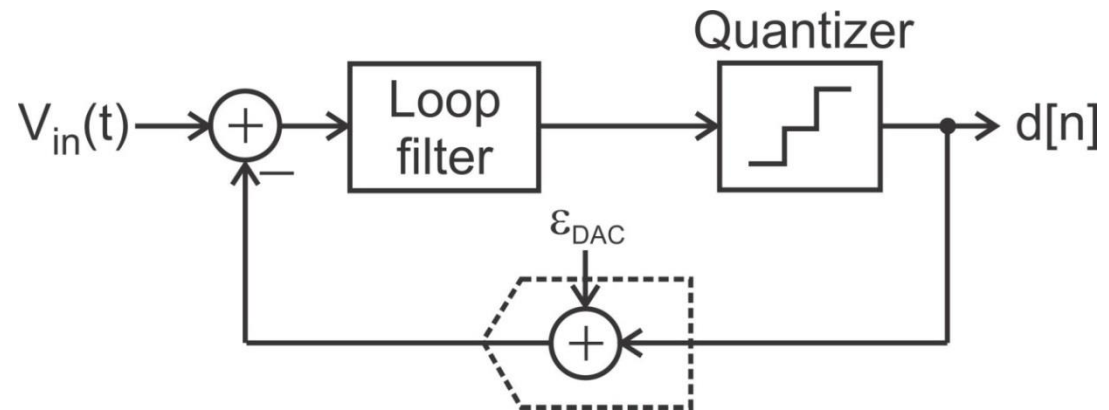
$$Y(z) = X(z) \frac{z^{-1}}{1 + \Delta z^{-1}} + E(z) \frac{1 - z^{-1}}{1 + \Delta z^{-1}} + V_{OS} \frac{z^{-1}}{1 + \Delta z^{-1}}$$

Constant gain and offset errors do not cause nonlinearity

Multibit DAC

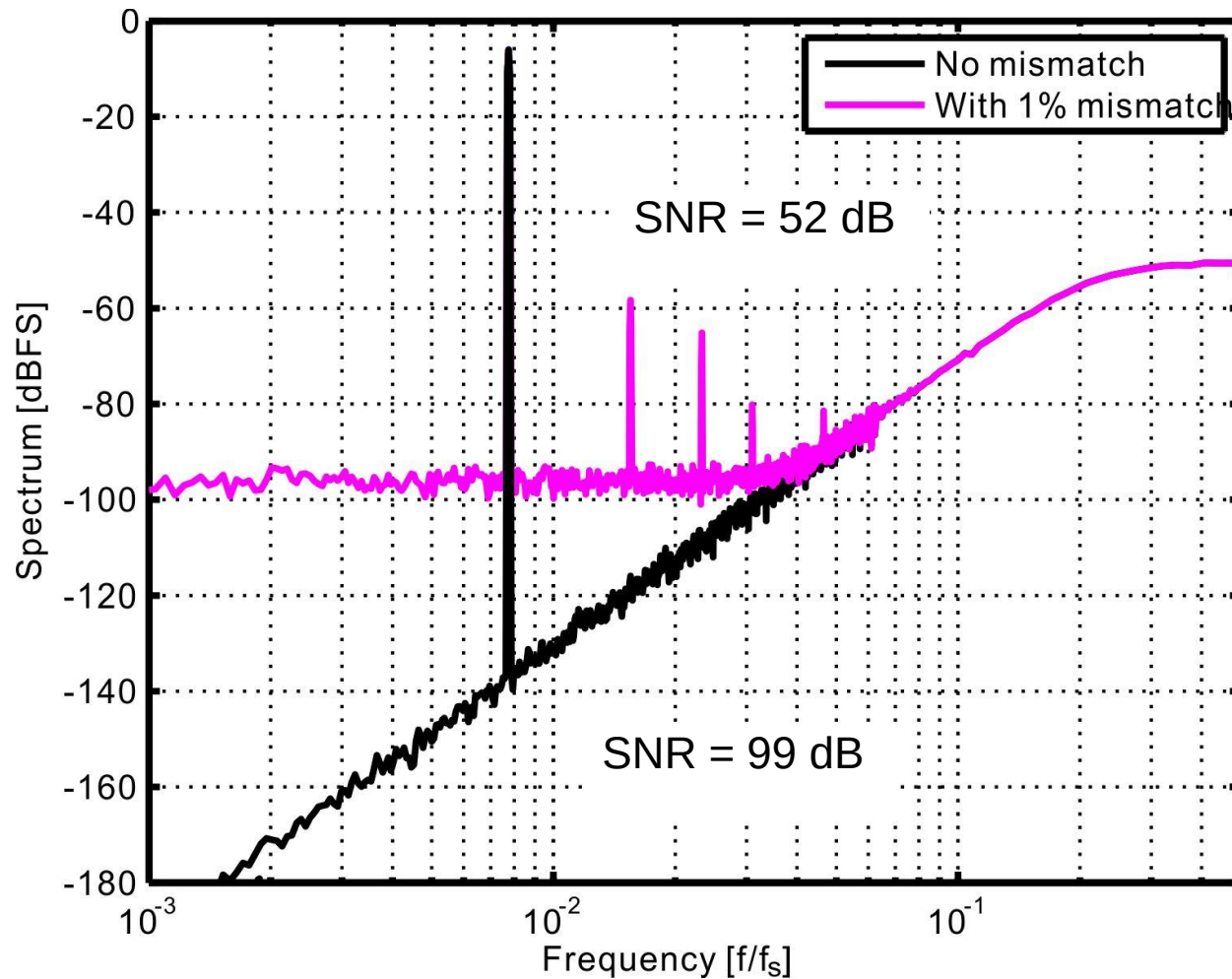


Multibit DAC is not linear
due to mismatches

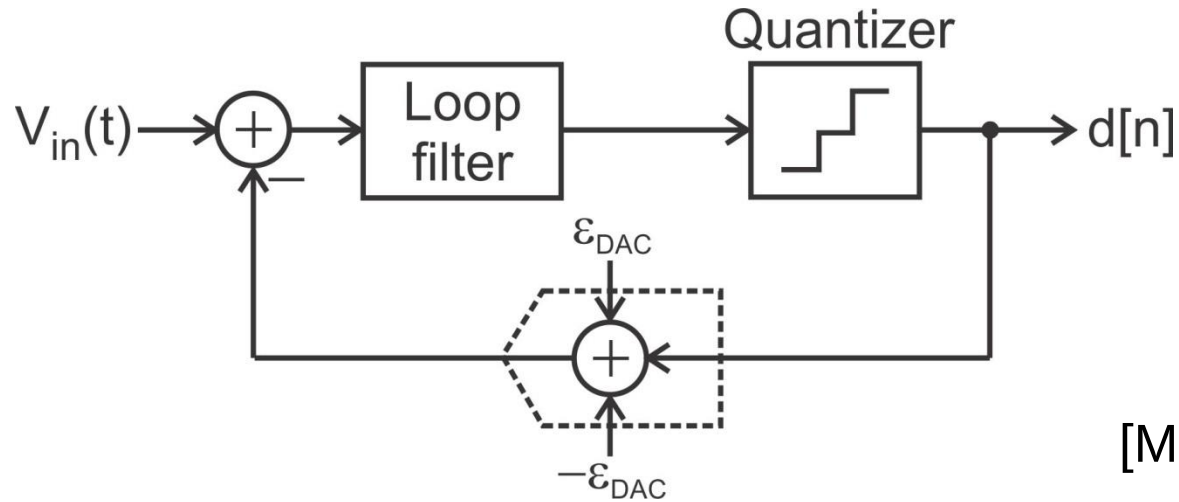


- The mismatch induced error ϵ_{DAC} directly adds to the input and is not shaped!
- It can seriously degrade the ADC performance

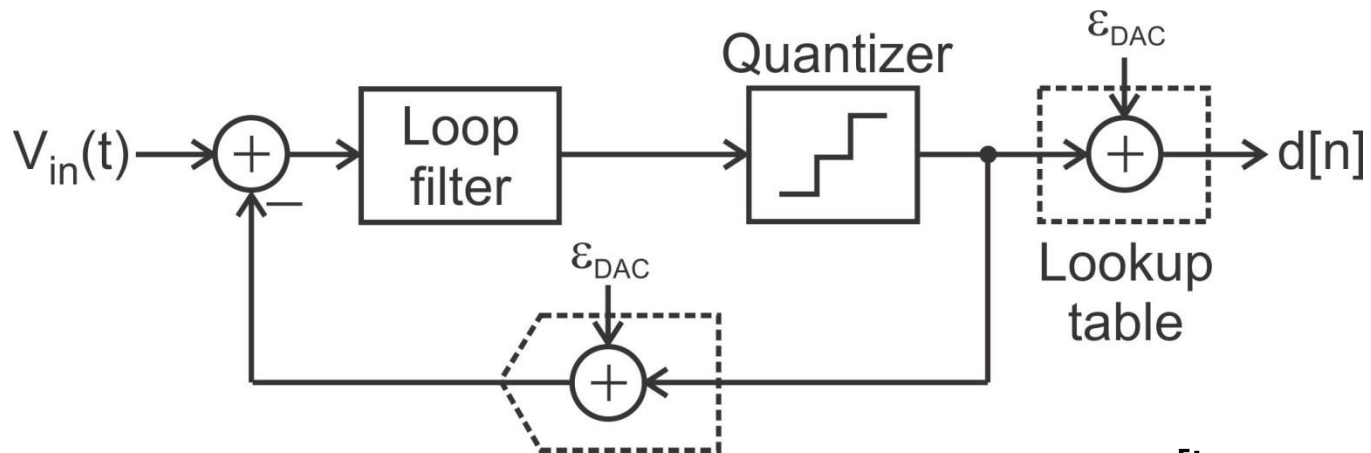
A 3rd-order $\Delta\Sigma$ ADC with 4-bit DAC



Calibration

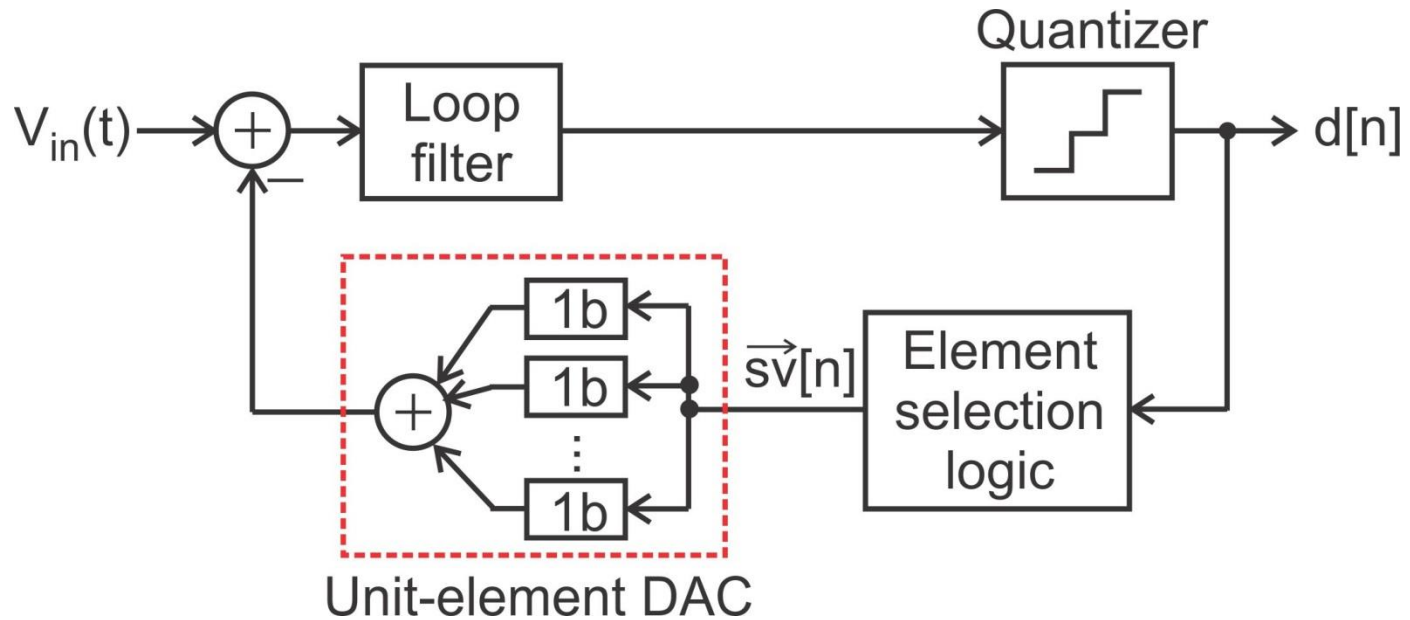


[Moon et al., 1999]

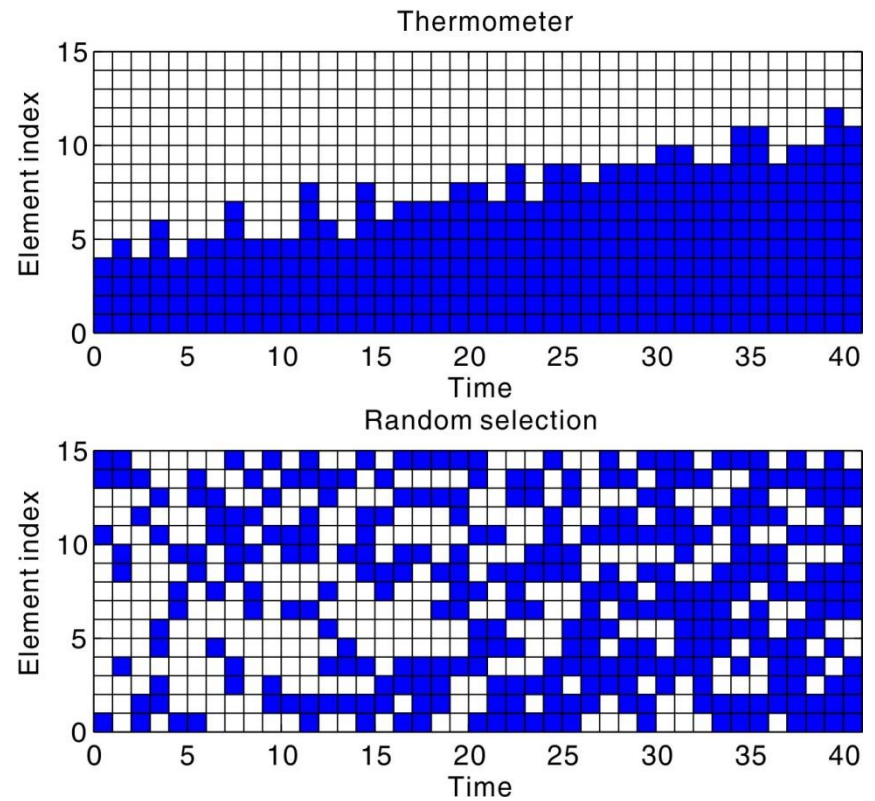
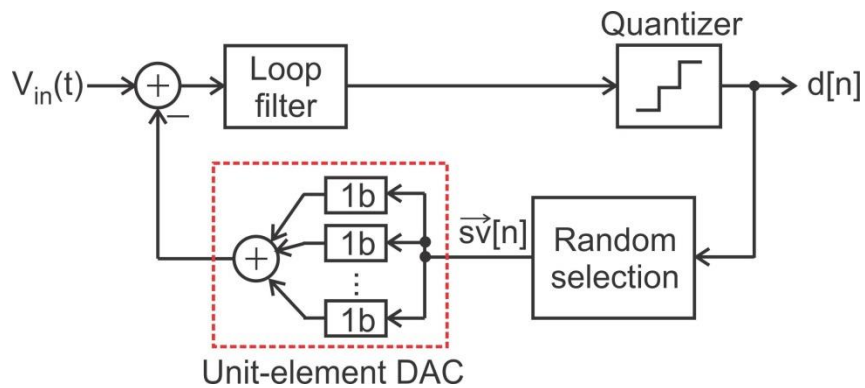


[Larson et al., 1988]

Selection Sequence Design

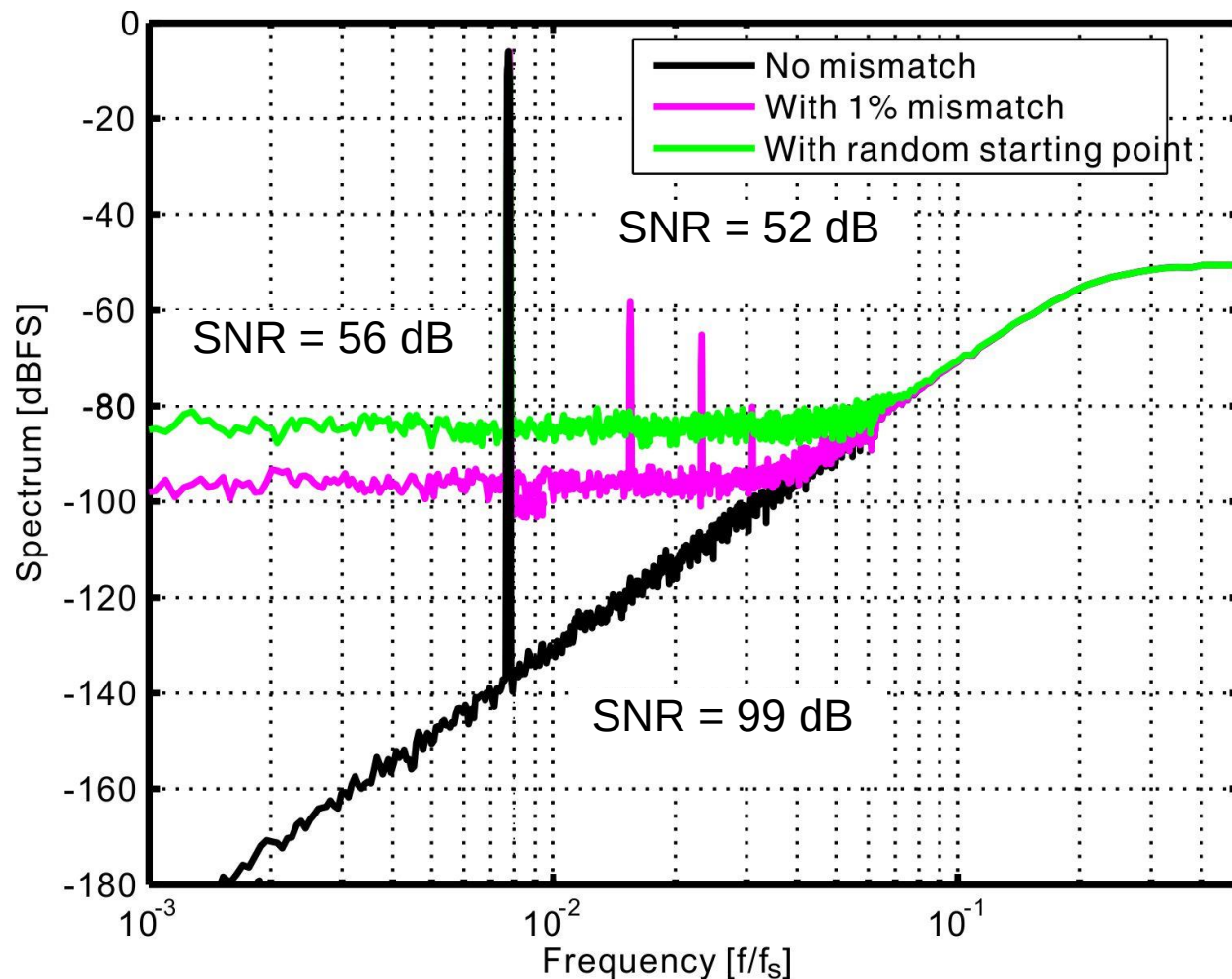


Random Selection

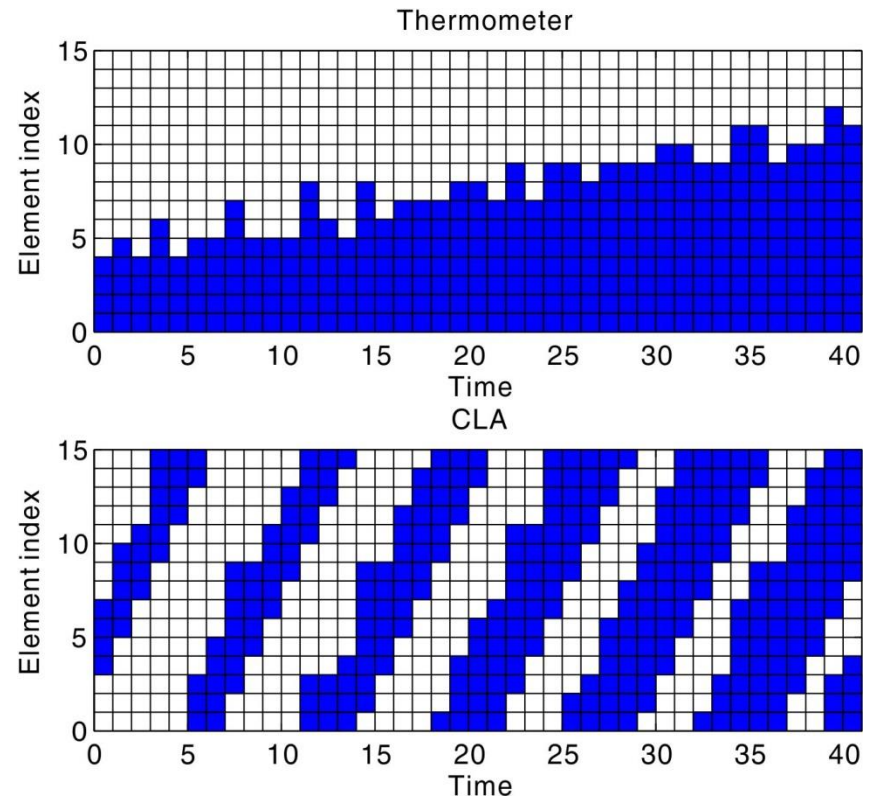
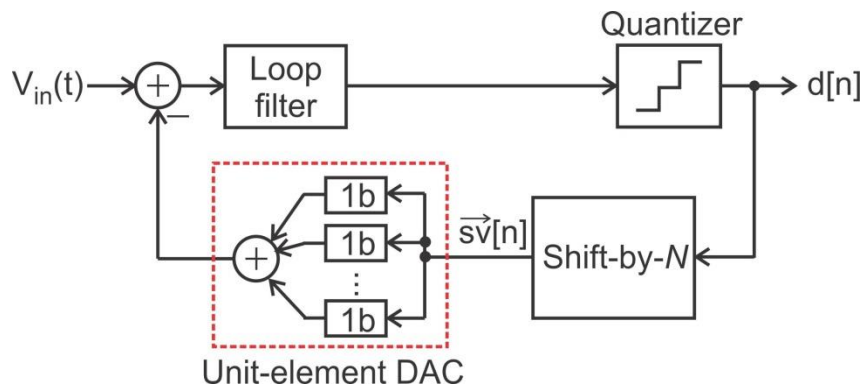


[Van de Plassche, 1976]
[Carley, 1989]

Tones Disappear but Noise Floor Rises

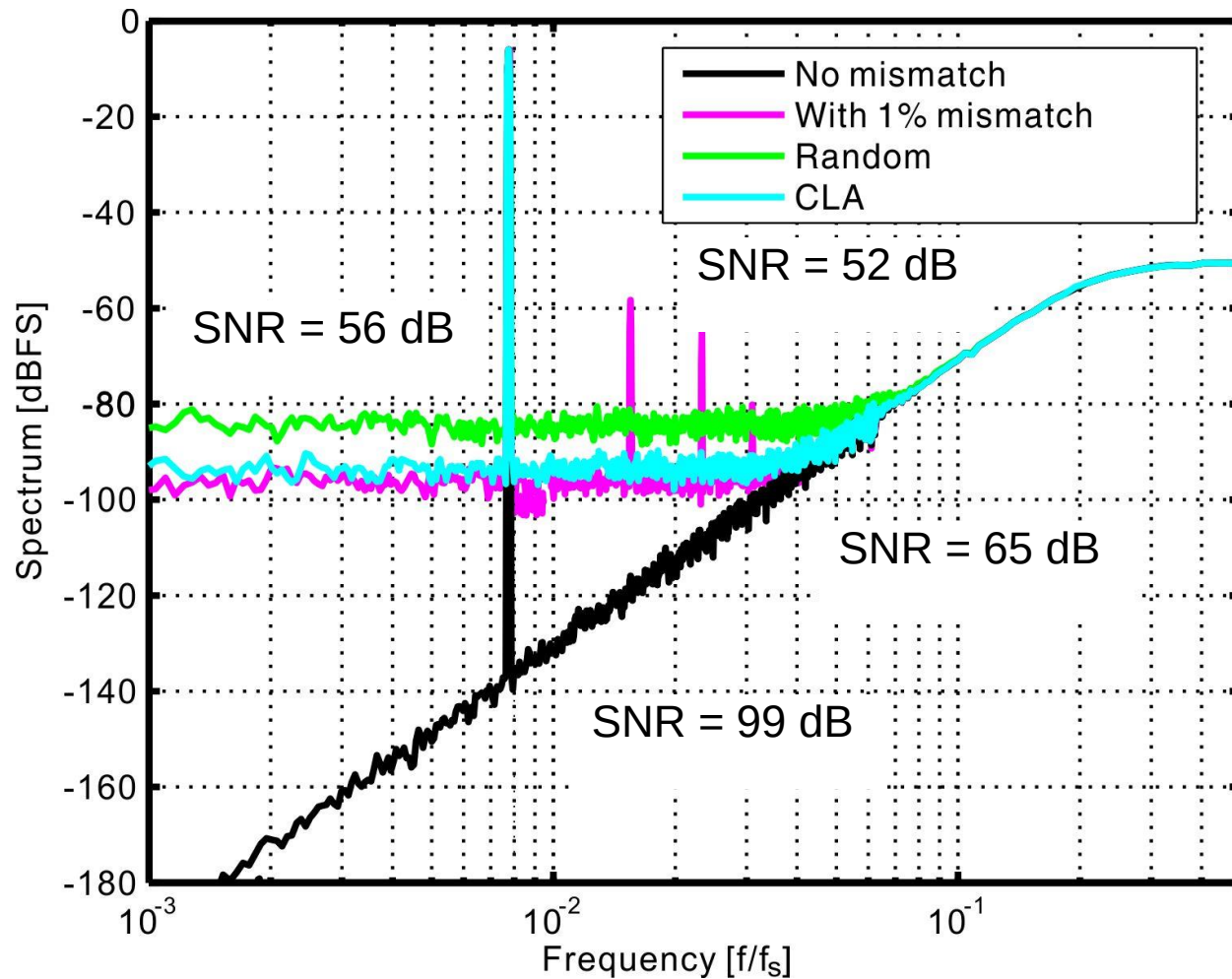


Deterministic Shifting (Clocked Averaging)

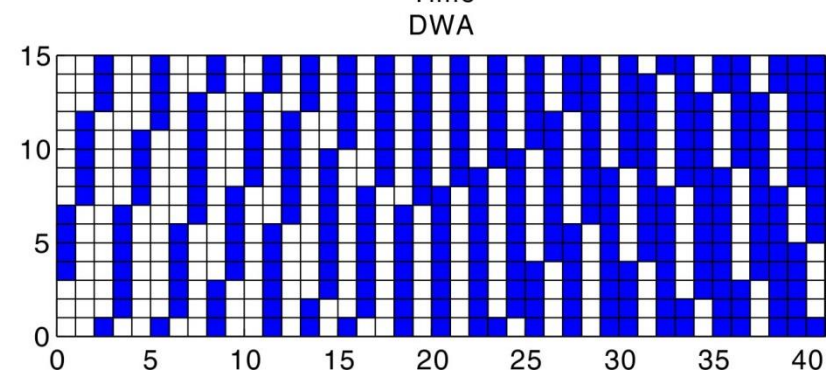
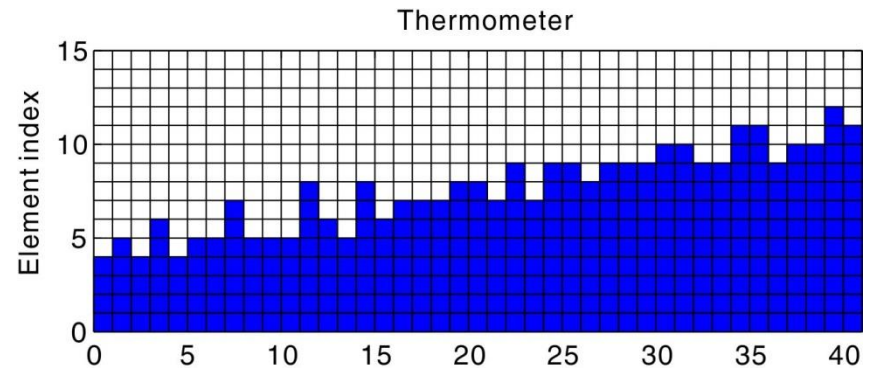
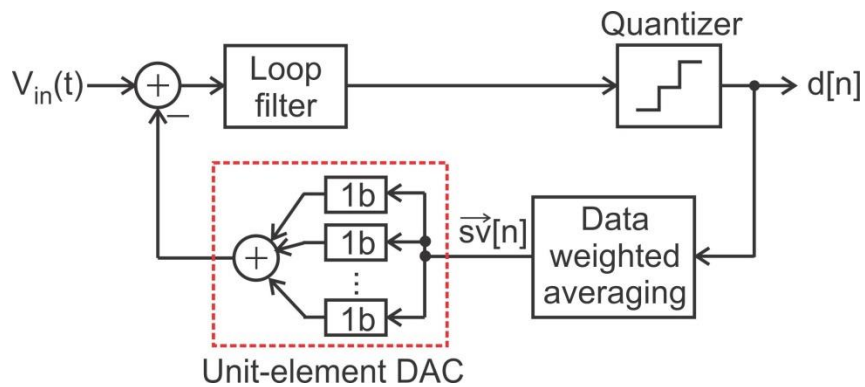


[Leung et al., 1992]

Noise Floor Lowers



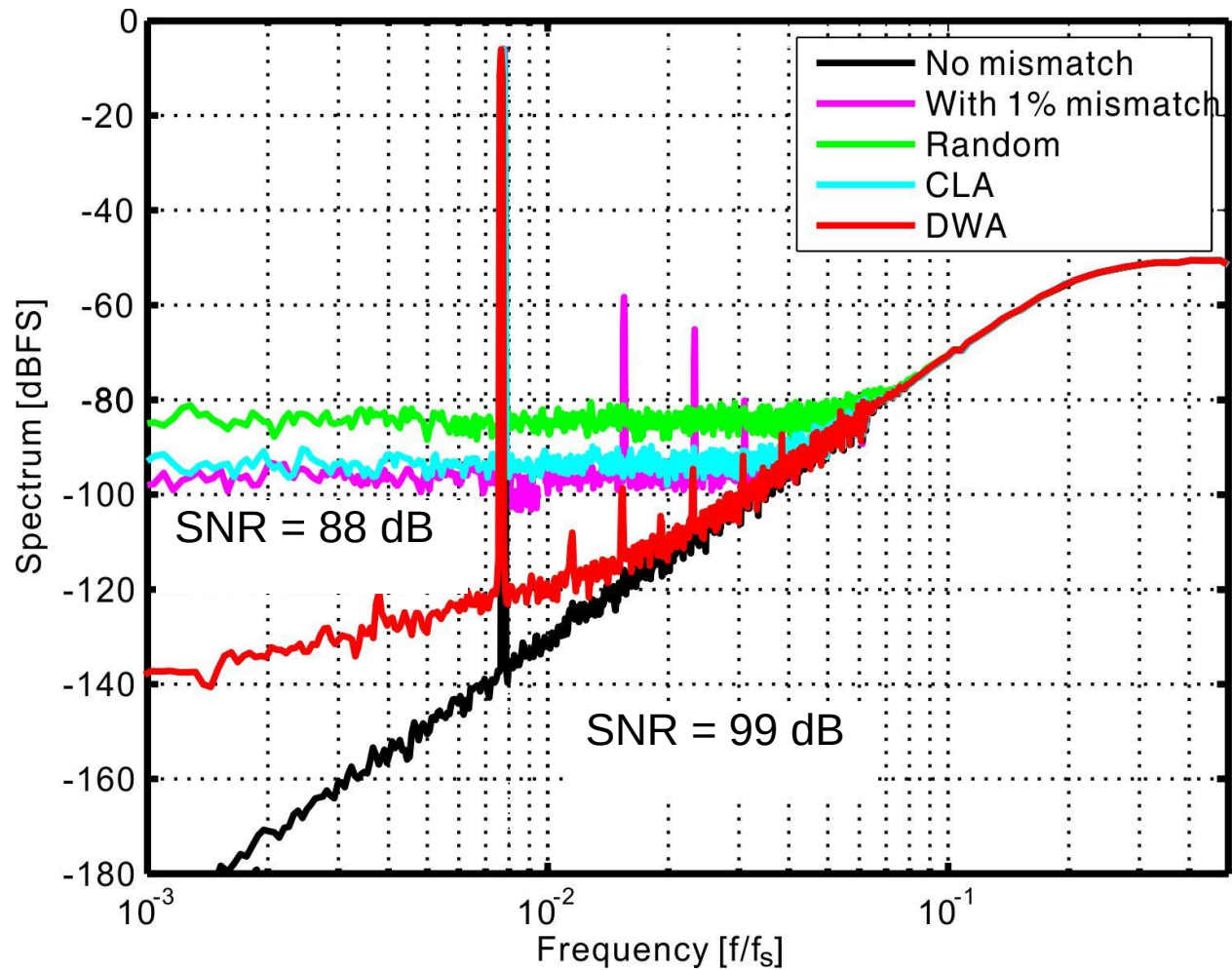
Data Weighted Averaging



[Jackson 1993]

[Baird & Fiez, 1995]

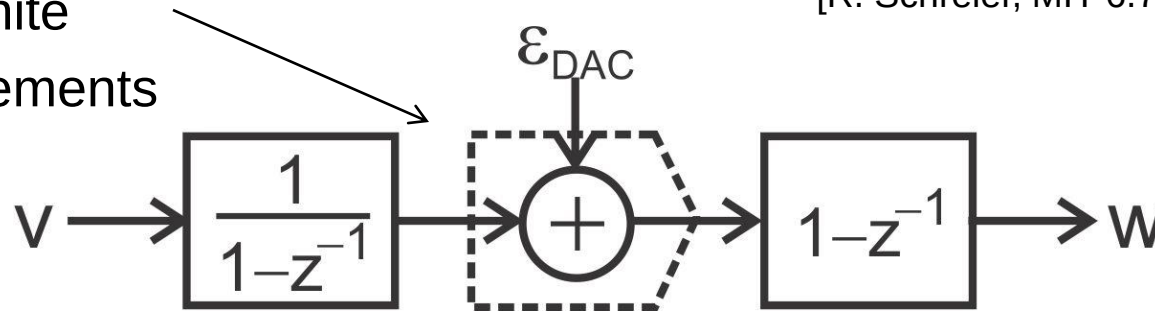
1st-order Mismatch Shaping



Why 1st-order Mismatch Shaping?

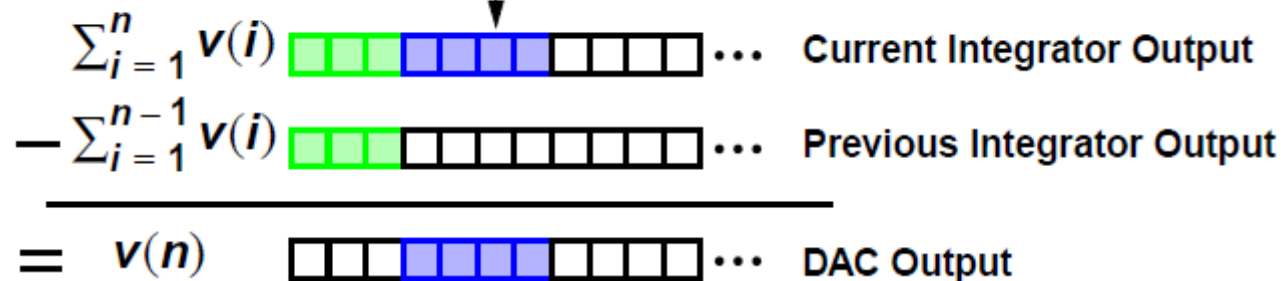
DAC has infinite number of elements

[R. Schreier, MIT 6.775 Lecture note]



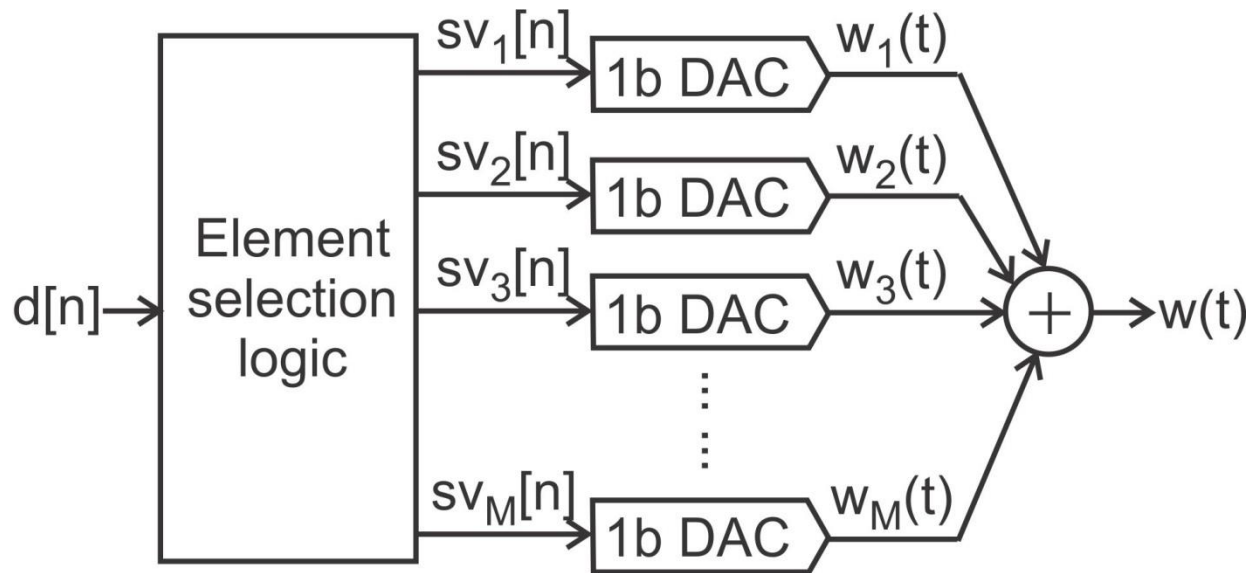
$$W = V + (1-z^{-1})\epsilon_{\text{DAC}}$$

Semi-Infinite Array of Unit Elements



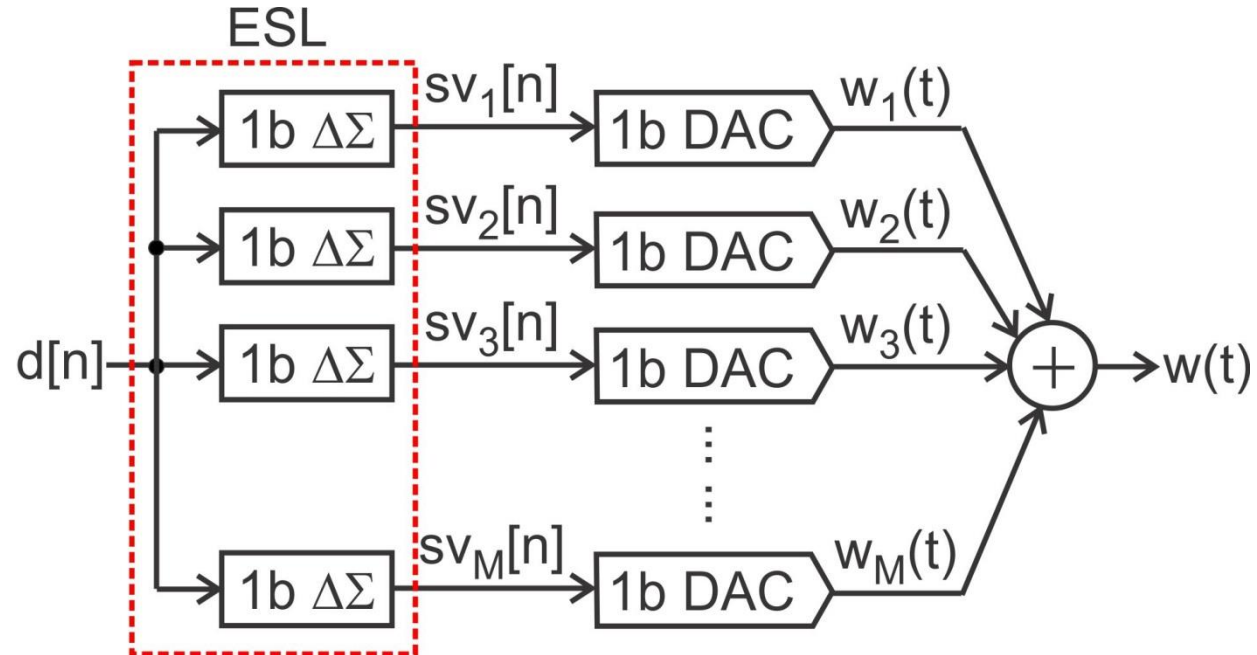
Use element in a rotation fashion

Necessary and Sufficient Conditions

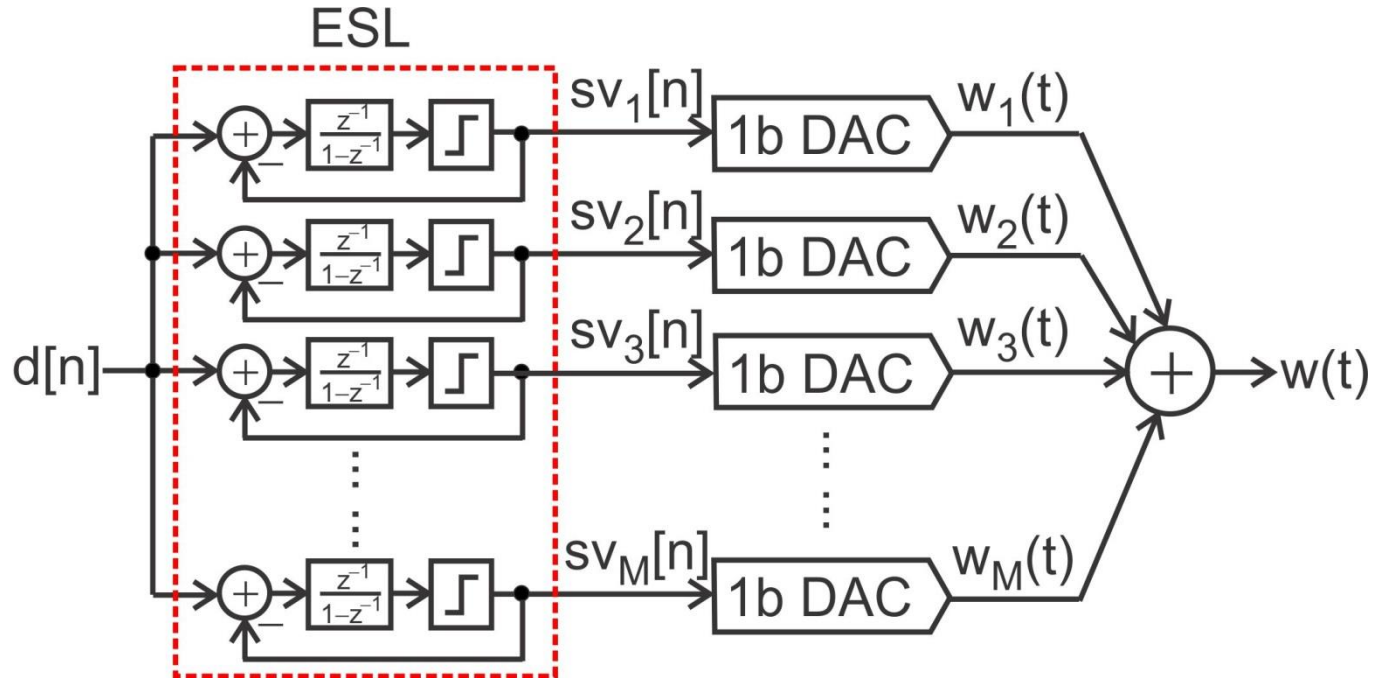


1. Each $sv_i[n]$ exhibits a shaped spectrum
2. $\sum sv_i[n] = d[n]$

Conceptual Implementation

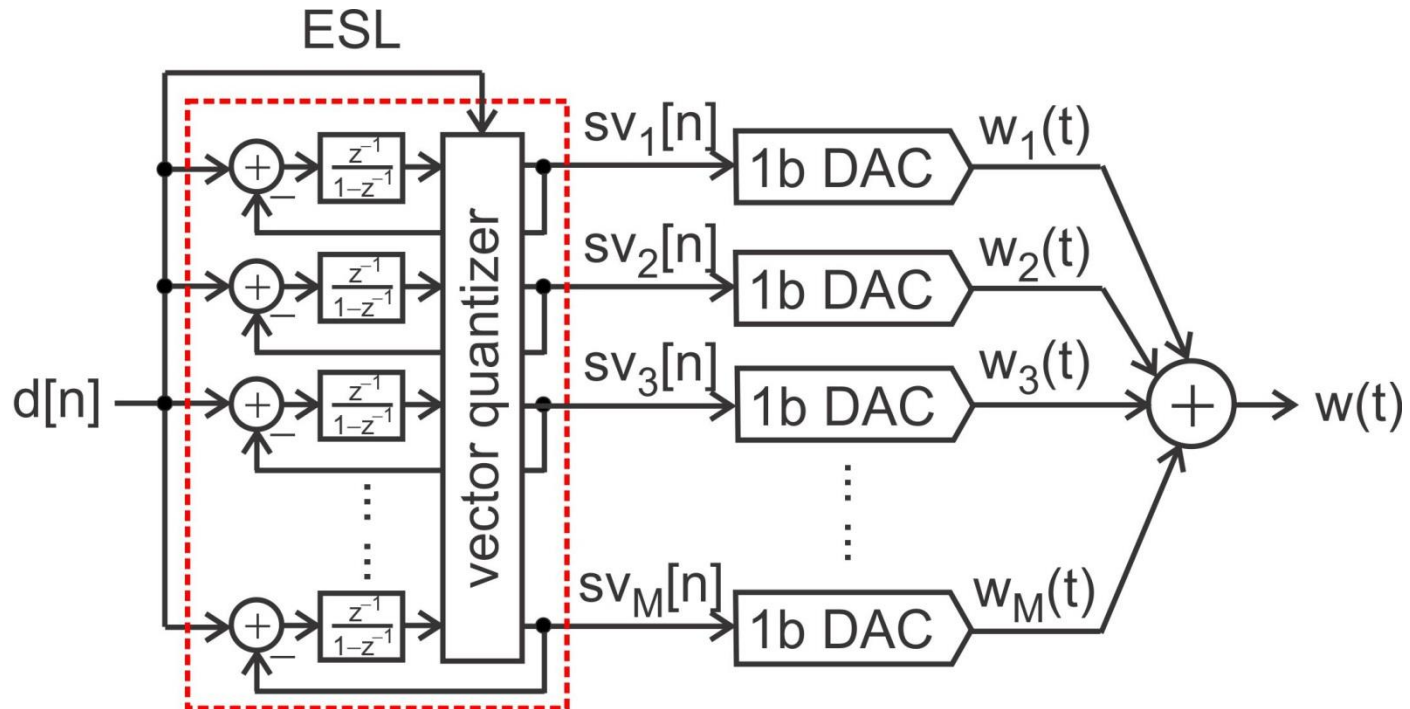


Vector-Quantizer-Based ESL (1)



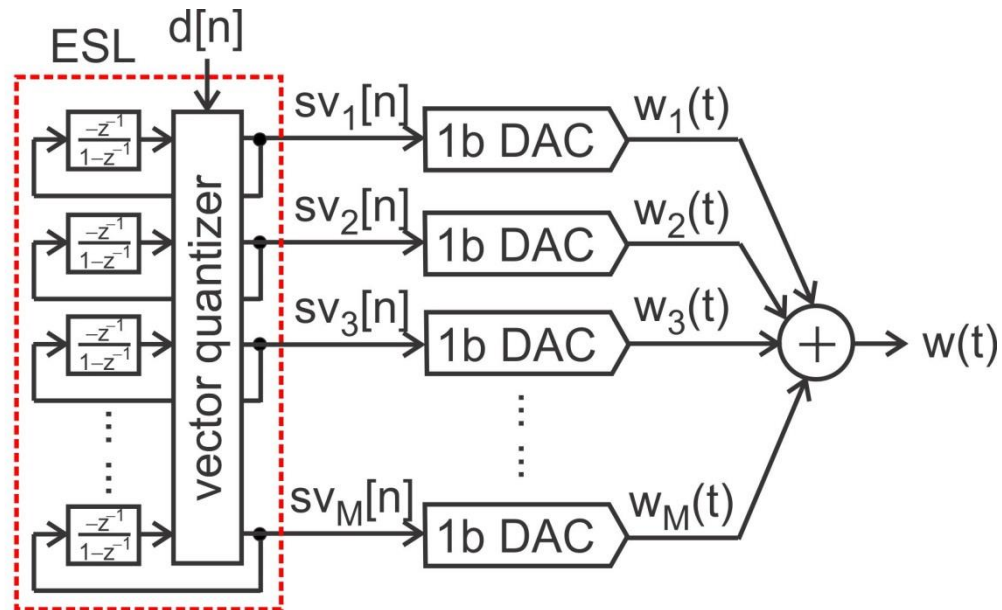
$\sum sv_i[n] \neq d[n]$, leading to large quantization noise

Vector-Quantizer-Based ESL (2)

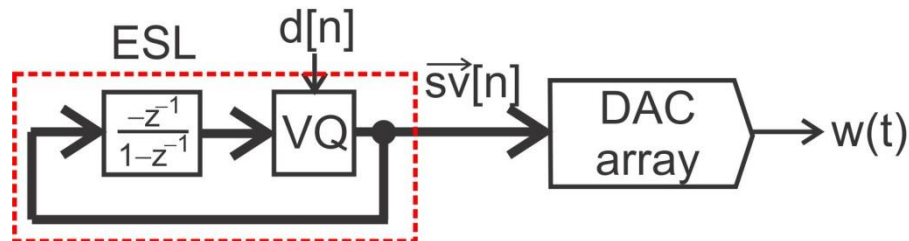


The vector quantizer (VQ) selects $d[n]$ largest components to be 1

Simplified VQ-Based ESL

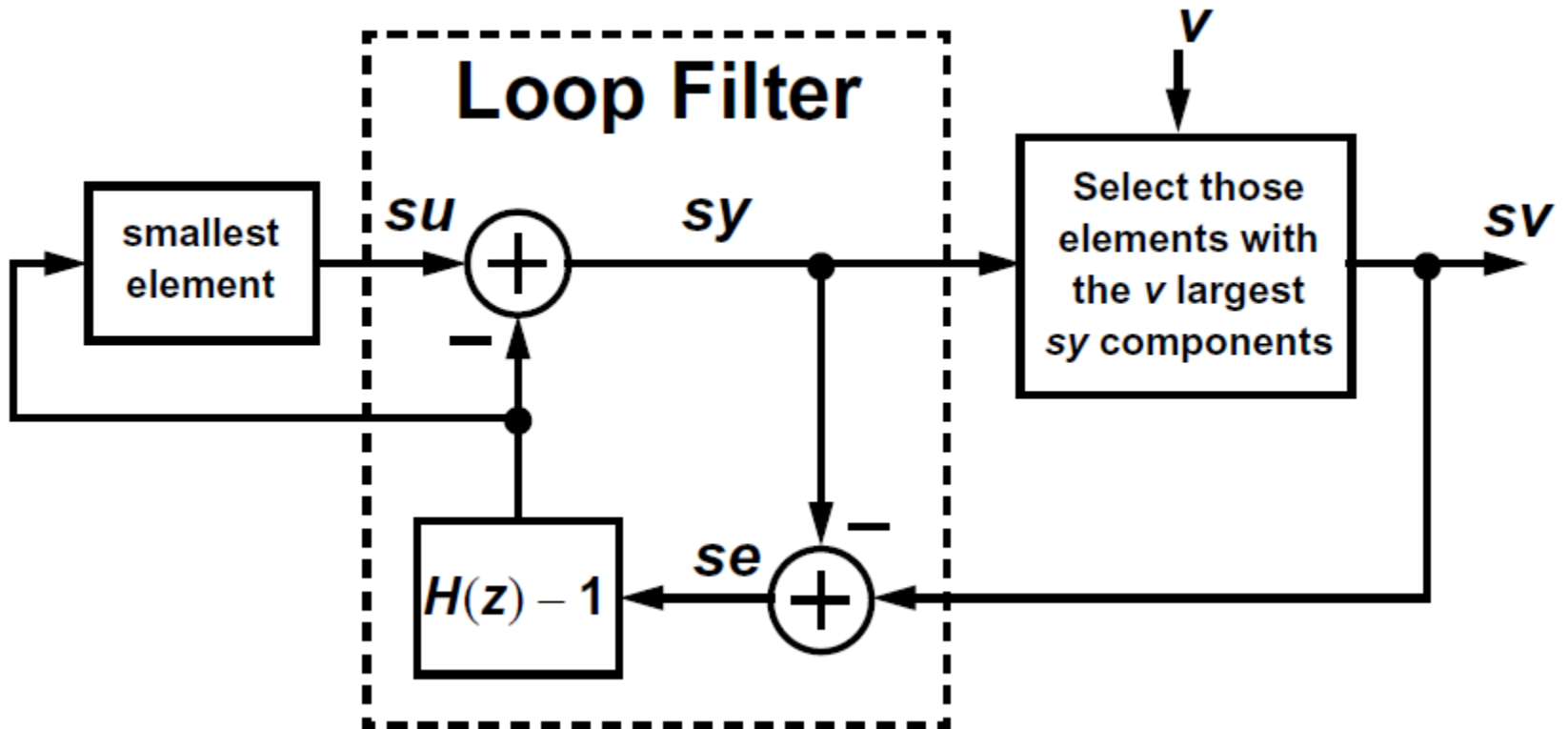


[Schreier & Zhang, 1995]

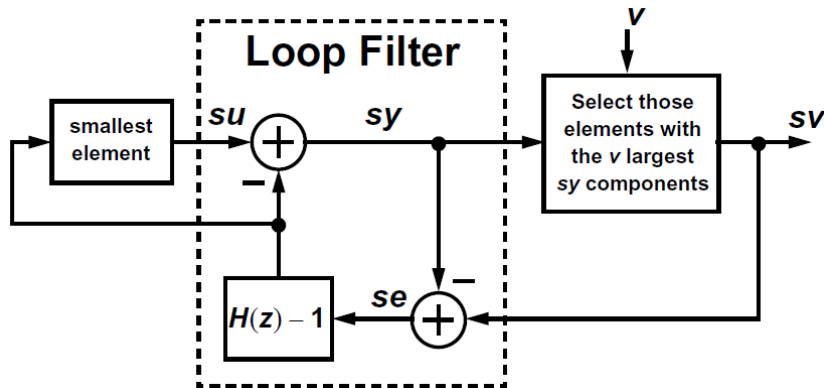


Canonical VQ-Based ESL

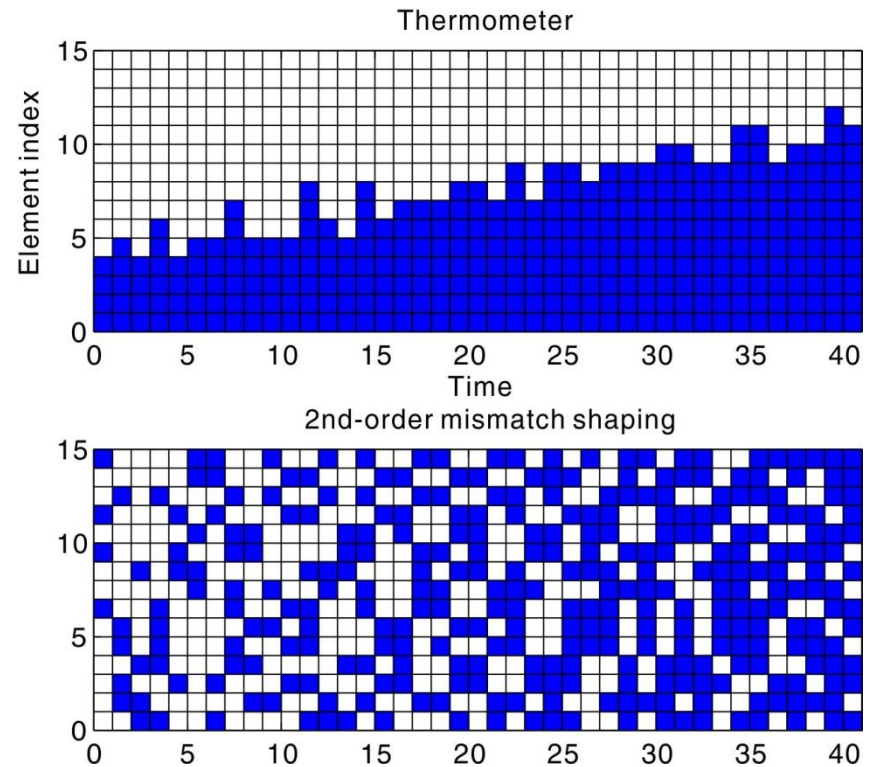
[Schreier & Zhang, 1995]



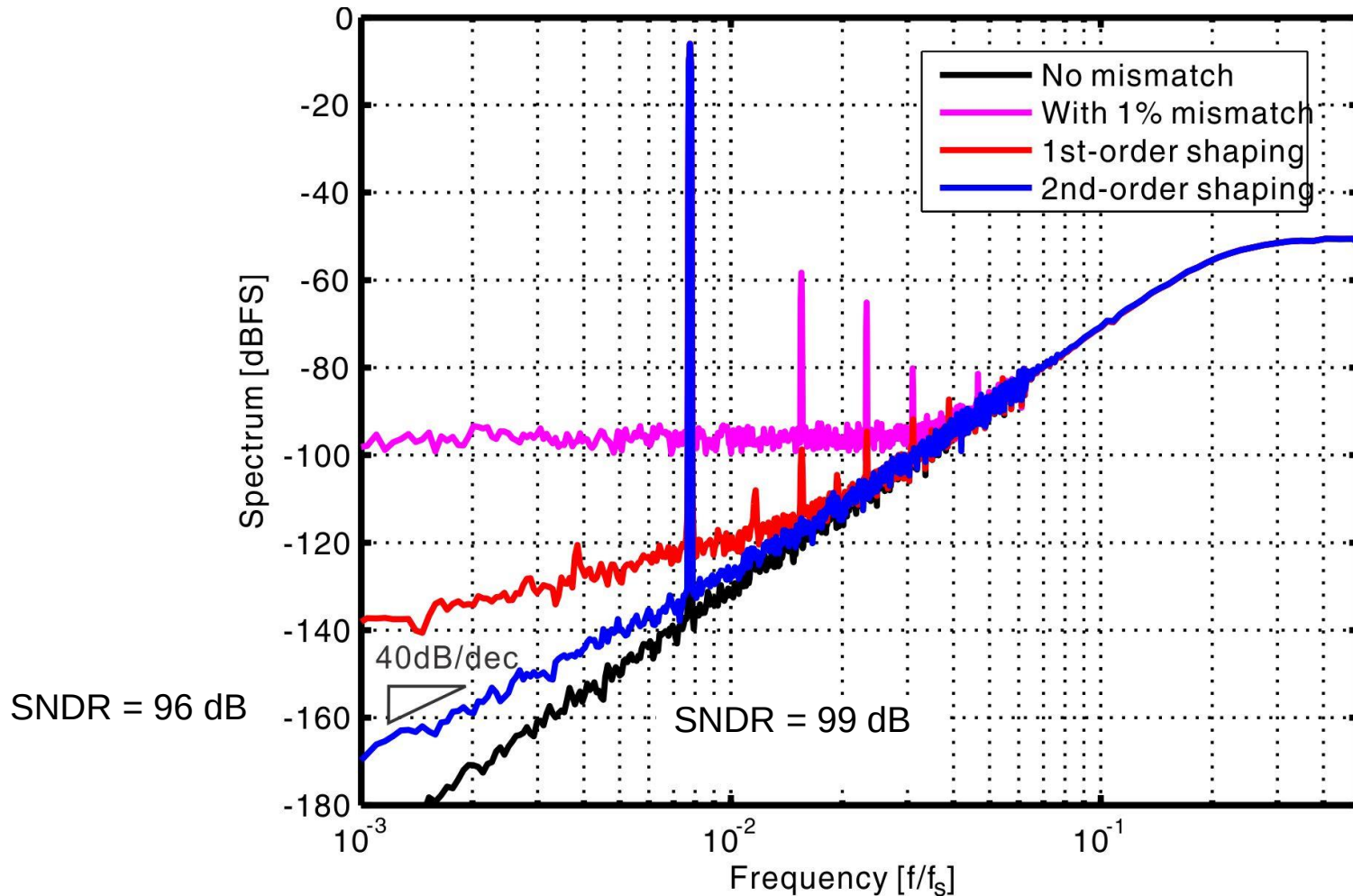
2nd-order Mismatch Shaping



$$H(z) = (1 - z^{-1})^2$$



Simulated Spectra



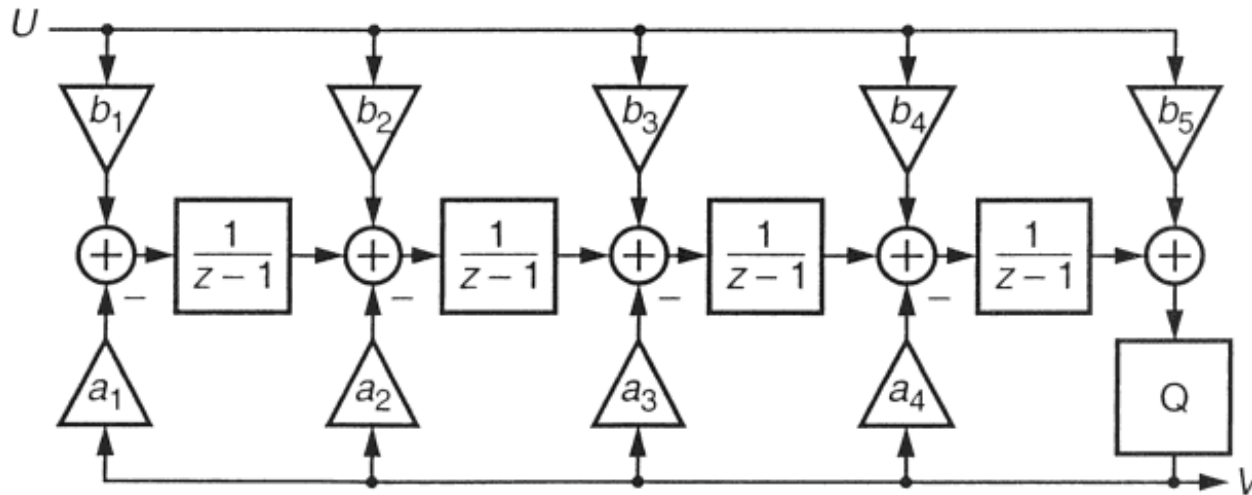
Research Directions

- Can we improve stability and achieve higher-order shaping?
- Can we reduce the complexity of the vector quantizer?
- Can we achieve shaping for segmented DAC?
- Can we reduce the delay in the ESL block?
- Can we apply simple and efficient dithering?
- Can we shape the dynamic ISI error?

Our Recent Publications

- Nan Sun, “High-order mismatch-shaped segmented multibit delta-sigma DACs with arbitrary unit weights,” *IEEE Transactions on Circuits and Systems – I*, vol. 59, no. 2, pp. 295-304, Feb. 2012.
- Nan Sun and Peiyan Cao, “Low-complexity high-order vector-based mismatch shaping in multi-bit $\Delta\Sigma$ ADCs,” *IEEE Transactions on Circuits and Systems – II*, vol. 58, no. 12, pp 872-876, Dec. 2011.
- Nan Sun, “High-order mismatch-shaping in multibit DACs,” *IEEE Transactions on Circuits and Systems – II*, vol. 58, no. 6, pp 346-350, Jun. 2011.
- Arindam Sanyal, Peijun Wang, and Nan Sun, “A thermometer-like mismatch shaping technique with minimum element transition activity for multi-bit delta-sigma DACs,” *IEEE Transactions on Circuits and Systems – II*, vol. 61, no. 7, pp. 461-465, Jul. 2014.
- Arindam Sanyal, Long Chen, and Nan Sun, “Dynamic element matching with signal-independent element transition rates for multibit delta sigma modulators,” *IEEE Transactions on Circuits and Systems – I*, accepted.
- Arindam Sanyal and Nan Sun, “A simple and efficient method for vector quantizer based mismatch-shaped $\Delta\Sigma$ DACs,” *IEEE Proceedings of International Symposium on Circuits and Systems*, pp. 528 – 531, 2011.
- Peijun Wang and Nan Sun, “A random DEM technique with minimal element transition rate for high-speed DACs,” *IEEE International Symposium on Circuits and Systems (ISCAS)*, pp. 1155-1158, 2014.

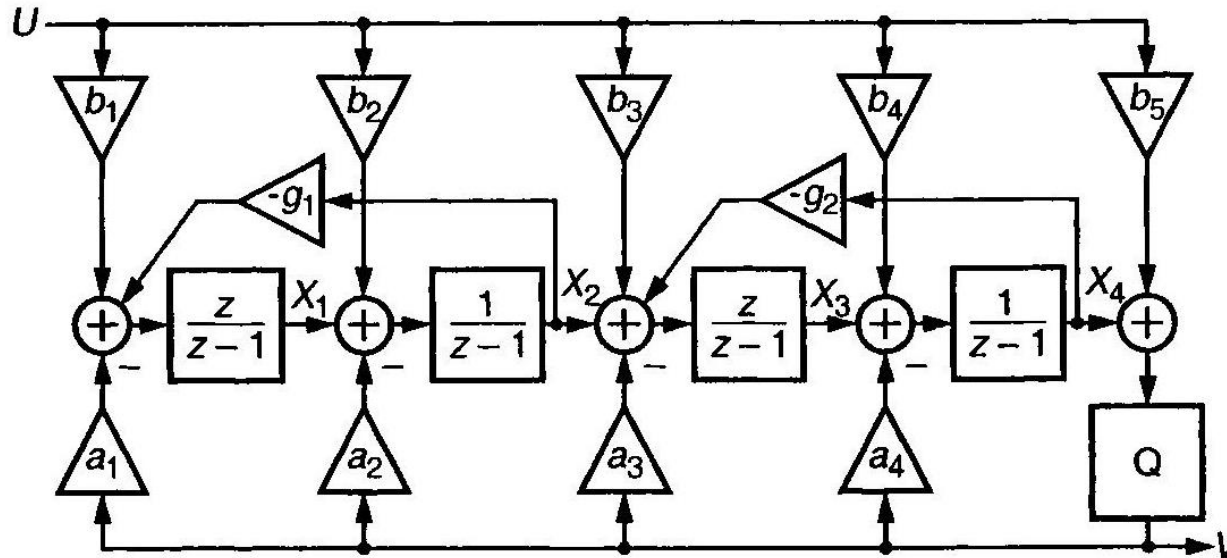
General CIFB Architecture



[Schreier, p. 115]

- CIFB - Cascade of Integrators with Feedback
- All zeros of NTF are at DC for this structure
 - $H_E(z) = (1-z^{-1})^N/D(z)$
- May not lead to minimum in-band quantization noise

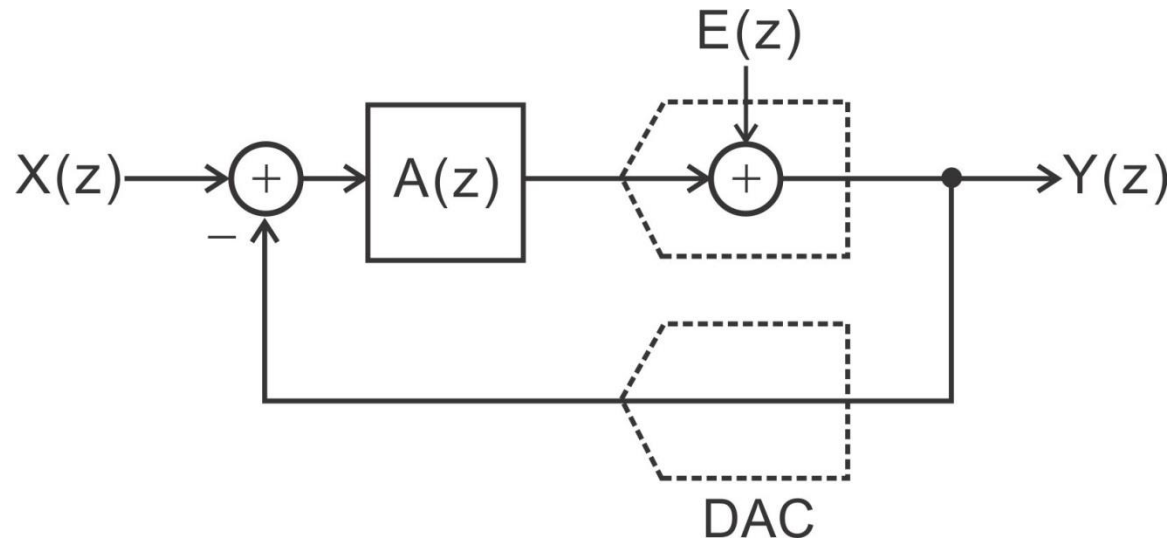
General CRFB Architecture



[Schreier, p. 118]

- CRFB - Cascade of Resonators with Feedback
- CRFB provides complex zeros

Single Feedback Loop Design Example



$$NTF = \frac{1}{1 + A(z)} = \frac{1}{1 + z^{-1} \frac{k_0 + k_1 z^{-1} \dots}{1 + m_1 z^{-1} + \dots}} = \frac{1 + m_1 z^{-1} + \dots}{1 + (k_0 + m_1) z^{-1} + \dots}$$

- $A(z)$ must have at least one delay inside for realizability $\rightarrow$ both the constant terms in NTF numerator and denominator should be 1
- Modulator design is essentially about the design of NTF and $A(z)$

NTF Design Procedure

- Standard design procedure:
 - Choose order based on desired SQNR and OSR
 - Design NTF using filter approximations (e.g. Chebyshev2)
 - Make sure that NTF is realizable and satisfies Lee's stability criterion
 - Derive $A(s)$ based on NTF
 - Evaluate performance and stability using time-domain simulations
 - Scale coefficients to limit integrator output swing
 - For details, see *Delta-Sigma Data Converters*, by Norsworthy, Schreier & Temes, Sections 4.4 and 5.6
- Can be automated
 - Delta-Sigma Toolbox for MATLAB by Richard Schreier
 - <http://www.mathworks.com/matlabcentral/fileexchange/19-delta-sigma-toolbox>

NTF Design Example in Matlab (1)

% Specify parameters

L = 4, OSR = 50, StopBand = 70

% NTF synthesize

[b,a] = cheby2(L, StopBand, 1/OSR, 'high')

% Scale b(1) to 1 to ensure realizability; a(1)=1 by default

b = b/b(1);

% Check stability using Lee's criteria

NTF = filt(b, a, 1)

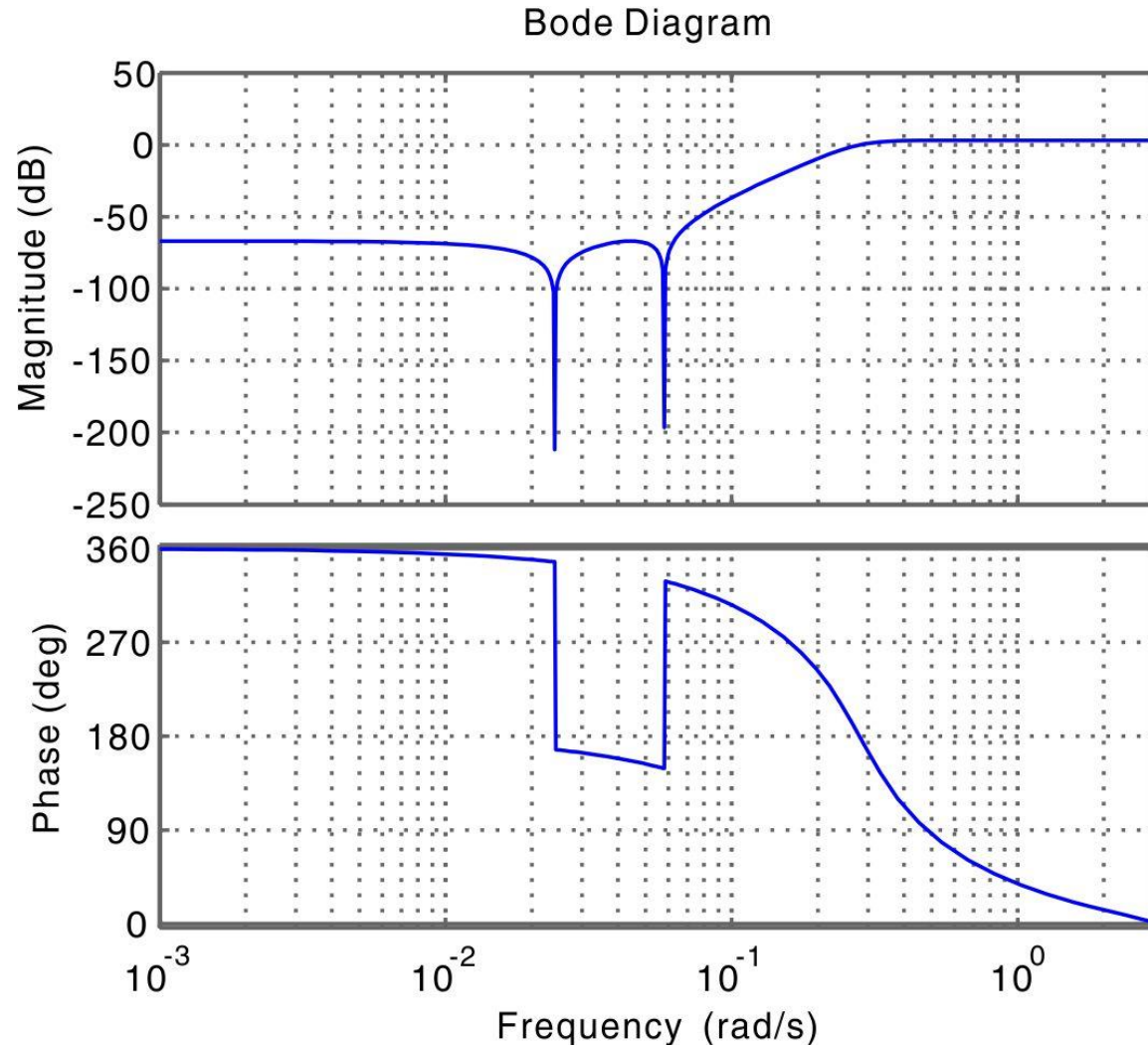
bode(NTF)

% Solve for A(z)

A = inv(NTF) - filt(1,1,1)

NTF Design Example in Matlab (2)

Inband slightly worse than -70dB due to scaling



$$\text{NTF}_{\text{max}} = 1.43$$

Commercial Example



Single-Supply 16-Bit $\Sigma\Delta$ Stereo ADC

AD1877*

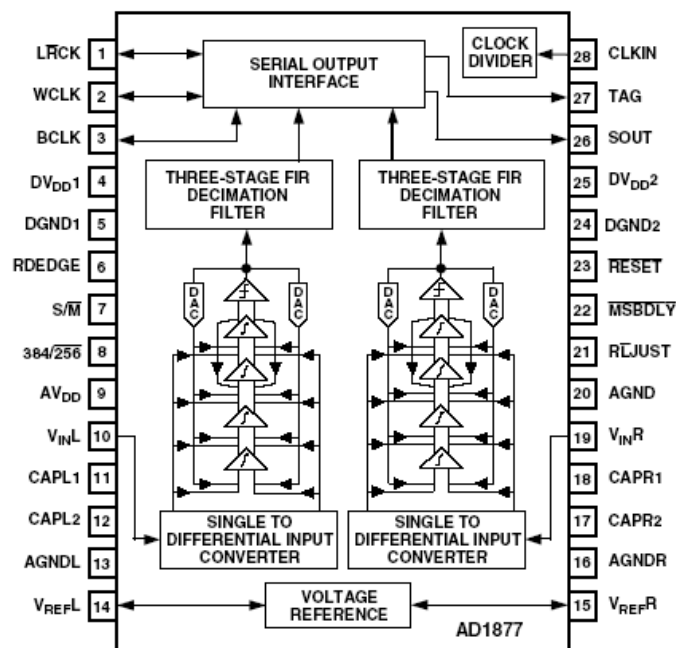
FEATURES

- Single 5 V Power Supply
- Single-Ended Dual-Channel Analog Inputs
- 92 dB (Typ) Dynamic Range
- 90 dB (Typ) S/(THD+N)
- 0.006 dB Decimator Passband Ripple
- Fourth-Order, 64-Times Oversampling $\Sigma\Delta$ Modulator
- Three-Stage, Linear-Phase Decimator
- $256 \times F_s$ or $384 \times F_s$ Input Clock
- Less than 100 μ W (Typ) Power-Down Mode
- Input Overrange Indication
- On-Chip Voltage Reference
- Flexible Serial Output Interface
- 28-Lead SOIC Package

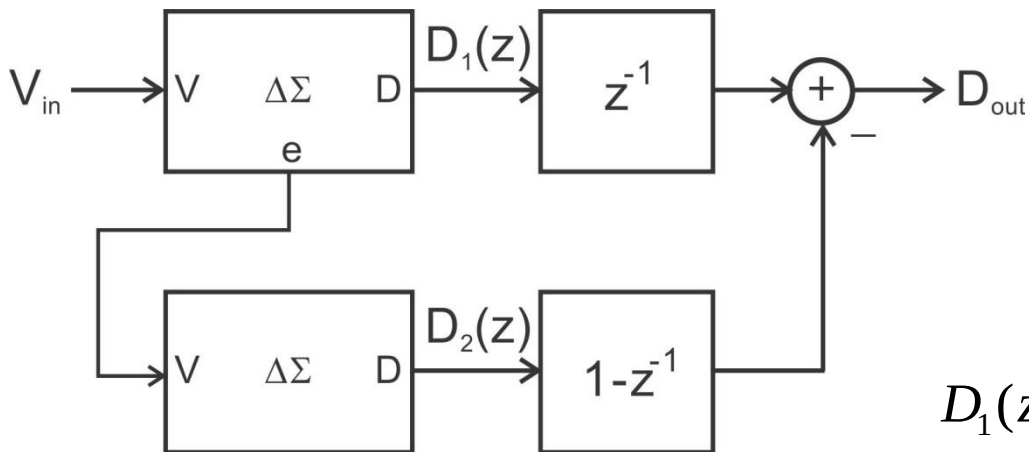
APPLICATIONS

- Consumer Digital Audio Receivers
- Digital Audio Recorders, Including Portables
 - CD-R, DCC, MD and DAT
- Multimedia and Consumer Electronic Equipment
- Sampling Music Synthesizers
- Digital Karaoke Systems

FUNCTIONAL BLOCK DIAGRAM



Multi-Stage Shaping (MASH)



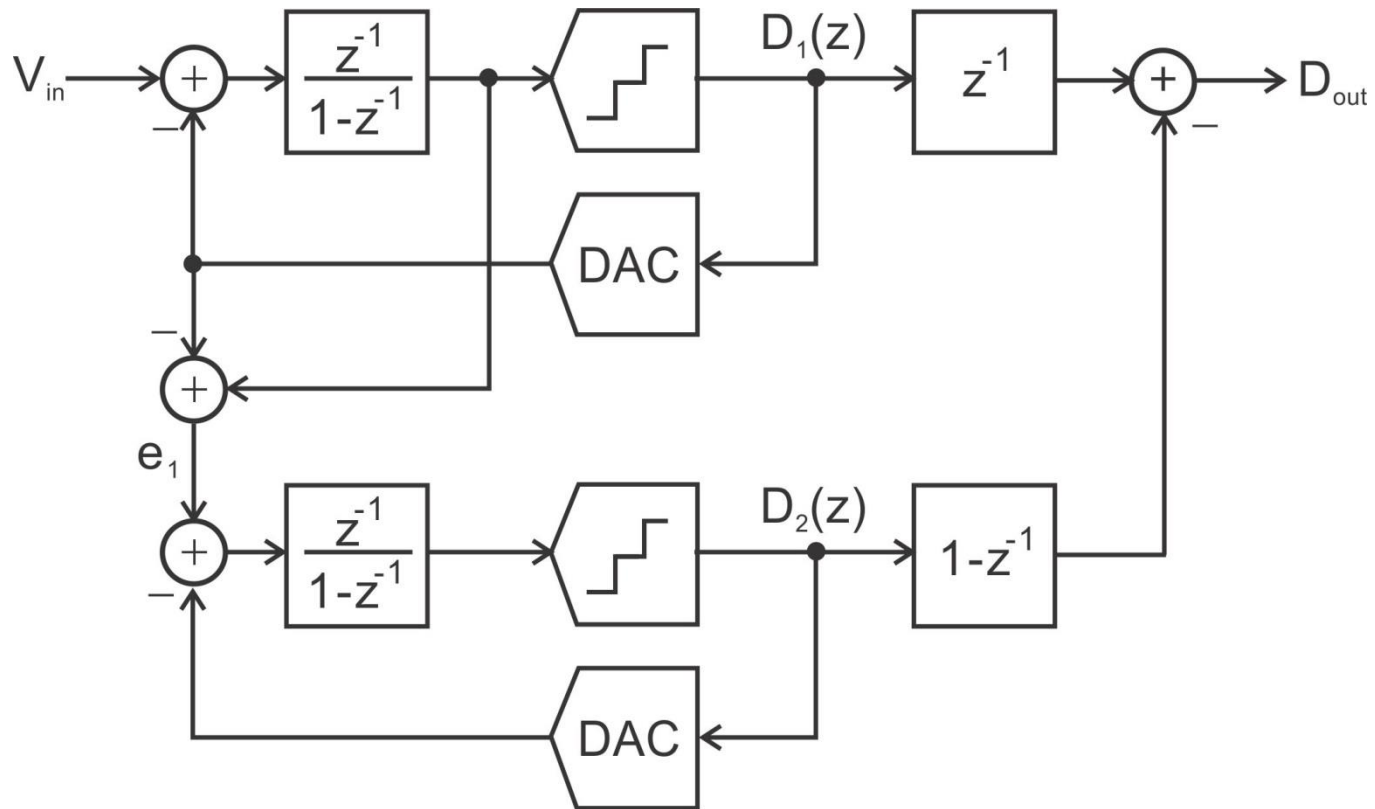
$$D_1(z) = V_{in}(z)z^{-1} + e_1(z)(1 - z^{-1})$$

$$D_2(z) = e_1(z)z^{-1} + e_2(z)(1 - z^{-1})$$

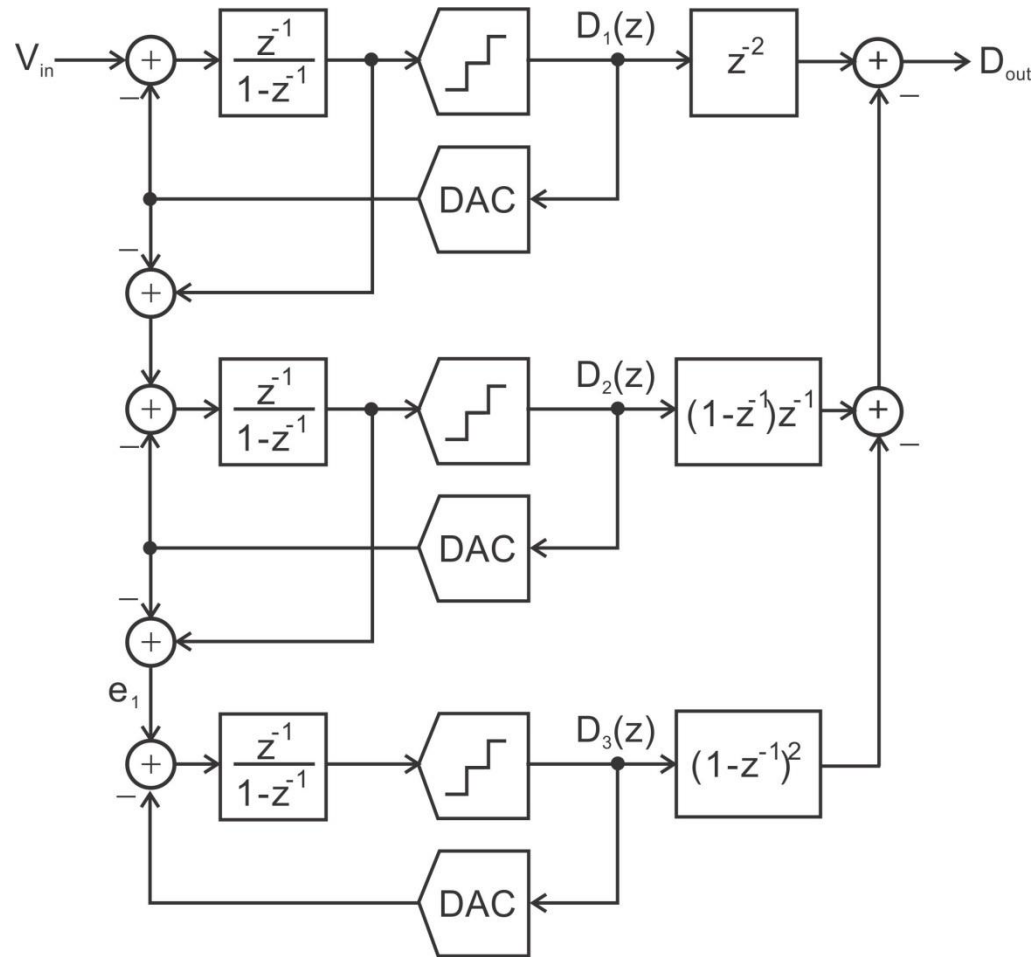
$$D_{out}(z) = D_1(z)z^{-1} - D_2(z)(1 - z^{-1}) = V_{in}(z)z^{-2} - e_2(z)(1 - z^{-1})^2$$

- Cascade of multiple low-order modulators to achieve high-order shaping
 - Better stability

Second Order (1-1) Cascade



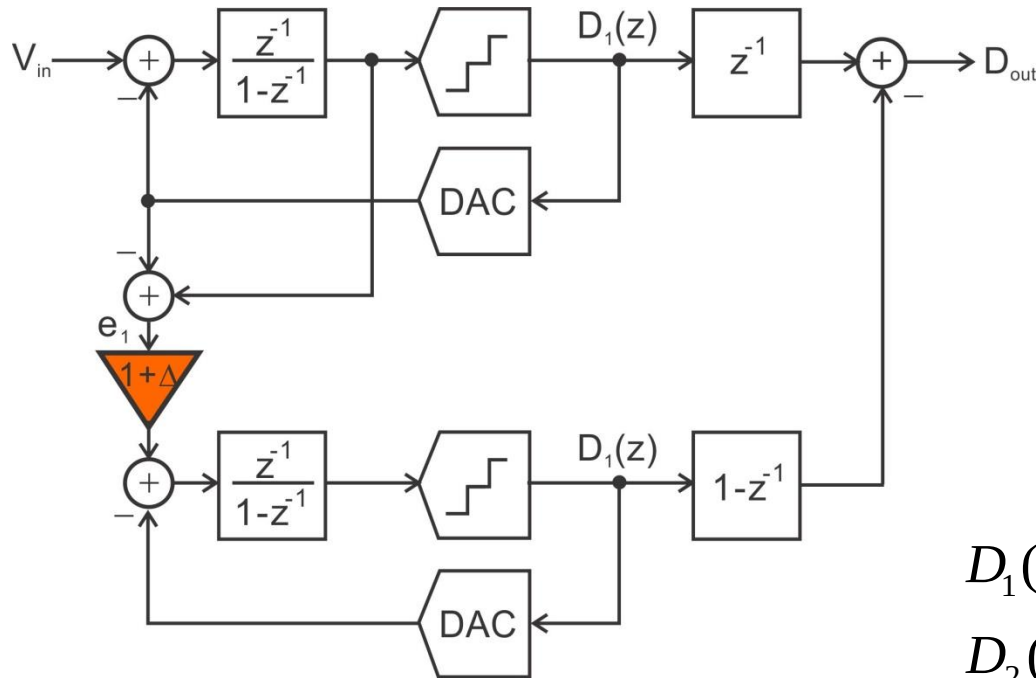
Third Order (1-1-1) Cascade



MASH Properties

- Overall noise shaping order = sum of all modulator orders
- Same benefits of higher-order noise shaping but without stability issues 😊
- However, cancellation of first stage quantization noise relies heavily on the accurate matching between the two modulators 😞

Mismatch Induced Noise Leakage



$$D_1(z) = V_{in}(z)z^{-1} + e_1(z)(1-z^{-1})$$

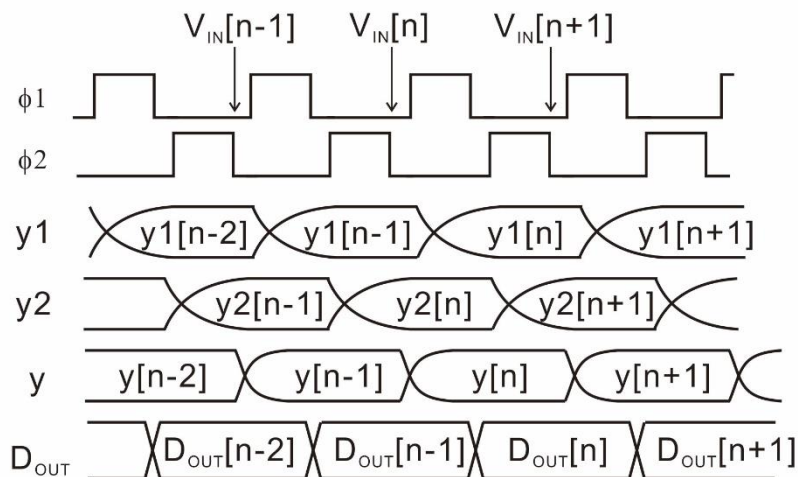
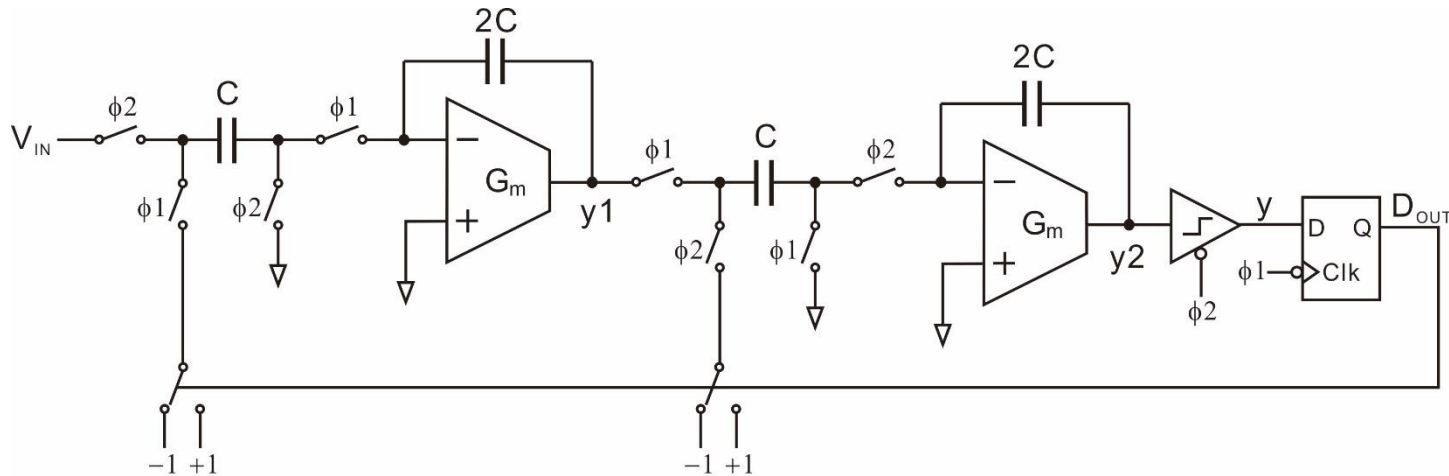
$$D_2(z) = (1+\Delta)e_1(z)z^{-1} + e_2(z)(1-z^{-1})$$

$$\begin{aligned} D_{out}(z) &= D_1(z)z^{-1} - D_2(z)(1-z^{-1}) \\ &= V_{in}(z)z^{-2} - e_1(z)\Delta(1-z^{-1})z^{-1} - e_2(z)(1-z^{-1})^2 \end{aligned}$$

Circuit Level Considerations

- Timing
- Noise
- Finite OTA gain
- OTA settling error
- OTA nonlinearity
- Others
 - Unwanted mixing effects
 - If V_{ref} contains $f_s/2$, out-of-band noise will be mixed down into the signal band!

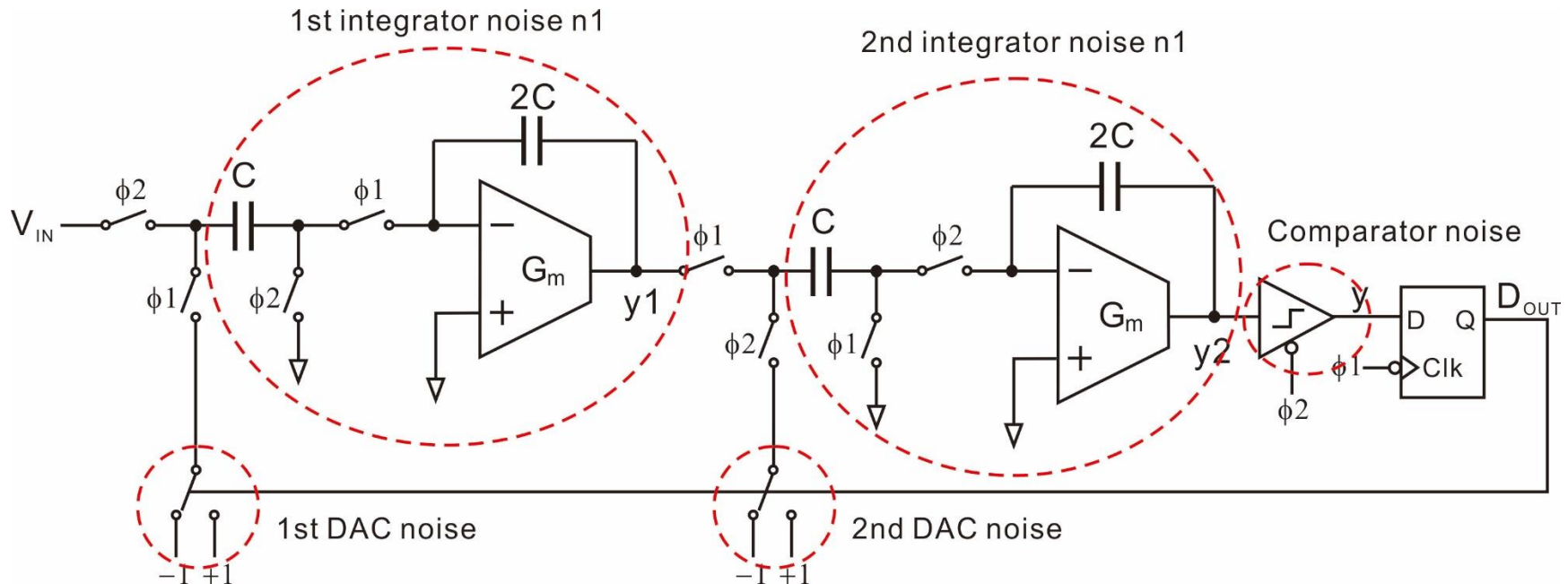
Example 2nd-order Modulator



$$\begin{cases} Y_1[n] = Y_1[n-1] + (V_{in}[n-1] - D_{out}[n-1]) / 2 \\ Y_2[n] = Y_2[n-1] + (Y_1[n-1] - D_{out}[n-1]) / 2 \\ D_{out}[n] = Y_2[n] > 0 \end{cases}$$

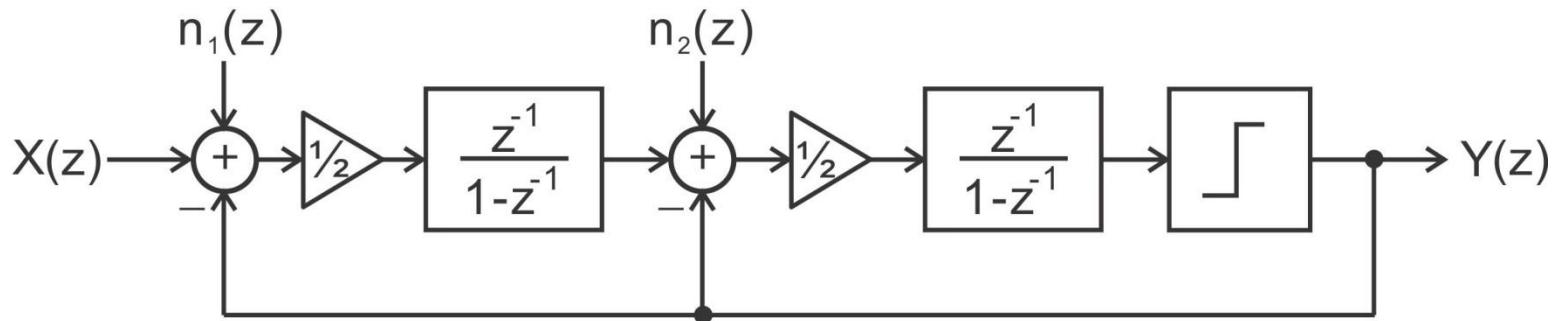
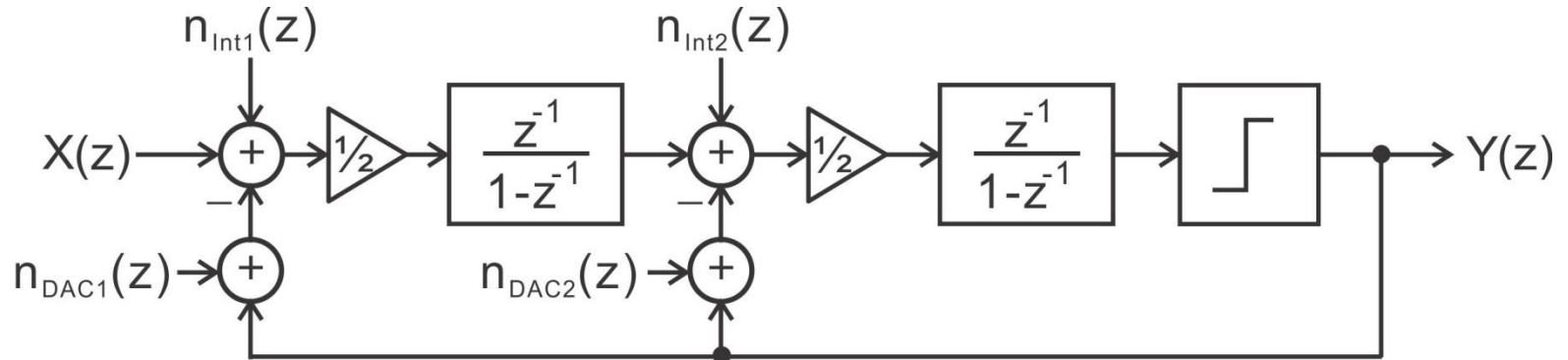
- OTAs and comparators all have entire half clock cycle to settle
- Not the only way

Electronic Noise (1)



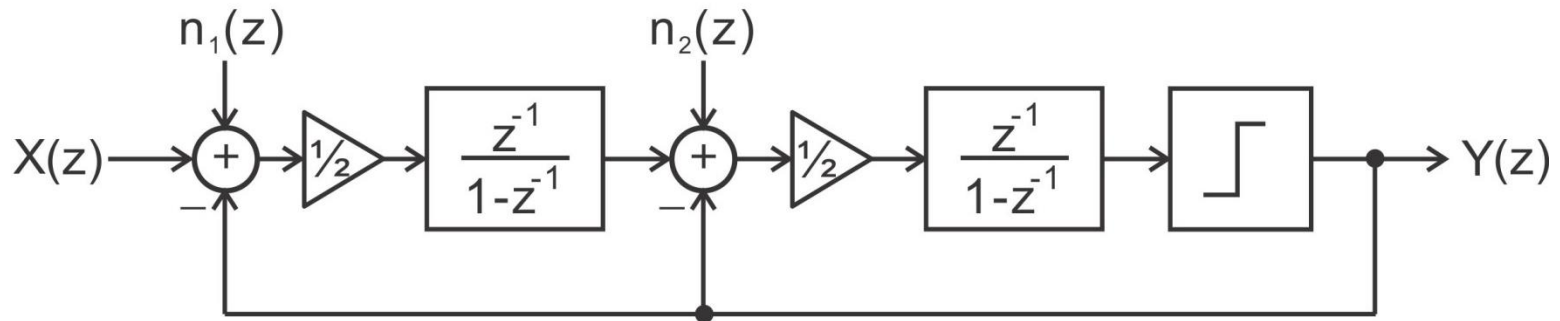
- Integrator electronic noise can be modeled as an input referred noise source
- Comparator electronic noise has the same effect as quantization noise
 - It is much smaller than quantization noise, and thus, can be ignored.
- DAC noise can be modeled as a noise source in the feedback path.⁸⁵

Electronic Noise (2)



$$\begin{cases} n_1(z) = n_{\text{int}1}(z) + n_{\text{DAC}1}(z) \\ n_2(z) = n_{\text{int}2}(z) + n_{\text{DAC}2}(z) \end{cases}$$

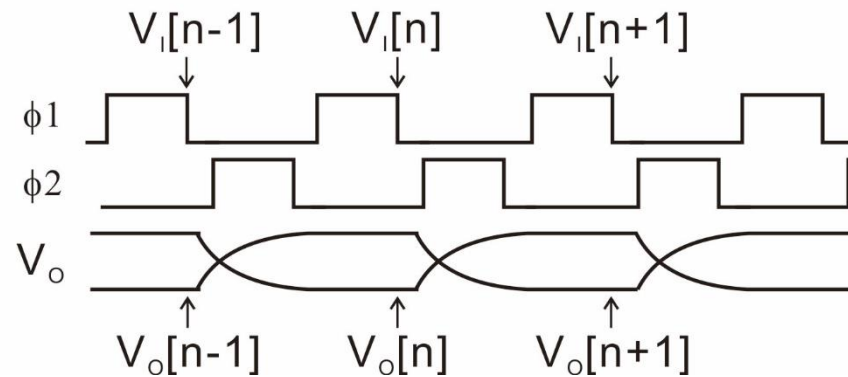
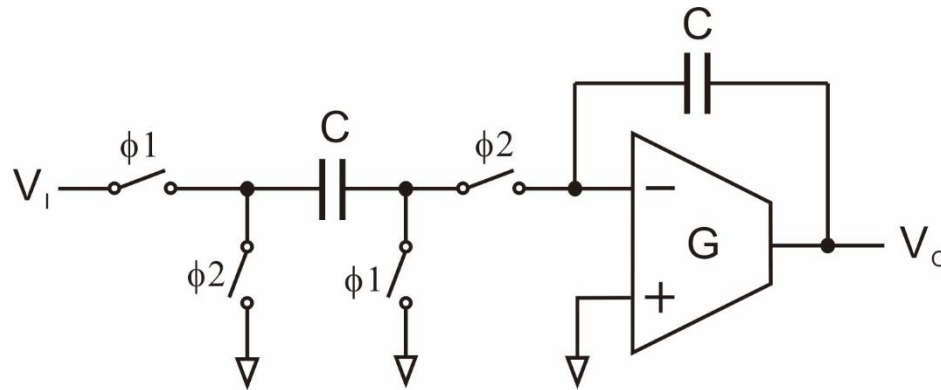
Electronic Noise (3)



$$\begin{cases} H_1(z) = STF = z^{-2} \\ H_2(z) = \frac{z^{-1}(1-z^{-1})}{D(z)} \end{cases}$$

- Noise from the 1st integrator and DAC adds directly to the output
 - Digital filter will reduce this noise by OSR
- Noise from the 2nd integrator and DAC is 1st-order shaped
 - Digital filter will remove most of it
- The key is the design of the 1st integrator and DAC

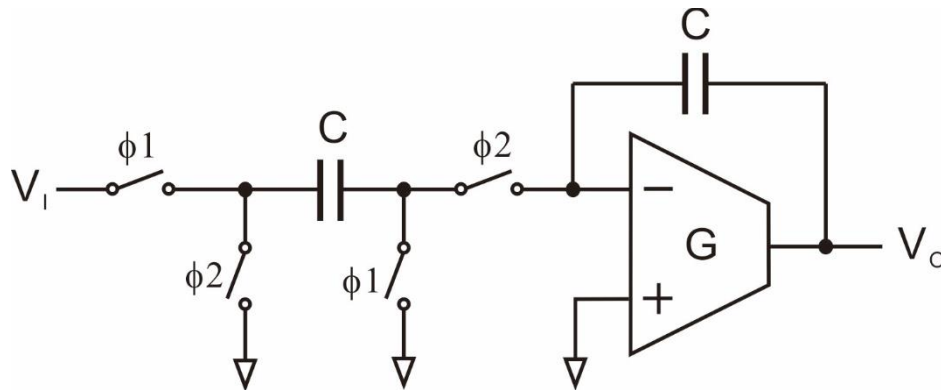
Integrator Analysis



$$V_o[n] = V_o[n-1] + V_i[n-1]$$

$$H(z) = \frac{z^{-1}}{1 - z^{-1}}$$

Finite OTA Gain (1)



$$\left(V_I(z) + V_O(z) \left(1 + \frac{1}{G} \right) \right) z^{-1} = V_O(z) \left(1 + \frac{2}{G} \right) \quad V_O[n] \approx V_O[n-1] \left(1 - \frac{1}{G} \right) + V_I[n-1] \left(1 - \frac{2}{G} \right)$$

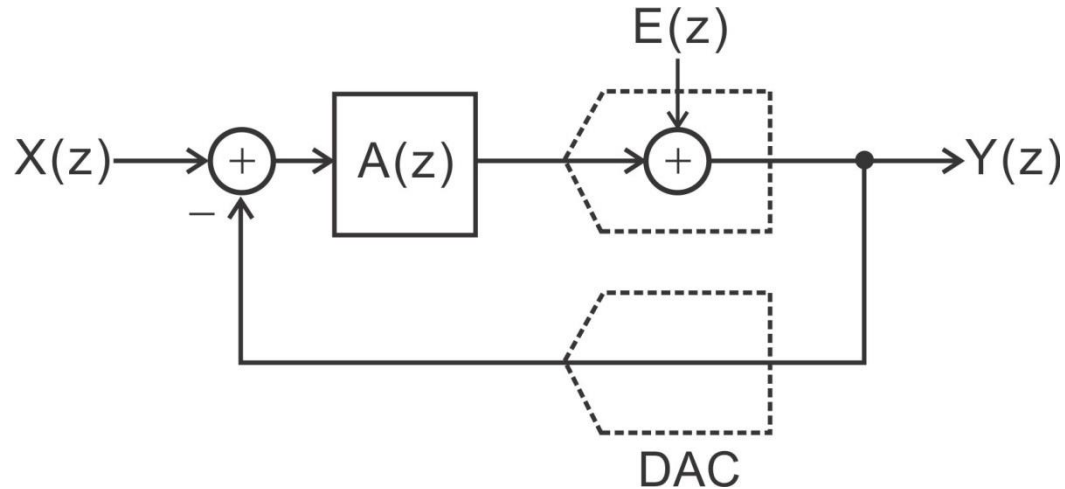
$$A(z) = \frac{z^{-1}}{1 + \frac{2}{G} - \left(1 + \frac{1}{G} \right) z^{-1}}$$

$$A(z=1) = G$$

- Does a finite gain cause nonlinearity?

Finite OTA Gain (2)

- Let us take a 1st-order modulator as an example

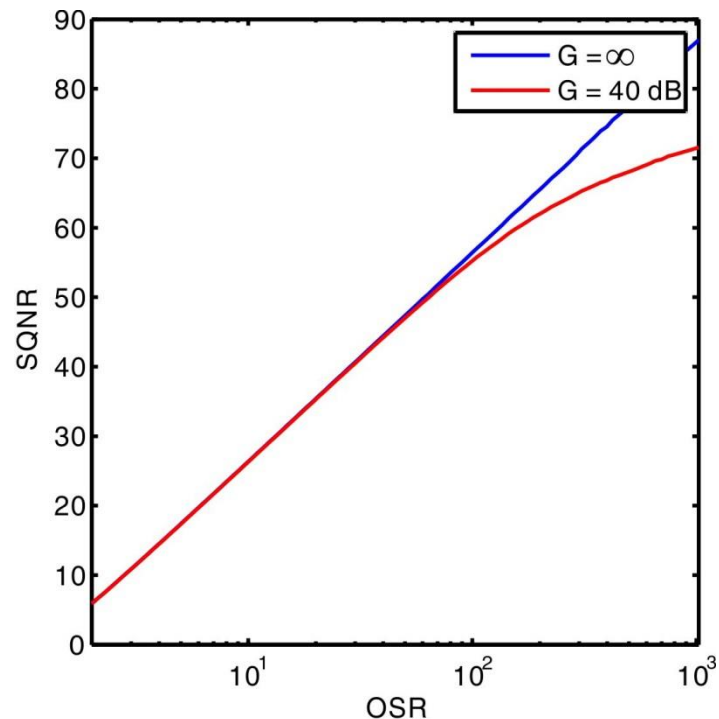
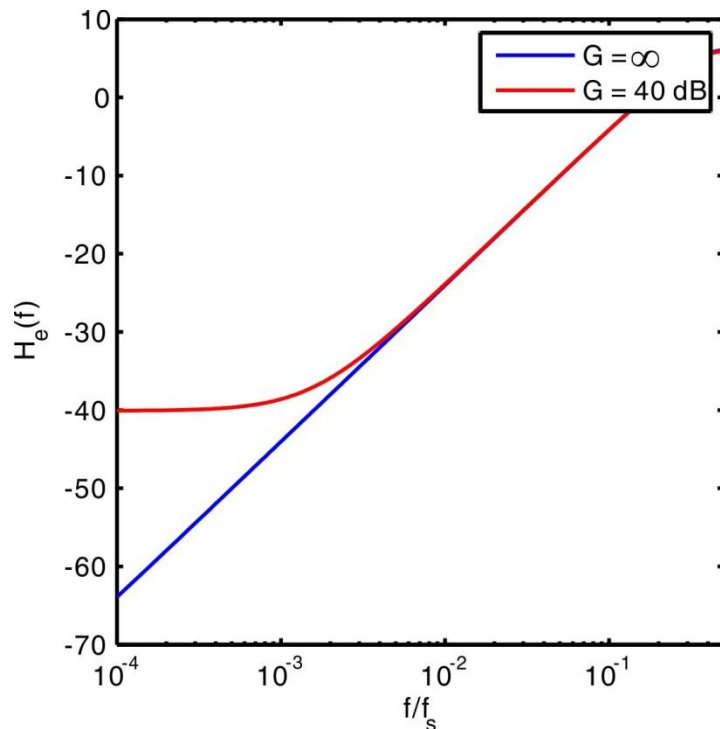


$$H_E(z) \big|_{G=\infty} = 1 - z^{-1}$$

$$H_E(z) = \frac{1 + \frac{2}{A} - \left(1 + \frac{1}{A}\right)z^{-1}}{1 + \frac{2}{A} - \frac{1}{A}z^{-1}}$$

- No distortion, but the effect of noise shaping is undermined

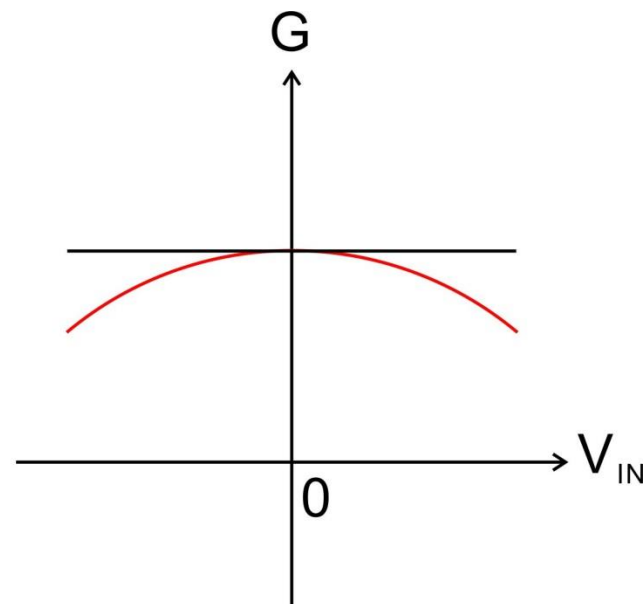
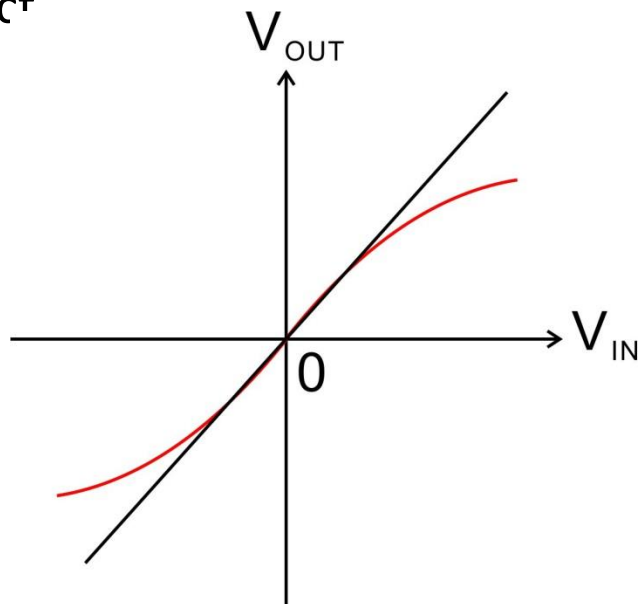
Finite OTA Gain (3)



- In general, we want $G \approx \text{OSR}$ to ensure good noise shaping
- E.g., for $\text{SQNR} = 60$ dB at $\text{OSR} = 128$, $G = 40$ dB is almost sufficient
- The requirement is not as strict as that for a pipelined ADC ($G \approx \text{SQNR}$)

Nonlinear OTA Gain

- A signal-dependent OTA gain is more problematic
- It causes $A(z)$ to be nonlinear, which causes the mixing of the quantization noise and the signal and the self mixing of the quantization noise, which leads to large in-band distortion and SQNR loss
- We usually over design the OTA gain so that its nonlinearity has negligible effect[†]

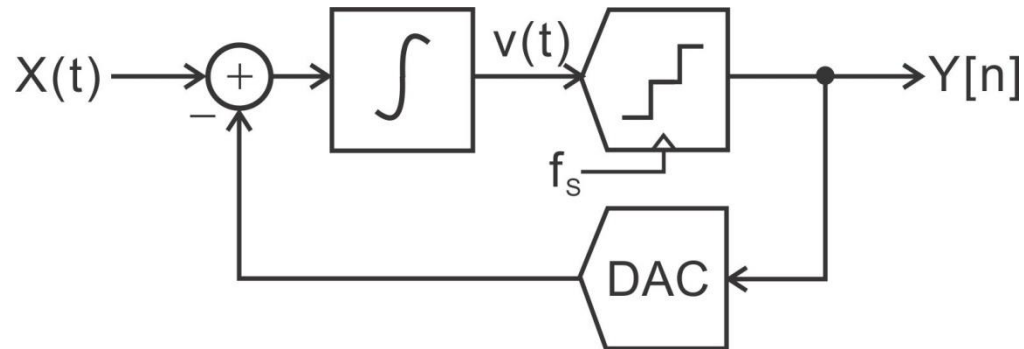


Continuous-Time $\Delta\Sigma$ ADC

- Advantages for discrete-time (DT) $\Delta\Sigma$ ADCs
 - The SC filters are very accurate and linear
 - The difference equations are independent of the clock rate. Transfer function naturally scales with the clock frequency
- Advantages for continuous-time (CT) $\Delta\Sigma$ ADCs
 - CT modulators have intrinsic anti-aliasing property
 - CT modulators can run at higher clock rate
 - It does not need to settle!

1st-order CT $\Delta\Sigma$ ADC

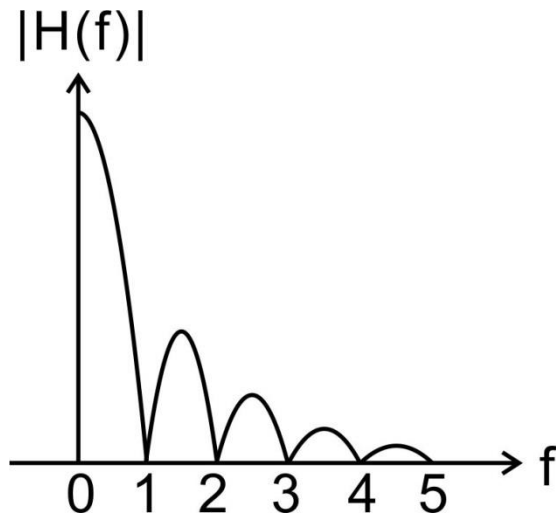
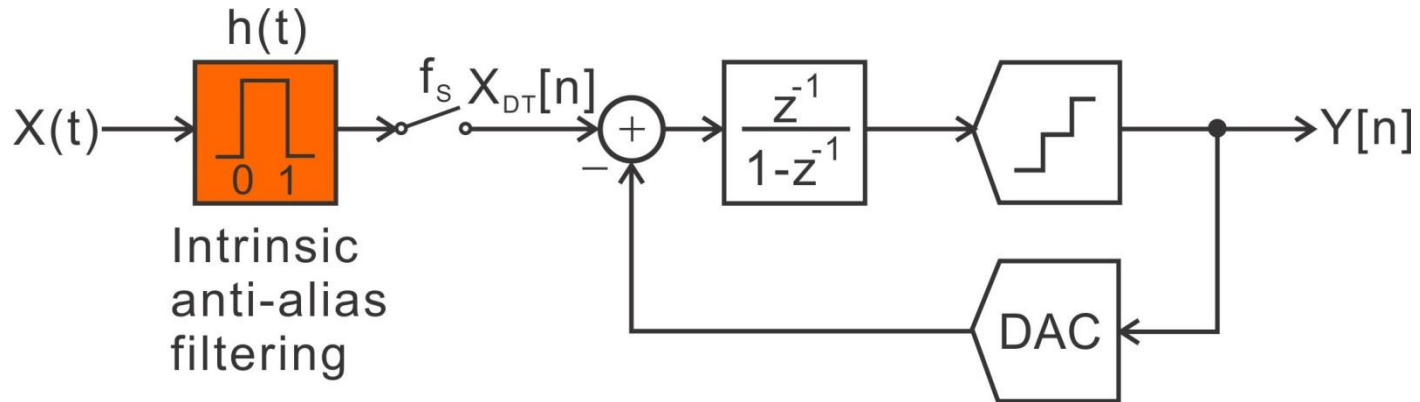
- Assuming sampling rate is 1 Hz



$$\begin{aligned} v[n] &= v[n-1] + \int_{n-1}^n (x(t) - y[n-1]) dt \\ &= v[n-1] + \int_{n-1}^n x(t) dt + y[n-1] \\ &= v[n-1] + x_{DT}[n-1] + y[n-1] \end{aligned}$$

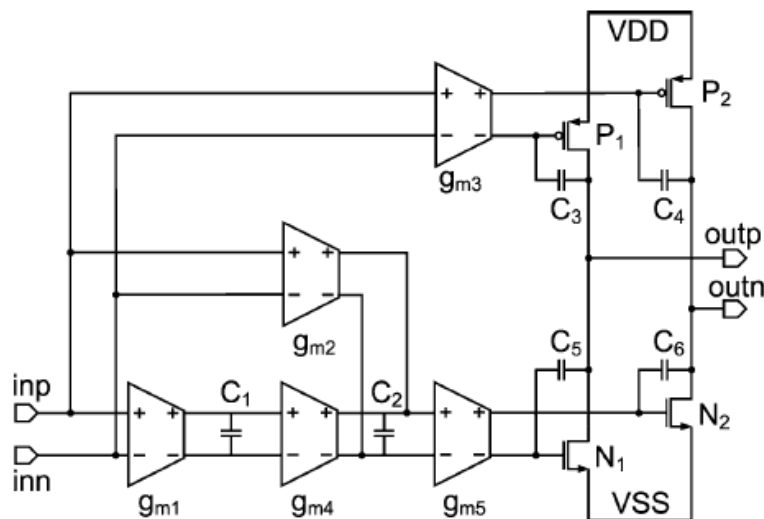
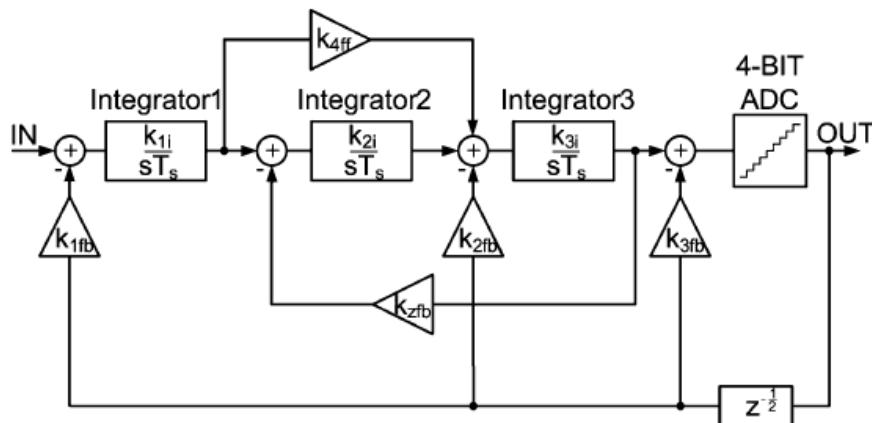
- The key is that the quantizer and the DAC operates in the DT fashion.

Equivalent DT Model



$$\begin{aligned}
 x_{DT}(t) &= x(t) * h(t) = x(t) * (u(t) - u(t-1)) \\
 &= \int_{-\infty}^{\infty} x(\tau) u(t-\tau) d\tau - \int_{-\infty}^{\infty} x(\tau) u(t-\tau-1) d\tau \\
 &= \int_{-\infty}^t x(\tau) d\tau - \int_{-\infty}^{t-1} x(\tau) d\tau \\
 &= \int_{t-1}^t x(\tau) d\tau
 \end{aligned}$$

CT $\Delta\Sigma$ ADC Example



1st Integrator 2nd Integrator 3rd Integrator Class-AB

[Mitteregger, ISSCC 2006]

Sampling Frequency		640MHz
Conversion Rate		40MS/s
Input Range		0-20MHz
Peak SNR		76dB
THD		-78dB
Peak SNDR		74dB
ENOB		12
Process		1.2V 130nm 1P8M CMOS
Chip Area		8.6mm ²
Power	Modulator	20mW
	Decimator 40MS/s	18mW
	PLL 2.56GHz	12mW
	I/O 1.8V	4mW

- Intrinsic anti-aliasing filter
- No need for settling
 - But OTA cannot slew
- Feedback DAC harder to design
- Clock jitter needs to be minimized